ColdBoxControlSystem_V18_Titan

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PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly]

PLC_DC/DC/RLY					
General\Project information					
Name	PLC_DC/DC/RLY	**Author**	MicheleBaschiera	**Comment**	This project is protected by CERN copyright, ensuring that all original content, including text, images, and designs, are legally safeguarded. Unauthorized reproduction, distribution, or use of any part of this project without explicit written permission from the copyright holder is strictly prohibited and may result in legal action.
Slot	1	**Rack**	0		
General\Catalog information					
Short designation	CPU 1214C DC/DC/Rly	**Description**	Work memory 150 KB; 24VDC power supply with DI14 x 24VDC SINK/SOURCE, DQ10 x relay and AI2 on board; 6 high-speed counters and 4 pulse outputs on-board; signal board expands on-board I/O; up to 3 communication modules for serial communication; up to 8 signal modules for I/O expansion; PROFINET IO controller, I-device, transport protocol TCP/IP, secure Open User Communication, S7 communication, Web server, OPC UA: Server DA	**Article number**	6ES7 214-1HG40-0XB0
Firmware version	V4.6		False		
General\Identification & Maintenance					
Plant designation		**Location identifier**		**Installation date**	2024-08-08 09:33:41.107
Additional information					
General\Checksums					
Text lists	FA 70 E8 75 1D 5A 8E 29	**Software**	BD 89 9D 84 69 26 21 A8		
PROFINET interface [X1]\General					
Name	PROFINET interface_1	**Author**	mbaschie	**Comment**	

PROFINET interface [X1]\General\Project information					
Name	DI 14/DQ 10_1	**Comment**		**Name**	AI 2_1
Comment					
PROFINET interface [X1]\Ethernet addresses\Interface networked with					
Subnet:	PN/IE_1				
PROFINET interface [X1]\Ethernet addresses\Internet protocol version 4 (IPv4)					
IP configuration	Set IP address in the project	**IP address:**	192.168.0.1	**Subnet mask:**	255.255.255.0
Use router	False				
PROFINET interface [X1]\Ethernet addresses\PROFINET					
PROFINET device name is set directly at the device	False	**Generate PROFINET device name automatically**	True	**PROFINET device name:**	plc_dc/dc/rly
Converted name:	plcxbdcxrdcxrrly34b4	**Device number:**	0		
PROFINET interface [X1]\Time synchronization					
Enable time synchronization via NTP server	Enable time synchronization via NTP server		IP addresses	**Server 1**	0.0.0.0
Server 2	0.0.0.0	**Server 3**	0.0.0.0	**Server 4**	0.0.0.0
Update interval	10sec			**CPU synchronizes the modules of the device.**	No synchronization
PROFINET interface [X1]\Digital inputs\Channel0					
Channel address	I0.0	**Input filters**	6.4 millisec	**Enable pulse catch**	0
PROFINET interface [X1]\Digital inputs\Channel0					
Enable rising edge detection	0	**Prefix Event Rising Edge**	49152	**Event name:**	0
Hardware interrupt:	0	**Rising edge0**	Rising edge0		
PROFINET interface [X1]\Digital inputs\Channel0					
Enable falling edge detection	0	**Prefix Event Falling Edge**	49280	**Event name:**	0
Hardware interrupt:	0	**Falling edge0**	Falling edge0		
PROFINET interface [X1]\Digital inputs\Channel1					
Channel address	I0.1	**Input filters**	6.4 millisec	**Enable pulse catch**	0
PROFINET interface [X1]\Digital inputs\Channel1					
Enable rising edge detection	0	**Prefix Event Rising Edge**	49153	**Event name:**	0
Hardware interrupt:	0	**Rising edge1**	Rising edge1		

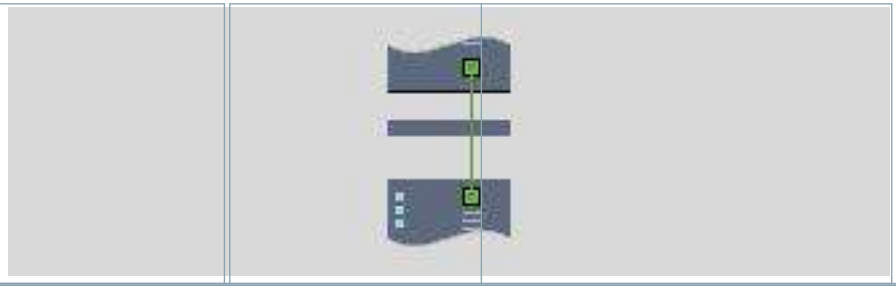
PROFINET interface [X1]\Digital inputs\Channel1					
Enable falling edge detection	0	**Prefix Event Falling Edge**	49281	**Event name:**	0
Hardware interrupt:	0	**Falling edge1**	Falling edge1		
PROFINET interface [X1]\Digital inputs\Channel2					
Channel address	I0.2	**Input filters**	6.4 millisec	**Enable pulse catch**	0
PROFINET interface [X1]\Digital inputs\Channel2					
Enable rising edge detection	0	**Prefix Event Rising Edge**	49154	**Event name:**	0
Hardware interrupt:	0	**Rising edge2**	Rising edge2		
PROFINET interface [X1]\Digital inputs\Channel2					
Enable falling edge detection	0	**Prefix Event Falling Edge**	49282	**Event name:**	0
Hardware interrupt:	0	**Falling edge2**	Falling edge2		
PROFINET interface [X1]\Digital inputs\Channel3					
Channel address	I0.3	**Input filters**	6.4 millisec	**Enable pulse catch**	0
PROFINET interface [X1]\Digital inputs\Channel3					
Enable rising edge detection	0	**Prefix Event Rising Edge**	49155	**Event name:**	0
Hardware interrupt:	0	**Rising edge3**	Rising edge3		
PROFINET interface [X1]\Digital inputs\Channel3					
Enable falling edge detection	0	**Prefix Event Falling Edge**	49283	**Event name:**	0
Hardware interrupt:	0	**Falling edge3**	Falling edge3		
PROFINET interface [X1]\Digital inputs\Channel4					
Channel address	I0.4	**Input filters**	6.4 millisec	**Enable pulse catch**	0
PROFINET interface [X1]\Digital inputs\Channel4					
Enable rising edge detection	0	**Prefix Event Rising Edge**	49156	**Event name:**	0
Hardware interrupt:	0	**Rising edge4**	Rising edge4		
PROFINET interface [X1]\Digital inputs\Channel4					
Enable falling edge detection	0	**Prefix Event Falling Edge**	49284	**Event name:**	0
Hardware interrupt:	0	**Falling edge4**	Falling edge4		
PROFINET interface [X1]\Digital inputs\Channel5					
Channel address	I0.5	**Input filters**	6.4 millisec	**Enable pulse catch**	0
PROFINET interface [X1]\Digital inputs\Channel5					
Enable rising edge detection	0	**Prefix Event Rising Edge**	49157	**Event name:**	0

Hardware interrupt:	0	**Rising edge5**	Rising edge5		
PROFINET interface [X1]\Digital inputs\Channel5					
Enable falling edge detection	0	**Prefix Event Falling Edge**	49285	**Event name:**	0
Hardware interrupt:	0	**Falling edge5**	Falling edge5		
PROFINET interface [X1]\Digital inputs\Channel6					
Channel address	I0.6	**Input filters**	6.4 millisec	**Enable pulse catch**	0
PROFINET interface [X1]\Digital inputs\Channel6					
Enable rising edge detection	0	**Prefix Event Rising Edge**	49158	**Event name:**	0
Hardware interrupt:	0	**Rising edge6**	Rising edge6		
PROFINET interface [X1]\Digital inputs\Channel6					
Enable falling edge detection	0	**Prefix Event Falling Edge**	49286	**Event name:**	0
Hardware interrupt:	0	**Falling edge6**	Falling edge6		
PROFINET interface [X1]\Digital inputs\Channel7					
Channel address	I0.7	**Input filters**	6.4 millisec	**Enable pulse catch**	0
PROFINET interface [X1]\Digital inputs\Channel7					
Enable rising edge detection	0	**Prefix Event Rising Edge**	49159	**Event name:**	0
Hardware interrupt:	0	**Rising edge7**	Rising edge7		
PROFINET interface [X1]\Digital inputs\Channel7					
Enable falling edge detection	0	**Prefix Event Falling Edge**	49287	**Event name:**	0
Hardware interrupt:	0	**Falling edge7**	Falling edge7		
PROFINET interface [X1]\Digital inputs\Channel8					
Channel address	I1.0	**Input filters**	6.4 millisec	**Enable pulse catch**	0
PROFINET interface [X1]\Digital inputs\Channel8					
Enable rising edge detection	0	**Prefix Event Rising Edge**	49160	**Event name:**	0
Hardware interrupt:	0	**Rising edge8**	Rising edge8		
PROFINET interface [X1]\Digital inputs\Channel8					
Enable falling edge detection	0	**Prefix Event Falling Edge**	49288	**Event name:**	0
Hardware interrupt:	0	**Falling edge8**	Falling edge8		
PROFINET interface [X1]\Digital inputs\Channel9					
Channel address	I1.1	**Input filters**	6.4 millisec	**Enable pulse catch**	0

PROFINET interface [X1]\Digital inputs\Channel9					
Enable rising edge detection	0	**Prefix Event Rising Edge**	49161	**Event name:**	0
Hardware interrupt:	0	**Rising edge9**	Rising edge9		
PROFINET interface [X1]\Digital inputs\Channel9					
Enable falling edge detection	0	**Prefix Event Falling Edge**	49289	**Event name:**	0
Hardware interrupt:	0	**Falling edge9**	Falling edge9		
PROFINET interface [X1]\Digital inputs\Channel10					
Channel address	I1.2	**Input filters**	6.4 millisec	**Enable pulse catch**	0
PROFINET interface [X1]\Digital inputs\Channel10					
Enable rising edge detection	0	**Prefix Event Rising Edge**	49162	**Event name:**	0
Hardware interrupt:	0	**Rising edge10**	Rising edge10		
PROFINET interface [X1]\Digital inputs\Channel10					
Enable falling edge detection	0	**Prefix Event Falling Edge**	49290	**Event name:**	0
Hardware interrupt:	0	**Falling edge10**	Falling edge10		
PROFINET interface [X1]\Digital inputs\Channel11					
Channel address	I1.3	**Input filters**	6.4 millisec	**Enable pulse catch**	0
PROFINET interface [X1]\Digital inputs\Channel11					
Enable rising edge detection	0	**Prefix Event Rising Edge**	49163	**Event name:**	0
Hardware interrupt:	0	**Rising edge11**	Rising edge11		
PROFINET interface [X1]\Digital inputs\Channel11					
Enable falling edge detection	0	**Prefix Event Falling Edge**	49291	**Event name:**	0
Hardware interrupt:	0	**Falling edge11**	Falling edge11		
PROFINET interface [X1]\Digital inputs\Channel12					
Channel address	I1.4	**Input filters**	6.4 millisec	**Enable pulse catch**	0
PROFINET interface [X1]\Digital inputs\Channel13					
Channel address	I1.5	**Input filters**	6.4 millisec	**Enable pulse catch**	0
PROFINET interface [X1]\Analog inputs\Noise reduction					
Integration time	50 Hz (20 ms)				
PROFINET interface [X1]\Analog inputs\Channel0					
Channel address	IW64	**Measurement type**	Voltage	**Voltage range**	0..10 V
Smoothing	Weak (4 cycles)			**Enable overflow diag-nostics**	1

PROFINET interface [X1]\Analog inputs\Channel1					
Channel address	IW66	**Measurement type**	Voltage	**Voltage range**	0..10 V
Smoothing	Weak (4 cycles)			**Enable overflow diagnostics**	1
PROFINET interface [X1]\Digital outputs					
Reaction to CPU STOP	Use substitute value				
PROFINET interface [X1]\Digital outputs\Channel0					
Channel address	Q0.0	**Substitute a value of 1 on a change from RUN to STOP.**	0		
PROFINET interface [X1]\Digital outputs\Channel1					
Channel address	Q0.1	**Substitute a value of 1 on a change from RUN to STOP.**	0		
PROFINET interface [X1]\Digital outputs\Channel2					
Channel address	Q0.2	**Substitute a value of 1 on a change from RUN to STOP.**	0		
PROFINET interface [X1]\Digital outputs\Channel3					
Channel address	Q0.3	**Substitute a value of 1 on a change from RUN to STOP.**	0		
PROFINET interface [X1]\Digital outputs\Channel4					
Channel address	Q0.4	**Substitute a value of 1 on a change from RUN to STOP.**	0		
PROFINET interface [X1]\Digital outputs\Channel5					
Channel address	Q0.5	**Substitute a value of 1 on a change from RUN to STOP.**	0		
PROFINET interface [X1]\Digital outputs\Channel6					
Channel address	Q0.6	**Substitute a value of 1 on a change from RUN to STOP.**	0		
PROFINET interface [X1]\Digital outputs\Channel7					
Channel address	Q0.7	**Substitute a value of 1 on a change from RUN to STOP.**	0		

PROFINET interface [X1]\Digital outputs\Channel8					
Channel address	Q1.0	**Substitute a value of 1 on a change from RUN to STOP.**	0		
PROFINET interface [X1]\Digital outputs\Channel9					
Channel address	Q1.1	**Substitute a value of 1 on a change from RUN to STOP.**	0		
PROFINET interface [X1]\Operating mode					
IO controller	True	**IO system**		**Device number**	0
IO device	False				
PROFINET interface [X1]\I/O addresses\Input addresses					
Start address	0.0	**End address**	1.7	**Organization block**	0
Process image	0				
PROFINET interface [X1]\I/O addresses\Input addresses					
Start address	64	**End address**	67	**Organization block**	0
Process image	0				
PROFINET interface [X1]\I/O addresses\Output addresses					
Start address	0.0	**End address**	1.7	**Organization block**	0
Process image	0				
PROFINET interface [X1]\Advanced options\Interface options					
Support device replacement without exchangeable medium	True	**Permit overwriting of device names of all assigned IO devices**	False	**Use IEC V2.2 LLDP mode**	False
Keep-Alive connection monitoring:	30s				
PROFINET interface [X1]\Advanced options\Real time settings\IO communication					
Send clock:	1.000ms				
PROFINET interface [X1]\Advanced options\Real time settings\Real time options					
Calculated bandwidth for cyclic IO data:	0.000ms	**Calculated bandwidth for cyclic IO data:**	0.000%		
PROFINET interface [X1]\Advanced options\Port [X1 P1]\General					
Name	Port_1	**Author**	mbaschie	**Comment**	
PROFINET interface [X1]\Advanced options\Port [X1 P1]\Port interconnection\Local port:					
Local port:	PLC_DC/DC/RLY\PROFINET interface_1 [X1]\Port_1 [X1 P1]	**Medium:**	Copper	**Cable name:**	---



PROFINET interface [X1]\Advanced options\Port [X1 P1]\Port interconnection\Partner port:					
	Monitoring of partner port is executed	**Partner port:**	HMI01.IE_CP_1\PROFINET Interface_1 [X1]\Port_1 [X1 P1 R]	**Medium:**	Copper
Cable length:	< 100 m				
PROFINET interface [X1]\Advanced options\Port [X1 P1]\Port options\Activate					
Activate this port for use	True				
PROFINET interface [X1]\Advanced options\Port [X1 P1]\Port options\Connection					
Transmission rate / duplex:	Automatic	**Monitor**	False	**Enable autonegotia-tion**	True
PROFINET interface [X1]\Advanced options\Port [X1 P1]\Port options\Boundaries					
End of detection of accessible devices	False	**End of topology dis-covery**	False	**End of the sync do-main**	False
PROFINET interface [X1]\Web server access					
Enable Web server for the IP address of this interface	False	**The Web server must also be activated in the properties of the PLC.**			
High speed counters (HSC)\HSC1\General\Enable					
Enable this high speed counter	0	**Enable this high speed counter**	0	**Enable this high speed counter**	0
Enable this high speed counter	0	**Enable this high speed counter**	0	**Enable this high speed counter**	0
High speed counters (HSC)\HSC1\General\Project information					
Name	HSC_1	**Comment**		**Name**	HSC_2
Comment		**Name**	HSC_3	**Comment**	
Name	HSC_4	**Comment**		**Name**	HSC_5
Comment		**Name**	HSC_6	**Comment**	
High speed counters (HSC)\HSC1\I/O addresses\Input addresses					
Start address	1000.0	**End address**	1003.7	**Start address**	1004.0

End address	1007.7	**Organization block**	0	**Start address**	1008.0
End address	1011.7	**Organization block**	0	**Process image**	0
Start address	1012.0	**End address**	1015.7	**Organization block**	0
Process image	0	**Start address**	1016.0	**End address**	1019.7
Organization block	0	**Process image**	0	**Start address**	1020.0
End address	1023.7	**Organization block**	0	**Process image**	0
Organization block	0	**Process image**	0	**Process image**	0
Pulse generators (PTO/PWM)\PTO1/PWM1\General\Enable					
Enable this pulse generator	0	**Enable this pulse generator**	0		
Pulse generators (PTO/PWM)\PTO1/PWM1\General\Project information					
Name	Pulse_1	**Comment**		**Name**	Pulse_2
Comment					
Pulse generators (PTO/PWM)\PTO1/PWM1\I/O addresses\Output addresses					
Start address	1000.0	**End address**	1001.7	**Start address**	1002.0
End address	1003.7	**Organization block**	0	**Organization block**	0
Process image	0	**Process image**	0		
Startup					
Startup after POWER ON	Warm restart - mode before POWER OFF	**Comparison preset to actual configuration**	Startup CPU even if mismatch	**Configuration time**	60000ms
OBs should be interruptible	0				
Cycle					
Cycle monitoring time [ms]	150ms			**Enable minimum cycle time for cyclic OBs**	0
Minimum cycle time	1ms				
Communication load					
Cycle load due to communication [%]	20%				
System and clock memory\System memory bits					
Enable the use of system memory byte	0	**Address of system memory byte (MBx)**	1	**First cycle**	
Diagnostic status changed		**Always 1 (high)**		**Always 0 (low)**	
System and clock memory\Clock memory bits					
Enable the use of clock memory byte	1	**Address of clock memory byte (MBx)**	30	**10 Hz clock**	%M30.0 (Clock_10Hz)
5 Hz clock	%M30.1 (Clock_5Hz)	**2.5 Hz clock**	%M30.2 (Clock_2.5Hz)	**2 Hz clock**	%M30.3 (Clock_2Hz)
1.25 Hz clock	%M30.4 (Clock_1.25Hz)	**1 Hz clock**	%M30.5 (Clock_1Hz)	**0.625 Hz clock**	%M30.6 (Clock_0.625Hz)

0.5 Hz clock	%M30.7 (Clock_0.5Hz)				
Web server\General					
Activate Web server on all modules of this device	False	**Permit access only with HTTPS**	False		
Web server\Automatic update					
Enable automatic update	True	**Update interval**	0s		
Web server\User management					
User name			**User rights**		
Everybody					
Web server\User-defined web pages					
Application name	**HTML source path**	**Default HTML page**	**Files with dynamic content**	**Web DB number**	**Fragment DB number**
		index.htm	.htm;.html	333	334
Web server\Overview of interfaces					
Device		**Interface**		**Enabled web server access**	
PLC_DC/DC/RLY		PROFINET interface_1		False	
User interface languages					
Assign project language			**User interface languages**		
None			German		
None			English		
None			French		
None			Spanish		
None			Italian		
None			Chinese (simplified)		
Time of day\Local time					
Time zone	(UTC +01:00) Berlin, Bern, Brussels, Rome, Stockholm, Vienna				
Time of day\Daylight saving time					
Activate daylight saving time	0	**Difference between standard and daylight saving time**	60min		
Time of day\Daylight saving time\Start of daylight saving time					
Starting week of the month:	Last		Sunday	**in**	March
at	1:00 a.m.				
Time of day\Daylight saving time\Start of standard time					
	Last		Sunday	**in**	October

at	2:00 a.m.				
Protection & Security					
Level of protection	No protection				
Protection & Security\Connection mechanisms					
Permit access with PUT/GET communication from remote partner	True				
Protection & Security\Security event					
Summarize diagnostics in case of high message volume	True	**Length of an interval**	20	**Unit**	seconds
Protection & Security\External load memory					
Disable copying from internal load memory to external load memory	False				
Advanced configuration\DnsParameterConfigurationMenu					
The Tree-Node DnsConfigurationMenu was not filled by some ACF					
Advanced configuration\Configuration control\Configuration control for central configuration					
Allow to reconfigure the device via the user program	0				
Advanced configuration\SNMP\SNMP configuration (Simple Network Management Protocol).					
Activate SNMP	True				
Configuration control\Configuration control for central configuration					
Allow to reconfigure the device via the user program	0				

Connection resources\				
	Station resources - Reserved - Maximum	**Station resources - Reserved - Configured**	**Station resources - Dynamic - Configured**	**Module resources - PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly] - Configured**
Maximum number of resources:		34	34	68
	Maximum	Configured	Configured	Configured
PG communication:	4	-	-	-
HMI communication:	12	2	0	2
S7 communication:	8	0	0	0
Open user communication:	8	0	0	0
Web communication:	2	-	-	-
OPC UA client/server communication:	0	-	-	-
Other communication:	-	-	0	0
Total resources used:		2	0	2
Available resources:		32	34	66

Overview of addresses\Overview of addresses\Overview of addresses					
Inputs	True	**Outputs**	True	**Address gaps**	False
Slot	True				

Type	Addr. from	Addr. to	Module	PIP	Device name	Device number	Size	Master / IO system	Rack	Slot
I	0	1	DI 14/DQ 10_1	Automatic update	PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly]	-	2 Bytes	-	0	1 1
O	0	1	DI 14/DQ 10_1	Automatic update	PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly]	-	2 Bytes	-	0	1 1
I	64	67	AI 2_1	Automatic update	PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly]	-	4 Bytes	-	0	1 2
I	1000	1003	HSC_1	Automatic update	PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly]	-	4 Bytes	-	0	1 16
I	1004	1007	HSC_2	Automatic update	PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly]	-	4 Bytes	-	0	1 17
I	1008	1011	HSC_3	Automatic update	PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly]	-	4 Bytes	-	0	1 18
I	1012	1015	HSC_4	Automatic update	PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly]	-	4 Bytes	-	0	1 19
I	1016	1019	HSC_5	Automatic update	PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly]	-	4 Bytes	-	0	1 20
I	1020	1023	HSC_6	Automatic update	PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly]	-	4 Bytes	-	0	1 21
O	1000	1001	Pulse_1	Automatic update	PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly]	-	2 Bytes	-	0	1 32
O	1002	1003	Pulse_2	Automatic update	PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly]	-	2 Bytes	-	0	1 33
O	1004	1005	Pulse_3	Automatic update	PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly]	-	2 Bytes	-	0	1 34
O	1006	1007	Pulse_4	Automatic update	PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly]	-	2 Bytes	-	0	1 35

Type	Addr. from	Addr. to	Module	PIP	Device name	Device number	Size	Master / IO system	Rack	Slot
I	96	111	A100	Automatic update	PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly]	-	16 Bytes	-	0	2
I	112	119	A101	Automatic update	PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly]	-	8 Bytes	-	0	3
O	112	115	A101	Automatic update	PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly]	-	4 Bytes	-	0	3

PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly] / Program blocks

Main OB1 [OB1]

Main OB1 Properties							
General							
Name	Main OB1	**Number**	1	**Type**	OB	**Language**	SCL
Numbering	Automatic						
Information							
Title	Main OB1	**Author**	MicheleBaschiera	**Comment**		**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Default value	Comment
Input			
Initial_Call	Bool		Initial call of this OB
Remanence	Bool		=True, if remanent data are available
Temp			
Constant			

```
0001 IF TRUE THEN
0002   "CBCS_MAIN_FC"();
0003 END_IF;
0004
0005
0006
0007
0008
```

Symbol	Address	Type	Comment

PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly] / Program blocks

Start [OB100] [OB100]

Start [OB100] Properties							
General							
Name	Start [OB100]	**Number**	100	**Type**	OB	**Language**	SCL
Numbering	Automatic						
Information							
Title	Complete Restart	**Author**	MicheleBaschiera	**Comment**	A "Startup" OB will execute one time when the operating mode of the PLC changes from STOP to RUN. After completion, the main "Program cycle" OB will begin executing.	**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Default value	Comment
Input			
LostRetentive	Bool		True if retentive data are lost
LostRTC	Bool		True if date and time are lost
Temp			
Constant			

```
(*
##################### SET HMI DEFAULTS ###########################
*)

 REGION SET_HMI_DEFAULTS

   //current settings for temp and dew for HMI
   "M/R/T_FC"(DATA_IN:="CBCS_MDB".System.MinTempValue,
        OFFSET:=10.0,
        DATA_OUT=>"HMI_DATAPROC_DB".System.ACT_MIN_TMP_SET);
   "M/R/T_FC"(DATA_IN:="CBCS_MDB".System.MaxTempValue,
        OFFSET:=10.0,
```

```
0013          DATA_OUT=>"HMI_DATAPROC_DB".System.ACT_MAX_TMP_SET);
0014     "M/R/T_FC"(DATA_IN:="CBCS_MDB".System.MinDewValue,
0015          OFFSET:=10.0,
0016          DATA_OUT=>"HMI_DATAPROC_DB".System.ACT_MIN_DEW_SET);
0017     "M/R/T_FC"(DATA_IN:="CBCS_MDB".System.MaxDewValue,
0018          OFFSET:=10.0,
0019          DATA_OUT=>"HMI_DATAPROC_DB".System.ACT_MAX_DEW_SET);
0020
0021     // set the new values to the current settings instead to be set all to zero for HMI
0022
0023     "M/R/T_FC"(DATA_IN:="CBCS_MDB".System.MinTempValue,
0024          OFFSET:=10,
0025          DATA_OUT=>"HMI_DATASET_DB".DAT_MIN_TMP);
0026
0027     "M/R/T_FC"(DATA_IN := "CBCS_MDB".System.MaxTempValue,
0028          OFFSET := 10,
0029          DATA_OUT => "HMI_DATASET_DB".DAT_MAX_TMP);
0030
0031     "M/R/T_FC"(DATA_IN := "CBCS_MDB".System.MinDewValue,
0032          OFFSET := 10,
0033          DATA_OUT => "HMI_DATASET_DB".DAT_MIN_DEW );
0034
0035     "M/R/T_FC"(DATA_IN := "CBCS_MDB".System.MaxDewValue,
0036          OFFSET := 10,
0037          DATA_OUT => "HMI_DATASET_DB".DAT_MAX_DEW);
0038
0039     // current settings for thermal cycle for HMI
0040     "HMI_DATAPROC_DB".System.COUNT_TCYC_AMOUNT_SET := "CBCS_MDB".System.Count_TCYC_Amount_Default;
0041     "HMI_DATASET_DB".DAT_COUNT_TCYC_AMOUNT_SET := "CBCS_MDB".System.Count_TCYC_Amount_Default;
0042
0043     //timer elapsed for HMI
0044     "HMI_DATAPROC_DB".System.COUNT_IDLE_STATMIN_MIN_SET := "CBCS_MDB".System.Interval_TCYC_MIN;
0045     "HMI_DATAPROC_DB".System.COUNT_IDLE_STATMAX_MIN_SET := "CBCS_MDB".System.Interval_TCYC_MIN;
0046     "HMI_DATASET_DB".DAT_COUNT_IDLEMIN_MIN_SET := "CBCS_MDB".System.Interval_TCYC_MIN;
0047     "HMI_DATASET_DB".DAT_COUNT_IDLEMAX_MIN_SET := "CBCS_MDB".System.Interval_TCYC_MIN;
0048
0049   END_REGION
0050
0051   REGION SET_MACHINE_DEFAULTS
```

```
0052     // temp and dew
0053    "CBCS_WDB".System.Temperature.TargetMinTemp := "CBCS_MDB".System.MinTempValue;
0054    "CBCS_WDB".System.Temperature.TargetMaxTemp := "CBCS_MDB".System.MaxTempValue;
0055    "CBCS_WDB".System.DewPoint.TargetMinDew := "CBCS_MDB".System.MinDewValue;
0056    "CBCS_WDB".System.DewPoint.TargetMaxDew := "CBCS_MDB".System.MaxDewValue;
0057    //thermal cycle
0058    "CBCS_WDB".System.COUNT_TCYC_AMOUNT_SET := "CBCS_MDB".System.Count_TCYC_Amount_Default;
0059    "CBCS_WDB".System.COUNT_IDLE_STATMIN_MIN_SET := "CBCS_MDB".System.Interval_TCYC_MIN;
0060    "CBCS_WDB".System.COUNT_IDLE_STATMAX_MIN_SET := "CBCS_MDB".System.Interval_TCYC_MIN;
0061
0062 END_REGION
0063
0064
0065 //RESET ALARMS AFTER RESTART
```

Symbol	Address	Type	Comment
"CBCS_MDB".System.Count_TCYC_Amount_Default		Int	Default amount of Thermal cycles
"CBCS_MDB".System.Interval_TCYC_MIN		Int	Default amount of minutes set to stay in idle state
"CBCS_MDB".System.MaxDewValue		Real	System defaults - max dew point
"CBCS_MDB".System.MaxTempValue		Real	Tray temperature - value for 20 mA from SCALE BLOCK
"CBCS_MDB".System.MinDewValue		Real	System defaults - min dew point
"CBCS_MDB".System.MinTempValue		Real	Tray temperature - value for 4 mA from SCALE BLOCK
"CBCS_WDB".System.COUNT_IDLE_STATMAX_MIN_SET	%DB15.DBW538	Int	Minutes Counter for Working Time idle warm state set from HMI
"CBCS_WDB".System.COUNT_IDLE_STATMIN_MIN_SET	%DB15.DBW536	Int	Minutes Counter for Working Time idle cold state set from HMI
"CBCS_WDB".System.COUNT_TCYC_AMOUNT_SET	%DB15.DBW548	Int	amount of thermal cycles set from HMI
"CBCS_WDB".System.DewPoint.TargetMaxDew	%DB15.DBD514	Real	Target max dew point
"CBCS_WDB".System.DewPoint.TargetMinDew	%DB15.DBD510	Real	Target min dew point
"CBCS_WDB".System.Temperature.TargetMaxTemp	%DB15.DBD472	Real	Max Target temperature
"CBCS_WDB".System.Temperature.TargetMinTemp	%DB15.DBD468	Real	Min Target temperature

Symbol	Address	Type	Comment
"HMI_DATAPROC_DB".System.ACT_MAX_DEW_SET		DInt	Max Dew setpoint
"HMI_DATAPROC_DB".System.ACT_MAX_TMP_SET		DInt	Max temp setpoin
"HMI_DATAPROC_DB".System.ACT_MIN_DEW_SET		DInt	Min Dew setpoint
"HMI_DATAPROC_DB".System.ACT_MIN_TMP_SET		DInt	MIn temp setpoin
"HMI_DATAPROC_DB".System.COUNT_IDLE_STATMAX_MIN_SET		Int	Minutes Counter for Working Time warm idle state SET
"HMI_DATAPROC_DB".System.COUNT_IDLE_STATMIN_MIN_SET		Int	Minutes Counter for Working Time cold idle state SET
"HMI_DATAPROC_DB".System.COUNT_TCYC_AMOUNT_SET		Int	current amount of t cycles set
"HMI_DATASET_DB".DAT_COUNT_IDLEMAX_MIN_SET		Int	Minutes set from HMI as idle state warm
"HMI_DATASET_DB".DAT_COUNT_IDLEMIN_MIN_SET		Int	Minutes set from HMI as idle state cold
"HMI_DATASET_DB".DAT_COUNT_TCYC_AMOUNT_SET		Int	Amount of thermal cycles set from HMI
"HMI_DATASET_DB".DAT_MAX_DEW		DInt	Max Dew set from HMI
"HMI_DATASET_DB".DAT_MAX_TMP		DInt	Max temp set from HMI
"HMI_DATASET_DB".DAT_MIN_DEW		DInt	Min Dew set from HMI
"HMI_DATASET_DB".DAT_MIN_TMP		DInt	MIn temp set from HMI

PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly] / Program blocks

AI_SCALE_FC [FC8]

AI_SCALE_FC Properties							
General							
Name	AI_SCALE_FC	**Number**	8	**Type**	FC	**Language**	SCL
Numbering	Manual						
Information							
Title	AI_SCALE_FC	**Author**	MicheleBaschiera	**Comment**	Funtion to scale analog input	**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Default value	Comment
Input			
IN_PTActive	Bool		Enable input
IN_PIW_Address	Int		Analog input address
IN_HI_LIM	Real		Output max value
IN_LO_LIM	Real		Output min value
IN_Safe_OUT	Real		Output safe value in case of any AI fault
IN_BIPOLAR	Bool		AI type: 0 = 0..10V, 0..5V, 4..20mA // 1 = +-10V, +-5V, +-20mA
IN_ACK	Bool		Alarms acknowledge
Output			
OUT_Value	Real		Output [engineering units]
InOut			
IN_OUT_Field_Fault	Bool		Field sensor fault or broken wire (latched)
IN_OUT_AI_CH_Fault	Bool		Analog input channel fault (latched)
Temp			
TMP_RET_VAL	Word		Return value from SCALE function
TMP_OUT	Real		OUT value from SCALE function
TMP_Field_Fault	Bool		Field sensor fault or broken wire
TMP_AI_CH_Fault	Bool		Analog input channel fault
TMP_RAW_AIN_VALUE	Real		Raw value from analog input
TMP_K1	Real		K1

Name	Data type	Default value	Comment
TMP_RAW_AIN_VALUE_DINT	DInt		Raw value from analog input, DINT format
TMP_PV_RANGE	Real		Process variable range [eng.units]
TMP_CARD_RANGE	Real		Card range [ticks]
TMP_PIW_PNT	DWord		Analog input pointer
TMP_AUX_REAL	Real		
Constant			
CARD_RNG_MN	Real	27648.0	
CARD_RNG_BI	Real	55296.0	
Return			
AI_SCALE_FC	Void		

```
0001 (*
0002 IF +-10mA OR +-10v OR +-5V range THEN card range = 27648*2 ticks, K1=-27648 // INBIPOLAR=1
0003 IF 4..20mA OR 0..10V OR 0..5V range THEN card range = 27648 ticks, K1=0 // INBIPOLAR=0
0004 *)
0005
0006 REGION INPUT TYPE SELECTION
0007   IF (#IN_BIPOLAR) THEN
0008     #TMP_CARD_RANGE := #CARD_RNG_BI;
0009     #TMP_K1 := (#CARD_RNG_MN * -1.0);
0010
0011   ELSE
0012     #TMP_CARD_RANGE := #CARD_RNG_MN;
0013     #TMP_K1 := 0.0;
0014   END_IF;
0015
0016
0017 END_REGION
0018
0019 REGION PROCESS VARIABLE RANGE CALCULATION
0020
0021   #TMP_PV_RANGE := #IN_HI_LIM - #IN_LO_LIM;
0022
0023 END_REGION
0024
0025 REGION ANALOG INPUT RAW VALUE
```

```
0026   #TMP_RAW_AIN_VALUE := INT_TO_REAL(#IN_PIW_Address);
0027   #TMP_RAW_AIN_VALUE_DINT := INT_TO_DINT(#IN_PIW_Address);
0028 END_REGION
0029
0030 REGION SCALING
0031 (*
0032  Process var.range[eng.units]
0033  ---------------------------- * (measure[ticks] - K1) + LO_LIM[eng.units]
0034  Card range[ticks]
0035 *)
0036
0037   #TMP_OUT := (((#TMP_RAW_AIN_VALUE - #TMP_K1) * #TMP_PV_RANGE)/#TMP_CARD_RANGE)+#IN_LO_LIM;
0038 END_REGION
0039
0040 REGION OUT OF RANGE TEST
0041
0042   IF (#TMP_RAW_AIN_VALUE_DINT >= 32767 OR #TMP_RAW_AIN_VALUE_DINT <= 1064) THEN
0043     #TMP_Field_Fault := TRUE;
0044
0045   ELSE
0046     #TMP_Field_Fault := FALSE;
0047
0048   END_IF;
0049
0050 END_REGION
0051
0052 REGION SAFE OUTPUT
0053 (*
0054  IF NOT ERRORS MOVE TMP TO OUT VALUE
0055 *)
0056   IF ( NOT #TMP_Field_Fault OR NOT #TMP_AI_CH_Fault AND #IN_PTActive)  THEN
0057     #OUT_Value := #TMP_OUT;
0058   ELSE
0059     #OUT_Value := #IN_Safe_OUT;
0060   END_IF;
0061
0062 END_REGION
0063
0064 REGION ALARM LATCHING FIELD
```

```
0065
0066   IF (#TMP_Field_Fault AND NOT #IN_ACK) THEN
0067     #IN_OUT_Field_Fault := TRUE;
0068
0069   ELSE
0070
0071     IF #IN_ACK THEN
0072       #IN_OUT_Field_Fault := FALSE;
0073     END_IF;
0074
0075   END_IF;
0076
0077 END_REGION
0078
0079 REGION ALARM LATCHING CHANNEL
0080
0081   IF (#TMP_AI_CH_Fault AND NOT #IN_ACK) THEN
0082     #IN_OUT_AI_CH_Fault := TRUE;
0083
0084   ELSE
0085
0086     IF #IN_ACK THEN
0087       #IN_OUT_AI_CH_Fault := FALSE;
0088     END_IF;
0089
0090   END_IF;
0091
0092 END_REGION
0093
0094 REGION END
0095   RETURN;
0096 END_REGION
0097
0098
```

Symbol	Address	Type	Comment
#CARD_RNG_BI	55296.0	Real	
#CARD_RNG_MN	27648.0	Real	
#IN_ACK		Bool	Alarms acknowledge

Symbol	Address	Type	Comment
#IN_BIPOLAR		Bool	AI type: 0 = 0..10V, 0..5V, 4..20mA // 1 = +-10V, +-5V, +-20mA
#IN_HI_LIM		Real	Output max value
#IN_LO_LIM		Real	Output min value
#IN_OUT_AI_CH_Fault		Bool	Analog input channel fault (latched)
#IN_OUT_Field_Fault		Bool	Field sensor fault or broken wire (latched)
#IN_PIW_Address		Int	Analog input address
#IN_PTActive		Bool	Enable input
#IN_Safe_OUT		Real	Output safe value in case of any AI fault
#OUT_Value		Real	Output [engineering units]
#TMP_AI_CH_Fault		Bool	Analog input channel fault
#TMP_CARD_RANGE		Real	Card range [ticks]
#TMP_Field_Fault		Bool	Field sensor fault or broken wire
#TMP_K1		Real	K1
#TMP_OUT		Real	OUT value from SCALE function
#TMP_PV_RANGE		Real	Process variable range [eng.units]
#TMP_RAW_AIN_VALUE		Real	Raw value from analog input
#TMP_RAW_AIN_VALUE_DINT		DInt	Raw value from analog input, DINT format

PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly] / Program blocks

ALRM_GEN_MAIN [FC14]

ALRM_GEN_MAIN Properties							
General							
Name	ALRM_GEN_MAIN	**Number**	14	**Type**	FC	**Language**	SCL
Numbering	Manual						
Information							
Title	ALRM_GEN_MAIN	**Author**	MicheleBaschiera	**Comment**	Function that generate the main alarms, set and reset	**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Default value	Comment
Input			
IN_ALRM	Bool		
Output			
InOut			
R_TRIG_ALRM	R_TRIG		
Input			
CLK	Bool		
Output			
Q	Bool		
InOut			
Static			
Stat_Bit	Bool		
INOUT_HMI_ALRM	Bool		
Temp			
TMP_R_TRIG_ALRM	Bool		
TMP_ALRM_PREV	Bool		
Constant			
Return			
ALRM_GEN_MAIN	Void		

```
0001 (*#R_TRIG_ALRM(CLK:=#IN_ALRM AND NOT #TMP_ALRM_PREV,
0002  Q=>#TMP_R_TRIG_ALRM);
0003
0004 IF #TMP_R_TRIG_ALRM THEN
0005  #INOUT_HMI_ALRM := TRUE;
0006
0007  #TMP_ALRM_PREV := NOT #TMP_R_TRIG_ALRM;
0008 END_IF;
0009 *)
0010
0011 IF #IN_ALRM THEN
0012   #INOUT_HMI_ALRM := true;
0013
0014 END_IF;
```

Symbol	Address	Type	Comment
#IN_ALRM		Bool	
#INOUT_HMI_ALRM		Bool	

PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly] / Program blocks

AO_UNSCALE_FC [FC9]

AO_UNSCALE_FC Properties							
General							
Name	AO_UNSCALE_FC	**Number**	9	**Type**	FC	**Language**	SCL
Numbering	Manual						
Information							
Title	AO_UNSCALE_FC	**Author**	MicheleBaschiera	**Comment**	Funtion to unscale analog output	**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Default value	Comment
Input			
IN_REAL	Real		Input value to be unscaled
IN_HI_LIM	Real		upper limit in engineering units
IN_LO_LIM	Real		lower limit in engineering units
IN_BIPOLAR	Bool		AI type: 0 = 0..10V, 0..5V, 4..20mA // 1 = +-10V, +-5V, +-20mA
Output			
OUT_Value	Int		Output Int unscale conversion
InOut			
IN_OUT_Fault	Bool		IN_REAL error
Temp			
TMP_PV_RANGE	Real		Process variable range [eng.units]
TMP_K2	Real		high limit for output value
TMP_K1	Real		low limit for output value
TMP_RSL	Real		temporary result
TMP_RET_VAL	Word		error temp
Constant			
CARD_RNG_MN	Real	27648.0	
Return			
AO_UNSCALE_FC	Void		

```
0001 (*
0002 IF +-10mA OR +-10v OR +-5V range THEN card range = 27648*2 ticks, K1=-27648 // INBIPOLAR=1
0003 IF 4..20mA OR 0..10V OR 0..5V range THEN card range = 27648 ticks, K1=0 // INBIPOLAR=0
0004 *)
0005
0006 REGION INPUT TYPE SELECTION
0007   IF (#IN_BIPOLAR) THEN
0008     #TMP_K2 := #CARD_RNG_MN;
0009     #TMP_K1 := (#CARD_RNG_MN * -1.0);
0010
0011   ELSE
0012     #TMP_K2 := #CARD_RNG_MN;
0013     #TMP_K1 := 0.0;
0014   END_IF;
0015
0016 END_REGION
0017
0018 REGION PROCESS VARIABLE RANGE CALCULATION
0019
0020   #TMP_PV_RANGE := #IN_HI_LIM - #IN_LO_LIM;
0021
0022 END_REGION
0023
0024 (*
0025  If the input value is outside the HI_LIM and LO_LIM range, the output
0026  is clamped to the nearer of either K1 or K2 and an error is logged.
0027  If the input value is exactly at a limit the output will be set to
0028  the computed limit with no error returned.
0029 *)
0030
0031 REGION REVERSE SCALING
0032
0033   //IF PV RANGE IS NEGATIVE
0034   IF (#TMP_PV_RANGE < 0.0) THEN
0035
0036     IF (#IN_REAL >= #IN_HI_LIM) THEN
0037       #TMP_RET_VAL := 8; // error
0038       #OUT_Value := REAL_TO_INT(#TMP_K1);
0039     ELSIF #IN_REAL <= #IN_LO_LIM THEN
```

```
0040        #TMP_RET_VAL := 8; // error
0041        #OUT_Value := REAL_TO_INT(#TMP_K2);
0042      ELSE
0043        // perform unscale calculation
0044        #TMP_RSL := #TMP_K2 - #TMP_K1;
0045        #OUT_Value := REAL_TO_INT(TRUNC(((#IN_REAL - #IN_LO_LIM) / #TMP_PV_RANGE) * #TMP_RSL + #TMP_K1));
0046      END_IF;
0047
0048    END_IF;
0049
0050  END_REGION
0051
0052
0053
0054  REGION SCALING
0055
0056    // CLAMP OUT VALUE
0057    IF #IN_REAL < #IN_LO_LIM THEN
0058      #TMP_RET_VAL := 8; // error
0059      #OUT_Value := REAL_TO_INT(#TMP_K1);
0060    ELSIF #IN_REAL > #IN_HI_LIM THEN
0061      #TMP_RET_VAL := 8; // error
0062      #OUT_Value := REAL_TO_INT(#TMP_K2);
0063    ELSE
0064      // perform unscale calculation
0065      #TMP_RSL := #TMP_K2 - #TMP_K1;
0066      #OUT_Value := REAL_TO_INT(TRUNC(((#IN_REAL - #IN_LO_LIM) / #TMP_PV_RANGE) * #TMP_RSL + #TMP_K1));
0067    END_IF;
0068
0069  END_REGION
0070
0071  REGION ALARM OUT SET
0072
0073    // set BR bit : no error-set BR bit to 1; with error-set BR bit to 0.
0074    IF #TMP_RET_VAL = 0 THEN
0075      #IN_OUT_Fault := FALSE;
0076    ELSE
0077      #IN_OUT_Fault := TRUE;
0078    END_IF;
```

```
0079
0080 END_REGION
0081
0082
0083 REGION END
0084   RETURN;
0085 END_REGION
0086
0087
```

Symbol	Address	Type	Comment
#CARD_RNG_MN	27648.0	Real	
#IN_BIPOLAR		Bool	AI type: 0 = 0..10V, 0..5V, 4..20mA // 1 = +-10V, +-5V, +-20mA
#IN_HI_LIM		Real	upper limit in engineering units
#IN_LO_LIM		Real	lower limit in engineering units
#IN_OUT_Fault		Bool	IN_REAL error
#IN_REAL		Real	Input value to be unscaled
#OUT_Value		Int	Output Int unscale conversion
#TMP_K1		Real	low limit for output value
#TMP_K2		Real	high limit for output value
#TMP_PV_RANGE		Real	Process variable range [eng.units]
#TMP_RET_VAL		Word	error temp
#TMP_RSL		Real	temporary result

PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly] / Program blocks

CBCS_MAIN_FC [FC20]

CBCS_MAIN_FC Properties							
General							
Name	CBCS_MAIN_FC	**Number**	20	**Type**	FC	**Language**	SCL
Numbering	Manual						
Information							
Title	CBCS_MAIN_FC	**Author**	MicheleBaschiera	**Comment**	Coldbox control system main function	**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Default value	Comment
Input			
Output			
InOut			
Temp			
Constant			
Return			
CBCS_MAIN_FC	Void		

```
0001 REGION COMMANDS FROM HMI
0002   IF TRUE THEN
0003
0004     "CBCS_WDB".HMI_CMD.ACT_DATA := "HMI_CMD_DB".HMI_CMD_DB.ACT_DATA;
0005     "CBCS_WDB".HMI_CMD.ALM_RES := "HMI_CMD_DB".HMI_CMD_DB.ALM_RES;
0006     "CBCS_WDB".HMI_CMD.DEWSEL1 := "HMI_CMD_DB".HMI_CMD_DB.DEWSEL1;
0007     "CBCS_WDB".HMI_CMD.DEWSEL2 := "HMI_CMD_DB".HMI_CMD_DB.DEWSEL2;
0008     "CBCS_WDB".HMI_CMD.LAMP_TST := "HMI_CMD_DB".HMI_CMD_DB.LAMP_TST;
0009     "CBCS_WDB".HMI_CMD.MAN := "HMI_CMD_DB".HMI_CMD_DB.MAN;
0010     "CBCS_WDB".HMI_CMD.OFF := "HMI_CMD_DB".HMI_CMD_DB.OFF;
0011     "CBCS_WDB".HMI_CMD.DECR_DW := "HMI_CMD_DB".HMI_CMD_DB.DECR_DW;
0012     "CBCS_WDB".HMI_CMD.INCR_UP := "HMI_CMD_DB".HMI_CMD_DB.INCR_UP;
0013     "CBCS_WDB".HMI_CMD.ON := "HMI_CMD_DB".HMI_CMD_DB.ON;
0014     "CBCS_WDB".HMI_CMD.PTSEL1 := "HMI_CMD_DB".HMI_CMD_DB.PTSEL1;
0015     "CBCS_WDB".HMI_CMD.PTSEL2 := "HMI_CMD_DB".HMI_CMD_DB.PTSEL2;
```

```
0016      "CBCS_WDB".HMI_CMD.PTSEL3 := "HMI_CMD_DB".HMI_CMD_DB.PTSEL3;
0017      "CBCS_WDB".HMI_CMD.PTSEL4 := "HMI_CMD_DB".HMI_CMD_DB.PTSEL4;
0018      "CBCS_WDB".HMI_CMD.PTSEL5 := "HMI_CMD_DB".HMI_CMD_DB.PTSEL5;
0019      "CBCS_WDB".HMI_CMD.PTSEL6 := "HMI_CMD_DB".HMI_CMD_DB.PTSEL6;
0020      "CBCS_WDB".HMI_CMD.PTSEL7 := "HMI_CMD_DB".HMI_CMD_DB.PTSEL7;
0021      "CBCS_WDB".HMI_CMD.PTSEL8 := "HMI_CMD_DB".HMI_CMD_DB.PTSEL8;
0022      "CBCS_WDB".HMI_CMD.STA := "HMI_CMD_DB".HMI_CMD_DB.STA;
0023      "CBCS_WDB".HMI_CMD.STP := "HMI_CMD_DB".HMI_CMD_DB.STP;
0024      "CBCS_WDB".HMI_CMD.SYNCH := "HMI_CMD_DB".HMI_CMD_DB.SYNCH;
0025      "CBCS_WDB".HMI_CMD.AUTO := "HMI_CMD_DB".HMI_CMD_DB.AUTO;
0026
0027    END_IF;
0028
0029  END_REGION
0030
0031
0032  REGION UNIT GENERAL STATUS - UNIT RUN OK
0033    // Statement section REGION
0034
0035  END_REGION
0036
0037  REGION DATA ACQUISITION
0038    "CBI_FB_DAQ_DB"();
0039  END_REGION
0040
0041  REGION COLDBOX MAIN CONTROLS
0042    "CBCS_FB_CTRL_DB"();
0043  END_REGION
0044
0045  REGION LAMP TEST
0046    // Statement section REGION
0047
0048  END_REGION
0049
0050  REGION GENERAL ALARM
0051    // Statement section REGION
0052
0053  END_REGION
0054
```

0055
0056
0057
0058

Symbol	Address	Type	Comment
"CBCS_WDB".HMI_CMD.ACT_DATA	%DB15.DBX557.5	Bool	Actualize Data
"CBCS_WDB".HMI_CMD.ALM_RES	%DB15.DBX556.5	Bool	Alarm reset
"CBCS_WDB".HMI_CMD.AUTO	%DB15.DBX556.1	Bool	Auto
"CBCS_WDB".HMI_CMD.DECR_DW	%DB15.DBX557.4	Bool	Decrease/Down
"CBCS_WDB".HMI_CMD.DEWSEL1	%DB15.DBX559.0	Bool	Dew selection 1
"CBCS_WDB".HMI_CMD.DEWSEL2	%DB15.DBX559.1	Bool	Dew selection 2
"CBCS_WDB".HMI_CMD.INCR_UP	%DB15.DBX557.3	Bool	Increase/Up
"CBCS_WDB".HMI_CMD.LAMP_TST	%DB15.DBX556.7	Bool	Lamp test
"CBCS_WDB".HMI_CMD.MAN	%DB15.DBX556.6	Bool	Manual
"CBCS_WDB".HMI_CMD.OFF	%DB15.DBX556.4	Bool	Off
"CBCS_WDB".HMI_CMD.ON	%DB15.DBX556.3	Bool	On
"CBCS_WDB".HMI_CMD.PTSEL1	%DB15.DBX558.0	Bool	PT Selection 1
"CBCS_WDB".HMI_CMD.PTSEL2	%DB15.DBX558.1	Bool	PT Selection 2
"CBCS_WDB".HMI_CMD.PTSEL3	%DB15.DBX558.2	Bool	PT Selection 3
"CBCS_WDB".HMI_CMD.PTSEL4	%DB15.DBX558.3	Bool	PT Selection 4
"CBCS_WDB".HMI_CMD.PTSEL5	%DB15.DBX558.4	Bool	PT Selection 5
"CBCS_WDB".HMI_CMD.PTSEL6	%DB15.DBX558.5	Bool	PT Selection 6
"CBCS_WDB".HMI_CMD.PTSEL7	%DB15.DBX558.6	Bool	PT Selection 7
"CBCS_WDB".HMI_CMD.PTSEL8	%DB15.DBX558.7	Bool	PT Selection 8
"CBCS_WDB".HMI_CMD.STA	%DB15.DBX556.0	Bool	Start
"CBCS_WDB".HMI_CMD.STP	%DB15.DBX556.2	Bool	Stop
"CBCS_WDB".HMI_CMD.SYNCH	%DB15.DBX557.6	Bool	Synchronization
"HMI_CMD_DB".HMI_CMD_DB.ACT_DATA		Bool	Actualize Data
"HMI_CMD_DB".HMI_CMD_DB.ALM_RES		Bool	Alarm reset
"HMI_CMD_DB".HMI_CMD_DB.AUTO		Bool	Auto
"HMI_CMD_DB".HMI_CMD_DB.DECR_DW		Bool	Decrease/Down
"HMI_CMD_DB".HMI_CMD_DB.DEW-SEL1		Bool	Dew selection 1
"HMI_CMD_DB".HMI_CMD_DB.DEW-SEL2		Bool	Dew selection 2

Symbol	Address	Type	Comment
"HMI_CMD_DB".HMI_CMD_DB.INCR_UP		Bool	Increase/Up
"HMI_CMD_DB".HMI_CMD_DB.LAMP_TST		Bool	Lamp test
"HMI_CMD_DB".HMI_CMD_DB.MAN		Bool	Manual
"HMI_CMD_DB".HMI_CMD_DB.OFF		Bool	Off
"HMI_CMD_DB".HMI_CMD_DB.ON		Bool	On
"HMI_CMD_DB".HMI_CMD_DB.PTSEL1		Bool	PT Selection 1
"HMI_CMD_DB".HMI_CMD_DB.PTSEL2		Bool	PT Selection 2
"HMI_CMD_DB".HMI_CMD_DB.PTSEL3		Bool	PT Selection 3
"HMI_CMD_DB".HMI_CMD_DB.PTSEL4		Bool	PT Selection 4
"HMI_CMD_DB".HMI_CMD_DB.PTSEL5		Bool	PT Selection 5
"HMI_CMD_DB".HMI_CMD_DB.PTSEL6		Bool	PT Selection 6
"HMI_CMD_DB".HMI_CMD_DB.PTSEL7		Bool	PT Selection 7
"HMI_CMD_DB".HMI_CMD_DB.PTSEL8		Bool	PT Selection 8
"HMI_CMD_DB".HMI_CMD_DB.STA		Bool	Start
"HMI_CMD_DB".HMI_CMD_DB.STP		Bool	Stop
"HMI_CMD_DB".HMI_CMD_DB.SYNCH		Bool	Synchronization

PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly] / Program blocks

COUNTER_IDLE_STATE [FC6]

COUNTER_IDLE_STATE Properties							
General							
Name	COUNTER_IDLE_STATE	**Number**	6	**Type**	FC	**Language**	SCL
Numbering	Manual						
Information							
Title	COUNTER_IDLE_STATE	**Author**	MicheleBaschiera	**Comment**	Function that return the amount of thermal cycles	**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Default value	Comment
Input			
IN_EN	Bool		enable input
IN_MIN_STAT_ELAPSED	Bool		min state time elapsed
IN_MAX_STAT_ELAPSED	Bool		max state time elapsed
IN_RST	Bool		reset input
Output			
OUT_COUNT	Int		counter amount
InOut			
CTU_TCYC	CTU_INT		
CU	Bool		
CD	Bool		
R	Bool		
LD	Bool		
QU	Bool		
QD	Bool		
PV	Int		
CV	Int		
CTU_INDEX	Int		
Temp			
Constant			

Name	Data type	Default value	Comment
Return			
COUNTER_IDLE_STATE	Void		

```
REGION COUNT THERMAL CYCLES AMOUNT

  IF #IN_EN THEN

    #CTU_TCYC(CU:=#IN_MIN_STAT_ELAPSED OR #IN_MAX_STAT_ELAPSED,
          R:= #IN_RST,
          PV:= #CTU_INDEX,
          CV=>#CTU_INDEX);

    #OUT_COUNT := #CTU_INDEX / 2;

  END_IF;


END_REGION

```

Symbol	Address	Type	Comment
#CTU_INDEX		Int	
#CTU_TCYC		IEC_Counter	
#IN_EN		Bool	enable input
#IN_MAX_STAT_ELAPSED		Bool	max state time elapsed
#IN_MIN_STAT_ELAPSED		Bool	min state time elapsed
#IN_RST		Bool	reset input
#OUT_COUNT		Int	counter amount

PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly] / Program blocks

IN_CNT_BOOL [FC7]

IN_CNT_BOOL Properties							
General							
Name	IN_CNT_BOOL	**Number**	7	**Type**	FC	**Language**	SCL
Numbering	Manual						
Information							
Title	IN_CNT_BOOL	**Author**	MicheleBaschiera	**Comment**	Return the amount of active input when are in TRUE state	**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Default value	Comment
Input			
IN_MB	Array[0..7] of Bool		
IN_MB[0]	Bool		
IN_MB[1]	Bool		
IN_MB[2]	Bool		
IN_MB[3]	Bool		
IN_MB[4]	Bool		
IN_MB[5]	Bool		
IN_MB[6]	Bool		
IN_MB[7]	Bool		
Output			
InOut			
INOUT_CNT	Int		
Temp			
TMP_CNT	Int		
TMP_RES	Int		
Constant			
Return			
IN_CNT_BOOL	Void		

```
0001
0002 FOR #TMP_CNT := 0 TO 7 DO
0003
0004   IF #IN_MB[#TMP_CNT] THEN
0005
0006     #TMP_RES := #TMP_RES + 1;
0007
0008   END_IF;
0009 END_FOR;
0010
0011 #INOUT_CNT := #TMP_RES;
0012 #TMP_RES := 0;
```

Symbol	Address	Type	Comment
#IN_MB[*]		Bool	
#INOUT_CNT		Int	
#TMP_CNT		Int	
#TMP_RES		Int	

PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly] / Program blocks

M/R/T_FC [FC4]

M/R/T_FC Properties							
General							
Name	M/R/T_FC	**Number**	4	**Type**	FC	**Language**	SCL
Numbering	Manual						
Information							
Title		**Author**	MicheleBaschiera	**Comment**	function to convert Real to DInt and log to HMI ProcData	**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Default value	Comment
Input			
DATA_IN	Real		
OFFSET	Real		
Output			
DATA_OUT	DInt		
InOut			
Temp			
Constant			
Return			
M/R/T_FC	Void		

```
0001 #DATA_OUT := ROUND(#DATA_IN * #OFFSET);
0002
```

Symbol	Address	Type	Comment
#DATA_IN		Real	
#DATA_OUT		DInt	
#OFFSET		Real	

PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly] / Program blocks

REF_AI_DINT_REAL [FC11]

REF_AI_DINT_REAL Properties							
General							
Name	REF_AI_DINT_REAL	**Number**	11	**Type**	FC	**Language**	SCL
Numbering	Manual						
Information							
Title	REF_AI_DINT_REAL	**Author**	MicheleBaschiera	**Comment**	Funtion to scale analog input to Real value	**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Default value	Comment
Input			
IN_int	DInt		
Output			
OUT_Real	Real		
Fault_Card	Bool		
InOut			
Temp			
OverflowValue	DInt		
Constant			
Return			
REF_AI_DINT_REAL	Void		

```
0001 (*
0002 CONVERT THE RAW VALUE FROM THE CARD TO REAL VALUE && CHECKS IS THE VALUE IS NOT OVERFLOWED
0003 RETURN
0004 *)
0005
0006 REGION CONV_DINT_REAL
0007
0008
0009   IF (NOT ((ABS(#IN_int)) >= #OverflowValue)) THEN
```

```
0010         // Statement section IF
0011        #OUT_Real := #IN_int * 0.01;
0012        #Fault_Card := FALSE;
0013    ELSE
0014        #OUT_Real := 0.0;
0015        #Fault_Card := TRUE;
0016
0017
0018    END_IF;
0019
0020
0021
0022
0023
0024
0025 END_REGION
0026
```

Symbol	Address	Type	Comment
#Fault_Card		Bool	
#IN_int		DInt	
#OUT_Real		Real	
#OverflowValue		DInt	

PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly] / Program blocks

REF_PT_INT_REAL [FC10]

REF_PT_INT_REAL Properties							
General							
Name	REF_PT_INT_REAL	**Number**	10	**Type**	FC	**Language**	SCL
Numbering	Manual						
Information							
Title	SCALE	**Author**	MicheleBaschiera	**Comment**	Scale funtion base on Siemens Standard	**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Default value	Comment
Input			
IN_PIW_Address	Int		Analog input address
IN_PTActive	Bool		Is PT Active
IN_Safe_OUT	Real		Output safe value in case of any AI fault
IN_ACK	Bool		Alarms acknowledge
Output			
OUT_Value	Real		Output [engineering units]
InOut			
IN_OUT_Field_Fault	Bool		Field sensor fault or broken wire (latched)
IN_OUT_AI_CH_Fault	Bool		Analog input channel fault (latched)
Temp			
TMP_RET_VAL	Word		Return value from SCALE function
TMP_OUT	Real		OUT value from SCALE function
TMP_Field_Fault	Bool		Field sensor fault or broken wire
TMP_AI_CH_Fault	Bool		Analog input channel fault
TMP_RAW_AIN_VALUE	Real		Raw value from analog input
TMP_K1	Real		K1
TMP_RAW_AIN_VALUE_DINT	DInt		Raw value from analog input, DINT format
TMP_PV_RANGE	Real		Process variable range [eng.units]
TMP_CARD_RANGE	Real		Card range [ticks]

Name	Data type	Default value	Comment
TMP_PIW_PNT	DWord		Analog input pointer
TMP_AUX_REAL	Real		
Temp_Value	Real		
Constant			
Return			
REF_PT_INT_REAL	Void		

```
0001 (*
0002 CONVERT THE RAW VALUE FROM THE CARD TO DEG CELSIUS && CHECKS IS THE VALUE IS NOT OVERFLOWED
0003 RETURN
0004 *)
0005
0006 REGION ANALOG INPUT RAW VALUE
0007   #TMP_RAW_AIN_VALUE := INT_TO_REAL(#IN_PIW_Address);
0008   #TMP_RAW_AIN_VALUE_DINT := INT_TO_DINT(#IN_PIW_Address);
0009 END_REGION
0010
0011 REGION SCALING
0012   #TMP_OUT := #TMP_RAW_AIN_VALUE * 0.1;
0013 END_REGION
0014
0015 REGION OUT OF RANGE TEST
0016
0017   IF (#TMP_RAW_AIN_VALUE_DINT >= 32767 OR #TMP_RAW_AIN_VALUE_DINT <= -32767) THEN
0018     #TMP_Field_Fault := TRUE;
0019
0020   ELSE
0021     #TMP_Field_Fault := FALSE;
0022
0023   END_IF;
0024
0025 END_REGION
0026
0027 REGION SAFE OUTPUT
0028 (*
0029  IF NOT ERRORS MOVE and PT IS ENABLED TMP TO OUT VALUE
0030 *)
```

```
0031   IF (#TMP_Field_Fault OR #TMP_AI_CH_Fault AND NOT #IN_PTActive) THEN
0032     #OUT_Value := #IN_Safe_OUT;
0033   ELSE
0034
0035     #OUT_Value := #TMP_OUT;
0036   END_IF;
0037
0038 END_REGION
0039
0040 REGION ALARM LATCHING FIELD
0041
0042   IF (#TMP_Field_Fault AND NOT #IN_ACK) THEN
0043     #IN_OUT_Field_Fault := TRUE;
0044
0045   ELSE
0046
0047     IF #IN_ACK THEN
0048       #IN_OUT_Field_Fault := FALSE;
0049     END_IF;
0050
0051   END_IF;
0052
0053 END_REGION
0054
0055 REGION ALARM LATCHING CHANNEL
0056
0057   IF (#TMP_AI_CH_Fault AND NOT #IN_ACK) THEN
0058     #IN_OUT_AI_CH_Fault := TRUE;
0059
0060   ELSE
0061
0062     IF #IN_ACK THEN
0063       #IN_OUT_AI_CH_Fault := FALSE;
0064     END_IF;
0065
0066   END_IF;
0067
0068 END_REGION
0069
```

```
0070
0071
0072 REGION END
0073   RETURN;
0074 END_REGION
0075
```

Symbol	Address	Type	Comment
#IN_ACK		Bool	Alarms acknowledge
#IN_OUT_AI_CH_Fault		Bool	Analog input channel fault (latched)
#IN_OUT_Field_Fault		Bool	Field sensor fault or broken wire (latched)
#IN_PIW_Address		Int	Analog input address
#IN_PTActive		Bool	Is PT Active
#IN_Safe_OUT		Real	Output safe value in case of any AI fault
#OUT_Value		Real	Output [engineering units]
#TMP_AI_CH_Fault		Bool	Analog input channel fault
#TMP_Field_Fault		Bool	Field sensor fault or broken wire
#TMP_OUT		Real	OUT value from SCALE function
#TMP_RAW_AIN_VALUE		Real	Raw value from analog input
#TMP_RAW_AIN_VALUE_DINT		DInt	Raw value from analog input, DINT format

PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly] / Program blocks

TEMP_LIMT_FC [FC12]

TEMP_LIMT_FC Properties							
General							
Name	TEMP_LIMT_FC	**Number**	12	**Type**	FC	**Language**	SCL
Numbering	Manual						
Information							
Title	TEMP_LIMT_FC	**Author**	MicheleBaschiera	**Comment**	Comparison function for temperature monitor	**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Default value	Comment
Input			
IN_TR_ActTemp	Real		Tray actual temperature
Output			
OUT_TempLimit_Hi	Bool		Temperature limits
OUT_TempLimit_Low	Bool		Temperature limits
InOut			
Temp			
Counter	Int		
Constant			
Return			
TEMP_LIMT_FC	Void		

```
0001
0002   IF (#IN_TR_ActTemp >= "CBCS_MDB".InnerColdbox.PT1000.SystemMaxTempThreshold) THEN
0003     #OUT_TempLimit_Hi := TRUE;
0004
0005   ELSIF (#IN_TR_ActTemp <= "CBCS_MDB".InnerColdbox.PT1000.SystemMinTempThreshold) THEN
0006     #OUT_TempLimit_Low := TRUE;
0007
0008   ELSE
0009     #OUT_TempLimit_Hi := FALSE;
```

```
0010      #OUT_TempLimit_Low := FALSE;
0011
0012    END_IF;
0013
0014
```

Symbol	Address	Type	Comment
"CBCS_MDB".InnerCold-box.PT1000.SystemMaxTempThres-hold		Real	System shutdown - max threshold temperature
"CBCS_MDB".InnerCold-box.PT1000.SystemMinTempThres-hold		Real	System shutdown - min threshold temperature
#IN_TR_ActTemp		Real	Tray actual temperature
#OUT_TempLimit_Hi		Bool	Temperature limits
#OUT_TempLimit_Low		Bool	Temperature limits

PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly] / Program blocks

TEMP_TARGET_FC [FC13]

TEMP_TARGET_FC Properties							
General							
Name	TEMP_TARGET_FC	**Number**	13	**Type**	FC	**Language**	SCL
Numbering	Manual						
Information							
Title	TEMP_TARGET_FC	**Author**	MicheleBaschiera	**Comment**	Determine the target temperatore to be send to the chiller	**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Default value	Comment
Input			
IN_ACT_TEMP	Real		actual system temperature
IN_HYSTERESIS	Real		hysteresis
IN_PT_AVG	Real		PT average temperature
IN_DEW_AVG	Real		Dew Point average temperature
IN_DLT_TEMP	Real		Dew Temp delta range
IN_MINTEMP	Real		min value to be passed to the chiller
IN_MAXTEMP	Real		max value to be passed to the chiller
IN_CMD_COLD	Bool		command going cold
IN_CMD_WARM	Bool		command going warm
Output			
OUT_TARGET_TEMP	Real		target temp to be send to the chiller
OUT_ACT_INMINTARGET	Bool		actual temperatute reach min target
OUT_ACT_INMAXTARGET	Bool		actual temperatute reach max target
InOut			
IO_MINTGRT_SET	Bool		min setpoint target set
IO_MAXTGRT_SET	Bool		max setpoint target set
Temp			
TMP_TEMP_LOTHRSLD_MIN	Real		MIN lower threshold

Name	Data type	Default value	Comment
TMP_TEMP_HITHRSLD_MIN	Real		MIN higher threshold
TMP_TEMP_LOTHRSLD_MAX	Real		MAX lower threshold
TMP_TEMP_HITHRSLD_MAX	Real		MAX higher threshold
TMP_ACT_DEWTEMP_DLT	Real		Actual delta DEW and PT Temperature
Constant			
▼ Return			
TEMP_TARGET_FC	Void		

```
0001 REGION HYSTERESIS RANGE VALUES
0002   #TMP_TEMP_LOTHRSLD_MIN := #IN_MINTEMP - #IN_HYSTERESIS;
0003   #TMP_TEMP_HITHRSLD_MIN := #IN_MINTEMP + #IN_HYSTERESIS;
0004   #TMP_TEMP_LOTHRSLD_MAX := #IN_MAXTEMP - #IN_HYSTERESIS;
0005   #TMP_TEMP_HITHRSLD_MAX := #IN_MAXTEMP + #IN_HYSTERESIS;
0006 END_REGION
0007
0008 REGION CALC DELTA DEW AND TEMP
0009
0010   #TMP_ACT_DEWTEMP_DLT := #IN_PT_AVG - #IN_DEW_AVG;
0011
0012 END_REGION
0013
0014
0015 REGION AUTO/MANUAL MODE TARGET TEMPERATURE SELECTION
0016
0017   IF TRUE THEN
0018
0019     IF #IN_CMD_COLD THEN
0020
0021       #IO_MINTGRT_SET := TRUE;
0022       #IO_MAXTGRT_SET := FALSE;
0023
0024     END_IF;
0025
0026     IF #IN_CMD_WARM THEN
0027
0028       #IO_MINTGRT_SET := FALSE;
0029       #IO_MAXTGRT_SET := TRUE;
```

```
0030
0031      END_IF;
0032
0033    ELSE
0034      #OUT_ACT_INMINTARGET := FALSE;
0035      #OUT_ACT_INMAXTARGET := FALSE;
0036
0037    END_IF;
0038
0039    // MAX SETPOINT REACH
0040    IF (#IN_ACT_TEMP <= #TMP_TEMP_HITHRSLD_MAX AND #IN_ACT_TEMP >= #TMP_TEMP_LOTHRSLD_MAX) THEN
0041
0042      #OUT_ACT_INMAXTARGET := TRUE;
0043
0044    // MIN SETPOINT REACH
0045    ELSIF (#IN_ACT_TEMP <= #TMP_TEMP_HITHRSLD_MIN AND #IN_ACT_TEMP >= #TMP_TEMP_LOTHRSLD_MIN) THEN
0046
0047      #OUT_ACT_INMINTARGET := TRUE;
0048
0049    // ACTUAL TEMP IS IN BETWEEN THE TWO SETPOINTS
0050    ELSIF (#IN_ACT_TEMP < #TMP_TEMP_LOTHRSLD_MAX AND #IN_ACT_TEMP > #TMP_TEMP_HITHRSLD_MIN) THEN
0051
0052      IF #IO_MINTGRT_SET THEN
0053        #OUT_TARGET_TEMP := #IN_MINTEMP;
0054
0055      ELSIF #IO_MAXTGRT_SET THEN
0056        #OUT_TARGET_TEMP := #IN_MAXTEMP;
0057      END_IF;
0058
0059    END_IF;
0060
0061  END_REGION
0062
0063  REGION SET OUTPUT TARGET TEMPERATURE
0064
0065        //SET TO NEAREST IF TARGETS ARE NOT SET
0066        IF (NOT #IO_MINTGRT_SET AND NOT #IO_MAXTGRT_SET) THEN
0067
0068          IF #IN_ACT_TEMP > 0 THEN
```

```
0069            #OUT_TARGET_TEMP := #IN_MAXTEMP;
0070            #IO_MINTGRT_SET := FALSE;
0071            #IO_MAXTGRT_SET := TRUE;
0072            //
0073          ELSE
0074            #OUT_TARGET_TEMP := #IN_MINTEMP;
0075            #IO_MINTGRT_SET := TRUE;
0076            #IO_MAXTGRT_SET := FALSE;
0077          END_IF;
0078
0079          //SET TO MIN TARGET TEMPERATURE IF IS SET TO MIN
0080        ELSIF #IO_MINTGRT_SET THEN
0081
0082          // INNER TEMP IN BELOW DEW POINT TO BE IMPROVED
0083          IF (#TMP_ACT_DEWTEMP_DLT < 0) THEN
0084
0085            #OUT_TARGET_TEMP := #IN_ACT_TEMP + (#IN_DLT_TEMP /2);
0086
0087          // INNER TEMP IS IN DELTA RANGE TO BE IMPROVED
0088          ELSIF (#TMP_ACT_DEWTEMP_DLT > 0 AND #TMP_ACT_DEWTEMP_DLT < #IN_DLT_TEMP) THEN
0089
0090            #OUT_TARGET_TEMP := #IN_ACT_TEMP;
0091
0092          ELSE
0093            #OUT_TARGET_TEMP := #IN_MINTEMP;
0094
0095          END_IF;
0096
0097          //SET TO MAX TARGET TEMPERATURE IF IS SET TO MAX
0098        ELSIF #IO_MAXTGRT_SET THEN
0099          #OUT_TARGET_TEMP := #IN_MAXTEMP;
0100
0101        END_IF;
0102
0103 END_REGION
0104
0105
0106
0107
```

0108

Symbol	Address	Type	Comment
#IN_ACT_TEMP		Real	actual system temperature
#IN_CMD_COLD		Bool	command going cold
#IN_CMD_WARM		Bool	command going warm
#IN_DEW_AVG		Real	Dew Point average temperature
#IN_DLT_TEMP		Real	Dew Temp delta range
#IN_HYSTERESIS		Real	hysteresis
#IN_MAXTEMP		Real	max value to be passed to the chiller
#IN_MINTEMP		Real	min value to be passed to the chiller
#IN_PT_AVG		Real	PT average temperature
#IO_MAXTGRT_SET		Bool	max setpoint target set
#IO_MINTGRT_SET		Bool	min setpoint target set
#OUT_ACT_INMAXTARGET		Bool	actual temperatute reach max target
#OUT_ACT_INMINTARGET		Bool	actual temperatute reach min target
#OUT_TARGET_TEMP		Real	target temp to be send to the chiller
#TMP_ACT_DEWTEMP_DLT		Real	Actual delta DEW and PT Temperature
#TMP_TEMP_HITHRSLD_MAX		Real	MAX higher threshold
#TMP_TEMP_HITHRSLD_MIN		Real	MIN higher threshold
#TMP_TEMP_LOTHRSLD_MAX		Real	MAX lower threshold
#TMP_TEMP_LOTHRSLD_MIN		Real	MIN lower threshold

PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly] / Program blocks

TIMER_IDLE_STATE [FC5]

TIMER_IDLE_STATE Properties							
General							
Name	TIMER_IDLE_STATE	**Number**	5	**Type**	FC	**Language**	SCL
Numbering	Manual						
Information							
Title	TIMER_IDLE_STATE	**Author**	MicheleBaschiera	**Comment**	Funtion that return the time elapsed during the thermal cycle idle state	**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Default value	Comment
Input			
IN_STA_COUNT	Bool		Input signal to start counting
Output			
InOut			
IO_SEC	Int		Elapsed time in seconds (0-59)
IO_MIN	Int		Elapsed time in minutes (0-59)
IO_HOUR	Int		Elapsed time in hours
IO_COUNT_RUN	Bool		Indicates if the counter is running
TP_COUNTER_IDLE_STATE	TP_TIME		
PT	Time		
ET	Time		
IN	Bool		
Q	Bool		
F_TRIG_IDLE_STATE	F_TRIG		
Input			
CLK	Bool		
Output			
Q	Bool		
InOut			

Name	Data type	Default value	Comment
Static			
Stat_Bit	Bool		
Temp			
TMP_F_TRIGGER	Bool		
Constant			
Return			
TIMER_IDLE_STATE	Void		

```
0001 REGION DETECT RISING EDGE AND RESET COUNTER
0002
0003   IF (#IN_STA_COUNT AND NOT #IO_COUNT_RUN) THEN
0004
0005     #IO_COUNT_RUN := TRUE;
0006     #IO_SEC := 0;
0007     #IO_MIN := 0;
0008     #IO_HOUR := 0;
0009
0010   END_IF;
0011
0012 END_REGION
0013
0014 REGION COUNT START
0015
0016   IF #IO_COUNT_RUN AND #IN_STA_COUNT THEN
0017
0018     #TP_COUNTER_IDLE_STATE(IN:=TRUE,
0019                  PT:=T#1S);
0020
0021     #F_TRIG_IDLE_STATE(CLK:=#TP_COUNTER_IDLE_STATE.Q,
0022            Q=>#TMP_F_TRIGGER);
0023
0024
0025      IF #TMP_F_TRIGGER THEN
0026
0027       #TP_COUNTER_IDLE_STATE.IN := FALSE; // Reset the timer
0028       //#TP_COUNTER_IDLE_STATE.IN := TRUE;
```

```
0029        #TMP_F_TRIGGER := FALSE;
0030
0031
0032      #IO_SEC := #IO_SEC + 1;
0033
0034        IF #IO_SEC >= 60 THEN
0035
0036          #IO_SEC := 0;
0037          #IO_MIN := #IO_MIN + 1;
0038
0039          IF #IO_MIN >= 60 THEN
0040
0041            #IO_MIN := 0;
0042            #IO_HOUR := #IO_HOUR + 1;
0043
0044          END_IF;
0045        END_IF;
0046      END_IF;
0047
0048    ELSE
0049
0050      #TP_COUNTER_IDLE_STATE.IN := FALSE;
0051      #IO_COUNT_RUN := FALSE;
0052
0053    END_IF;
0054
0055
0056  END_REGION
0057
0058
```

Symbol	Address	Type	Comment
#F_TRIG_IDLE_STATE		Multi_FB	
#IN_STA_COUNT		Bool	Input signal to start counting
#IO_COUNT_RUN		Bool	Indicates if the counter is running
#IO_HOUR		Int	Elapsed time in hours
#IO_MIN		Int	Elapsed time in minutes (0-59)
#IO_SEC		Int	Elapsed time in seconds (0-59)
#TMP_F_TRIGGER		Bool	

Symbol	Address	Type	Comment
#TP_COUNTER_IDLE_STATE		IEC_Timer	
#TP_COUNTER_IDLE_STATE.IN		Bool	
#TP_COUNTER_IDLE_STATE.Q		Bool	

PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly] / Program blocks

CBCS_FB_CTRL [FB21]

CBCS_FB_CTRL Properties							
General							
Name	CBCS_FB_CTRL	**Number**	21	**Type**	FB	**Language**	SCL
Numbering	Manual						
Information							
Title	CBCS_FB_CTRL	**Author**	MicheleBaschiera	**Comment**	Coldbox control system main function block	**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Default value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
Input									
Output									
InOut									
Static									
TOFF_SW1	TOF_TIME		Non-retain	True	True	True	True		Timer filter door switch 1
PT	Time	T#0ms	Non-retain	True	True	True	False		
ET	Time	T#0ms	Non-retain	True	False	True	False		
IN	Bool	false	Non-retain	True	True	True	False		
Q	Bool	false	Non-retain	True	False	True	False		
TOFF_SW2	TOF_TIME		Non-retain	True	True	True	True		Timer filter door switch 2
PT	Time	T#0ms	Non-retain	True	True	True	False		
ET	Time	T#0ms	Non-retain	True	False	True	False		
IN	Bool	false	Non-retain	True	True	True	False		
Q	Bool	false	Non-retain	True	False	True	False		

Name	Data type	Default value	Retain	Accessible from HMI/OPC UA/Web API	Writ-able from HMI/ OPC UA/ Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion				Comment
TOFF_CHL_SYS_WRN	TOF_TIME		Non-retain	True	True	True	True					Timer filter chiller alarm out
PT	Time	T#0ms	Non-retain	True	True	True	False					
ET	Time	T#0ms	Non-retain	True	False	True	False					
IN	Bool	false	Non-retain	True	True	True	False					
Q	Bool	false	Non-retain	True	False	True	False					
R_TRIG_MAN_STA	R_TRIG			True	True	True	True					Manual Start cmd rising edge detect
Input												
CLK	Bool	false	Non-retain	True	True	True	False					
Output												
Q	Bool	false	Non-retain	True	True	True	False					
InOut												
Static												
Stat_Bit	Bool	false	Non-retain	True	True	True	False					
R_TRIG_MAN_STP	R_TRIG			True	True	True	True					Manual Stop cmd rising edge detect
Input												
CLK	Bool	false	Non-retain	True	True	True	False					
Output												
Q	Bool	false	Non-retain	True	True	True	False					
InOut												
Static												
Stat_Bit	Bool	false	Non-retain	True	True	True	False					
R_TRIG_ALRM	R_TRIG			True	True	True	True					
Input												
CLK	Bool	false	Non-retain	True	True	True	False					

Name	Data type	Default value	Retain	Accessible from HMI/OPC UA/Web API	Writ-able from HMI/ OPC UA/ Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion				Comment
Output												
Q	Bool	false	Non-retain	True	True	True	False					
InOut												
Static												
Stat_Bit	Bool	false	Non-retain	True	True	True	False					
R_TRIG_ALRM2	R_TRIG			True	True	True	True					
Input												
CLK	Bool	false	Non-retain	True	True	True	False					
Output												
Q	Bool	false	Non-retain	True	True	True	False					
InOut												
Static												
Stat_Bit	Bool	false	Non-retain	True	True	True	False					
TP_MAN_GO_COLD	TP_TIME		Non-retain	True	True	True	True					
PT	Time	T#0ms	Non-retain	True	True	True	False					
ET	Time	T#0ms	Non-retain	True	False	True	False					
IN	Bool	false	Non-retain	True	True	True	False					
Q	Bool	false	Non-retain	True	False	True	False					
TP_MAN_GO_WARM	TP_TIME		Non-retain	True	True	True	True					
PT	Time	T#0ms	Non-retain	True	True	True	False					
ET	Time	T#0ms	Non-retain	True	False	True	False					
IN	Bool	false	Non-retain	True	True	True	False					
Q	Bool	false	Non-retain	True	False	True	False					
F_TRIG_MAN_GO_COLD	F_TRIG			True	True	True	True					
Input												
CLK	Bool	false	Non-retain	True	True	True	False					

Name	Data type	Default value	Retain	Accessible from HMI/OPC UA/Web API	Writ-able from HMI/ OPC UA/ Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion				Comment
Output												
Q	Bool	false	Non-retain	True	True	True	False					
InOut												
Static												
Stat_Bit	Bool	false	Non-retain	True	True	True	False					
F_TRIG_MAN_GO_WARM	F_TRIG			True	True	True	True					
Input												
CLK	Bool	false	Non-retain	True	True	True	False					
Output												
Q	Bool	false	Non-retain	True	True	True	False					
InOut												
Static												
Stat_Bit	Bool	false	Non-retain	True	True	True	False					
CTU_TCYC	CTU_INT		Retain	True	True	True	True					Cycle count
CU	Bool	false	Retain	True	True	True	False					
CD	Bool	false	Retain	True	True	True	False					
R	Bool	false	Retain	True	True	True	False					
LD	Bool	false	Retain	True	True	True	False					
QU	Bool	false	Retain	True	True	True	False					
QD	Bool	false	Retain	True	True	True	False					
PV	Int	0	Retain	True	True	True	False					
CV	Int	0	Retain	True	True	True	False					
CTU_TCYC_TOT	CTU_INT		Retain	True	True	True	True					Session cycle count
CU	Bool	false	Retain	True	True	True	False					
CD	Bool	false	Retain	True	True	True	False					
R	Bool	false	Retain	True	True	True	False					

Name	Data type	Default value	Retain	Accessible from HMI/OPC UA/Web API	Writ-able from HMI/ OPC UA/ Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion	Comment
LD	Bool	false	Retain	True	True	True	False		
QU	Bool	false	Retain	True	True	True	False		
QD	Bool	false	Retain	True	True	True	False		
PV	Int	0	Retain	True	True	True	False		
CV	Int	0	Retain	True	True	True	False		
▼ Temp									
TMP_R_TRIG_MAN_STA	Bool								Manual Start cmd rising edge
TMP_R_TRIG_MAN_STP	Bool								Manual Stop cmd rising edge
TMP_MAN_STA_PREV	Bool								Previous manual start state
TMP_MAN_STP_PREV	Bool								Previous manual stop state
Constant									

```
(*
################## MAIN COLDBOX CONTROL SYSTEM ####################
*)

REGION READ_DR_SENS

  //READ OUT DOOR SENSORS AND APPLY SOME FILTER IN CASE SENSOR IS FLICKERING
  #TOFF_SW1(IN := "CBS_DR_SW1",
        PT := T#150MS,
        Q => "CBS_DR_SW1_CLS");
  #TOFF_SW2(IN := "CBS_DR_SW2",
        PT := T#150MS,
        Q => "CBS_DR_SW2_CLS");

END_REGION

REGION CHILLER_READY
```

```
0018
0019   //READ OUT CHILLER STATE FOR WARNING STATE EVENT
0020   #TOFF_CHL_SYS_WRN(IN := NOT "CHL_SYS_WRN",
0021             PT := T#250MS,
0022             Q => "CHL_SYS_RDY");
0023
0024 END_REGION
0025
0026 REGION SYSTEM STATUS FROM FIELD
0027
0028   // DOOR CLOSED STATE
0029   IF ("CBS_DR_SW1_CLS" AND "CBS_DR_SW2_CLS") THEN
0030     "CBS_DR_STAT_CLS" := TRUE;
0031     "CTRL_TCYC_SETUP" := FALSE;
0032     "HMI_SYS_ALARM".%X0 := FALSE;
0033   ELSE
0034     "CBS_DR_STAT_CLS" := FALSE;
0035     "CTRL_TCYC_SETUP" := TRUE;
0036     "HMI_SYS_ALARM".%X0 := TRUE;
0037
0038   END_IF;
0039
0040   // CHILLER STATE NOT READY
0041   IF NOT "CHL_SYS_RDY" THEN
0042     "CTRL_TCYC_FLT" := TRUE;
0043     "CTRL_TCYC_RUN" := FALSE;
0044     "CTRL_TCYC_GO_WARM" := FALSE;
0045     "CTRL_TCYC_GO_COLD" := FALSE;
0046   ELSE
0047     "CTRL_TCYC_FLT" := FALSE;
0048   END_IF;
0049
0050 END_REGION
0051
0052
0053
0054 (*
0055 ################ FEEDBACKS AND ALARMS #################
0056 *)
```

```
0057
0058
0059 REGION ALARMS
0060
0061   // MAIN SYSTEM ALARMS
0062   IF "CTRL_TCYC_RUN" THEN
0063
0064     IF NOT "CBS_DR_STAT_CLS" THEN
0065
0066       "CTRL_TCYC_WRN" := TRUE;
0067
0068
0069     ELSIF "CBCS_WDB".Chiller.Temperature.ActTemp >= "CBCS_MDB".System.SystemMaxTempLimThreshold OR "CBCS_WDB".Chill-
     er.Temperature.ActTemp <= "CBCS_MDB".System.SystemMinTempLimThreshold THEN
0070       "CTRL_TCYC_WRN" := TRUE;
0071
0072     ELSE
0073       //"CTRL_TCYC_WRN" := FALSE
0074       ;
0075     END_IF;
0076
0077
0078   ELSE
0079     "CTRL_TCYC_WRN" := FALSE;
0080
0081   END_IF;
0082
0083 END_REGION
0084
0085 REGION ENABLE SYSTEM READY STATE
0086
0087   //ENABLE FLAGS ACCORDING TO SYSTEM STATE - DOOR CLOSED AND CHILLER READY
0088   IF NOT "CTRL_TCYC_FLT" AND NOT "CTRL_ALRM_MAIN" THEN
0089
0090     IF ("CBS_DR_STAT_CLS" AND "CHL_SYS_RDY") THEN
0091       "CTRL_TCYC_RDY" := TRUE;
0092       "CHL_EN_RUN" := TRUE;
0093     ELSE
0094       "CHL_EN_RUN" := FALSE;
```

```
0095        "CTRL_TCYC_RDY" := FALSE;
0096        "CTRL_TCYC_RUN" := FALSE;
0097      END_IF;
0098
0099    END_IF;
0100
0101 END_REGION
0102
0103 REGION SYSTEM MAIN STATES FOR HMI
0104
0105    // SINGLE STATE FOR HMI DISPLAY
0106    "HMI_SYS_STATES" := 16#0;
0107
0108    IF NOT "CTRL_TCYC_FLT" THEN
0109
0110      // PRE-RUN MACHINE STATES
0111      IF NOT "CTRL_TCYC_RUN" THEN
0112
0113        IF "CBS_DR_STAT_CLS" AND "CHL_SYS_RDY" AND NOT "CTRL_ALRM_MAIN" THEN
0114          "HMI_SYS_STATES".%X1 := TRUE; // MACHITE STATE - READY
0115          "HMI_SYS_STATES".%X7 := FALSE; // MACHINE STATE - SETTING UP
0116
0117        ELSE
0118          "HMI_SYS_STATES".%X7 := TRUE; // MACHINE STATE - SETTING UP
0119          "HMI_SYS_STATES".%X1 := FALSE; // MACHITE STATE - READY
0120        END_IF;
0121
0122      END_IF;
0123
0124      // DURING RUNNING MACHINE STATES
0125      IF "CTRL_TCYC_RUN" THEN
0126
0127        // GOING COLD MACHINE STATE
0128        IF "CTRL_TCYC_GO_COLD" AND NOT "CTRL_TCYC_MINTRG_REACH" THEN
0129          "HMI_SYS_STATES".%X4 := TRUE; // MACHINE STATE - GOING COLD
0130        ELSE
0131          "HMI_SYS_STATES".%X4 := FALSE; // MACHINE STATE - GOING COLD
0132        END_IF;
0133
```

```
0134        // GOING WARM MACHINE STATE
0135        IF "CTRL_TCYC_GO_WARM" AND NOT "CTRL_TCYC_MAXTRG_REACH" THEN
0136          "HMI_SYS_STATES".%X3 := TRUE; // MACHINE STATE - GOING WARM
0137        ELSE
0138          "HMI_SYS_STATES".%X3 := FALSE; // MACHINE STATE - GOING WARM
0139        END_IF;
0140
0141        IF "CTRL_TCYC_MINTRG_REACH" AND NOT "CTRL_TCYC_GO_WARM" THEN
0142          "HMI_SYS_STATES".%X10 := TRUE; // STATE WHEN SYSTEM IS HOLDING IN COLD STATE
0143        ELSIF "CTRL_TCYC_MAXTRG_REACH" AND NOT "CTRL_TCYC_GO_COLD" THEN
0144          "HMI_SYS_STATES".%X11 := TRUE; // STATE WHEN SYSTEM IS HOLDING IN WARM STATE
0145        END_IF;
0146
0147      END_IF;
0148    END_IF;
0149
0150    IF "CTRL_TCYC_FLT" THEN
0151      "HMI_SYS_STATES".%X6 := TRUE; // MACHINE STATE - FAULT
0152    ELSE
0153      "HMI_SYS_STATES".%X6 := FALSE; // MACHINE STATE - FAULT
0154    END_IF;
0155
0156    IF FALSE THEN
0157      "HMI_SYS_STATES".%X5 := TRUE; // MACHINE STATE - WARNING
0158    END_IF;
0159
0160  END_REGION
0161
0162  REGION THERMAL CYCLE AUTO/MANUAL MODE
0163    //TO DO IMPLEMENT TIME OUT IF MANUAL MODE IS KEPT TOO LONG
0164    // MODE SELECTION FOR AUTOMATIC AND MANUAL - COLD AND WARM COMMANDS
0165    IF "CBCS_WDB".HMI_CMD.AUTO AND NOT "CBCS_WDB".HMI_CMD.MAN THEN
0166      "CTRL_TCYC_AUTO" := TRUE;
0167      "CTRL_TCYC_MAN" := FALSE;
0168    ELSE
0169      "CTRL_TCYC_AUTO" := FALSE;
0170      "CTRL_TCYC_MAN" := TRUE;
0171
0172    END_IF;
```

```
0173
0174   // SET CONTROL CMD MANUAL AND AUTO
0175   IF "CTRL_TCYC_RDY" THEN
0176
0177     //detect positive edge START CMD / GO COLD
0178     #R_TRIG_MAN_STA(CLK := "CBCS_WDB".HMI_CMD.STA AND NOT #TMP_MAN_STA_PREV,
0179             Q => #TMP_R_TRIG_MAN_STA);
0180
0181     // set temp start omd -> set Go cold CMD
0182     IF #TMP_R_TRIG_MAN_STA THEN
0183
0184       //SET MANUAL GO COLD
0185       IF "CTRL_TCYC_MAN" THEN
0186         "CTRL_TCYC_GO_COLD" := NOT "CTRL_TCYC_GO_COLD";
0187         "CTRL_TCYC_GO_WARM" := FALSE;
0188
0189         // SET AUTO MODE ON
0190       ELSIF "CTRL_TCYC_AUTO" THEN
0191
0192         "CTRL_SYS_AUTO_SET" := NOT "CTRL_SYS_AUTO_SET";
0193         "CTRL_TCYC_GO_WARM" := FALSE;
0194
0195       END_IF;
0196
0197       #TMP_MAN_STA_PREV := NOT "CTRL_TCYC_GO_COLD";
0198
0199     END_IF;
0200
0201     //detect positive edge STOP CMD / GO WARM
0202     #R_TRIG_MAN_STP(CLK := "CBCS_WDB".HMI_CMD.STP AND NOT #TMP_MAN_STP_PREV,
0203             Q => #TMP_R_TRIG_MAN_STP);
0204
0205
0206     // set temp start cmd -> set Go warm CMD
0207     IF #TMP_R_TRIG_MAN_STP THEN
0208
0209       //SET MANUAL GO WARM
0210       IF "CTRL_TCYC_MAN" THEN
0211         "CTRL_TCYC_GO_WARM" := NOT "CTRL_TCYC_GO_WARM";
```

```
0212          "CTRL_TCYC_GO_COLD" := FALSE;
0213
0214          // SET AUTO MODE OFF
0215        ELSIF "CTRL_TCYC_AUTO" THEN
0216
0217          "CTRL_SYS_AUTO_SET" := FALSE;
0218          "CTRL_TCYC_GO_COLD" := FALSE;
0219
0220        END_IF;
0221
0222        #TMP_MAN_STP_PREV := NOT "CTRL_TCYC_GO_WARM";
0223
0224      END_IF;
0225
0226    END_IF;
0227
0228  END_REGION
0229
0230
0231
0232  (*
0233  ################ OUTPUT CONTROL #################
0234  *)
0235
0236  REGION OUTPUT_VISUAL_SIGNAL
0237
0238     //OUTPUT COMMANDS FOR THE SEMAPHORE ( TO BE IMPROVED)
0239
0240    IF ("CTRL_TCYC_SETUP") THEN
0241      "CBS_DR_OS_GN" := TRUE;
0242
0243    ELSIF ("CTRL_TCYC_RDY"
0244      AND NOT "CTRL_TCYC_GO_COLD") THEN
0245      "CBS_DR_OS_GN" := "Clock_0.5Hz";
0246
0247    ELSE
0248      "CBS_DR_OS_GN" := FALSE;
0249
0250    END_IF;
```

```
0251
0252   IF ("CTRL_TCYC_RUN") THEN
0253     "CBS_DR_OS_RD" := TRUE;
0254
0255   ELSIF ("CTRL_TCYC_GO_WARM" OR "CTRL_TCYC_GO_COLD") THEN
0256     "CBS_DR_OS_RD" := "Clock_0.5Hz";
0257
0258   ELSE
0259     "CBS_DR_OS_RD" := FALSE;
0260
0261   END_IF;
0262
0263   IF ("CTRL_TCYC_WRN") THEN
0264     "CBS_DR_OS_OR" := "Clock_0.5Hz";
0265
0266   ELSIF ("CTRL_TCYC_FLT") THEN
0267     "CBS_DR_OS_OR" := TRUE;
0268
0269   ELSE
0270     "CBS_DR_OS_OR" := FALSE;
0271
0272   END_IF;
0273
0274 END_REGION
0275
0276 (*
0277  #################### OUT COMMAND TO CHILLER ################
0278 *)
0279
0280 REGION CHILLER ENABLE COMMAND OUTPUT
0281
0282   // ENABLE THE CHILLER BY SETTING Stand-By-input
0283   IF "CHL_EN_RUN" THEN
0284     "CHL_EN_RUN_CMD" := TRUE;
0285     // TODO: SET A MESSAGE ON SCREEN TO TELL THE USER TO START CHILLER
0286   ELSE
0287     "CHL_EN_RUN_CMD" := FALSE;
0288   END_IF;
0289
```

```
0290 END_REGION
0291
0292 REGION CHILLER STANDBY
0293
0294   // TEMPERATURE SET TO BE AMBIENT WHEN SYSTEM IS IN STANDBY
0295   IF ("CTRL_TCYC_RDY" OR "CTRL_TCYC_SETUP") AND NOT "CTRL_TCYC_RUN" THEN
0296
0297     IF NOT "CTRL_TCYC_GO_COLD" AND NOT "CTRL_TCYC_GO_WARM" THEN
0298
0299       "TEMP_TARGET_FC"(IN_ACT_TEMP := "CBCS_WDB".Chiller.Temperature.ActTemp,
0300                IN_HYSTERESIS := "CBCS_MDB".System.Hysteresis,
0301                IN_PT_AVG:=0.0,
0302                IN_DEW_AVG:=0.0,
0303                IN_DLT_TEMP:=0.0,
0304                IN_MINTEMP := "CBCS_MDB".System.MaxTempValue,
0305                IN_MAXTEMP := "CBCS_MDB".System.MaxTempValue,
0306                IN_CMD_COLD := FALSE,
0307                IN_CMD_WARM := FALSE,
0308                OUT_TARGET_TEMP => "CBCS_WDB".Chiller.Temperature.ActualTargetTemp,
0309                OUT_ACT_INMINTARGET => "CTRL_TCYC_MINTRG_REACH",
0310                OUT_ACT_INMAXTARGET => "CTRL_TCYC_MAXTRG_REACH",
0311                IO_MINTGRT_SET := "CTRL_SYS_MINTRG_SET",
0312                IO_MAXTGRT_SET := "CTRL_SYS_MAXTRG_SET");
0313
0314       "AO_UNSCALE_FC"(IN_REAL := "CBCS_WDB".Chiller.Temperature.ActualTargetTemp,
0315               IN_HI_LIM := "CBCS_MDB".Chiller.Temperature.Chiller_HiLim,
0316               IN_LO_LIM := "CBCS_MDB".Chiller.Temperature.Chiller_LowLim,
0317               IN_BIPOLAR := FALSE,
0318               OUT_Value => "CHL_CTRL_TEMP_CMD",
0319               IN_OUT_Fault := "CBCS_WDB".Chiller.Temperature.CHILLER_FIELD_FAULT);
0320
0321     END_IF;
0322
0323     "CBCS_WDB".System.COUNT_IDLE_STAT_SEC := 0;
0324     "CBCS_WDB".System.COUNT_IDLE_STAT_MIN := 0;
0325     "CBCS_WDB".System.COUNT_IDLE_STAT_HOUR := 0;
0326
0327   END_IF;
0328
```

```
0329
0330
0331 END_REGION
0332
0333 REGION AUTO MODE TEMPERATURE CONTROL
0334
0335   // MANAGE TEMPERATURES SWAPT WHEN IT IS AUTO
0336   IF "CTRL_SYS_AUTO_SET" THEN
0337
0338     //IF IN READY STATE SET THE TEMP MIN AS DEFAULT
0339     IF "CTRL_TCYC_RDY" AND NOT "CTRL_TCYC_RUN" THEN
0340       "CTRL_TCYC_GO_WARM" := FALSE;
0341       "CTRL_TCYC_GO_COLD" := TRUE;
0342
0343       // IF SYSTEM IS RUNNING CYCLE BETWEEN COLD AND WARM STATE
0344     ELSIF "CTRL_TCYC_RUN" THEN
0345
0346       IF "CBCS_WDB".System.COUNT_TCYC_AMOUNT < "CBCS_WDB".System.COUNT_TCYC_AMOUNT_SET THEN
0347
0348         IF "CTRL_TCYC_GO_COLD" THEN
0349
0350           IF "CTRL_TCYC_MINTRG_REACH" AND "CTRL_SYS_TIM_ELAP_MIN" THEN
0351
0352             "CTRL_TCYC_GO_COLD" := FALSE;
0353             "CTRL_TCYC_GO_WARM" := TRUE;
0354
0355           END_IF;
0356
0357         ELSIF "CTRL_TCYC_GO_WARM" THEN
0358
0359           IF "CTRL_TCYC_MAXTRG_REACH" AND "CTRL_SYS_TIM_ELAP_MAX" THEN
0360
0361             "CTRL_TCYC_GO_WARM" := FALSE;
0362             "CTRL_TCYC_GO_COLD" := TRUE;
0363
0364           END_IF;
0365         END_IF;
0366
0367       ELSE
```

```
0368          "CTRL_SYS_AUTO_SET" := FALSE;
0369          "CTRL_TCYC_GO_COLD" := FALSE;
0370          "CTRL_TCYC_GO_WARM" := FALSE;
0371        END_IF;
0372      END_IF;
0373
0374    ELSIF NOT "CTRL_SYS_AUTO_SET" AND "CTRL_TCYC_AUTO" THEN
0375
0376      "CTRL_TCYC_GO_COLD" := FALSE;
0377      "CTRL_TCYC_GO_WARM" := FALSE;
0378
0379    END_IF;
0380
0381 END_REGION
0382
0383
0384 REGION CHILLER THERMAL CYCLE MODE GO COLD
0385
0386    //OUTPUT CMD TO CHILLER TO GO COLD WHEN IN AUTO O MANUAL MODE
0387    IF "CHL_EN_RUN" THEN
0388
0389      IF "CTRL_TCYC_GO_COLD" AND NOT "CTRL_TCYC_GO_WARM" THEN
0390
0391
0392        // origin "CBCS_WDB".Chiller.Temperature.ActTemp
0393        "TEMP_TARGET_FC"(IN_ACT_TEMP := "CBCS_WDB".InnerColdbox.CBI_PT_AVG_ACT_TEMP,
0394                 IN_HYSTERESIS := "CBCS_MDB".System.Hysteresis,
0395                 IN_PT_AVG:="CBCS_WDB".InnerColdbox.CBI_PT_AVG_ACT_TEMP,
0396                 IN_DEW_AVG:="CBCS_WDB".InnerColdbox.CBI_DEW_AVG_ACT_TEMP,
0397                 IN_DLT_TEMP:="CBCS_MDB".System.DewTempDelta_Range,
0398                 IN_MINTEMP := "CBCS_WDB".System.Temperature.TargetMinTemp,
0399                 IN_MAXTEMP := "CBCS_WDB".System.Temperature.TargetMaxTemp,
0400                 IN_CMD_COLD := "CTRL_TCYC_GO_COLD",
0401                 IN_CMD_WARM := "CTRL_TCYC_GO_WARM",
0402                 OUT_TARGET_TEMP => "CBCS_WDB".Chiller.Temperature.ActualTargetTemp,
0403                 OUT_ACT_INMINTARGET => "CTRL_TCYC_MINTRG_REACH",
0404                 OUT_ACT_INMAXTARGET => "CTRL_TCYC_MAXTRG_REACH",
0405                 IO_MINTGRT_SET := "CTRL_SYS_MINTRG_SET",
0406                 IO_MAXTGRT_SET := "CTRL_SYS_MAXTRG_SET");
```

```
0407
0408        "AO_UNSCALE_FC"(IN_REAL := "CBCS_WDB".Chiller.Temperature.ActualTargetTemp,
0409                IN_HI_LIM := "CBCS_MDB".Chiller.Temperature.Chiller_HiLim,
0410                IN_LO_LIM := "CBCS_MDB".Chiller.Temperature.Chiller_LowLim,
0411                IN_BIPOLAR := FALSE,
0412                OUT_Value => "CHL_CTRL_TEMP_CMD",
0413                IN_OUT_Fault := "CBCS_WDB".Chiller.Temperature.CHILLER_FIELD_FAULT);
0414
0415        "TIMER_IDLE_STATE"(IN_STA_COUNT := "CTRL_TCYC_MINTRG_REACH" AND "CTRL_TCYC_GO_COLD",
0416                    IO_COUNT_RUN := "CBCS_WDB".System.COUNT_IDLE_STAT_RUN,
0417                    IO_SEC := "CBCS_WDB".System.COUNT_IDLE_STAT_SEC,
0418                    IO_MIN := "CBCS_WDB".System.COUNT_IDLE_STAT_MIN,
0419                    IO_HOUR := "CBCS_WDB".System.COUNT_IDLE_STAT_HOUR,
0420                    TP_COUNTER_IDLE_STATE := #TP_MAN_GO_COLD,
0421                    F_TRIG_IDLE_STATE := #F_TRIG_MAN_GO_COLD);
0422
0423        "CTRL_TCYC_RUN" := TRUE;
0424
0425      ELSIF NOT "CTRL_TCYC_GO_WARM" AND NOT "CTRL_TCYC_GO_COLD" THEN
0426        "CTRL_TCYC_RUN" := FALSE;
0427
0428      END_IF;
0429
0430
0431
0432      IF "CTRL_TCYC_MINTRG_REACH" OR "CTRL_TCYC_MAXTRG_REACH" THEN
0433        "CTRL_TCYC_IDLE" := TRUE;
0434      ELSE
0435        "CTRL_TCYC_IDLE" := FALSE;
0436      END_IF;
0437
0438
0439    ELSE
0440      "CTRL_TCYC_GO_COLD" := FALSE;
0441
0442    END_IF;
0443
0444
0445
```

```
END_REGION

REGION CHILLER THERMAL CYCLE MODE GO WARM

  //PUTPUT COMMAND TO CHILLER TO GO WARM WHEN IS AUTO O MANUAL MODE
  IF "CHL_EN_RUN" THEN

    IF "CTRL_TCYC_GO_WARM" AND NOT "CTRL_TCYC_GO_COLD" THEN

      //origin "CBCS_WDB".Chiller.Temperature.ActTemp
      "TEMP_TARGET_FC"(IN_ACT_TEMP := "CBCS_WDB".InnerColdbox.CBI_PT_AVG_ACT_TEMP,
               IN_HYSTERESIS := "CBCS_MDB".System.Hysteresis,
               IN_PT_AVG:="CBCS_WDB".InnerColdbox.CBI_PT_AVG_ACT_TEMP,
               IN_DEW_AVG:="CBCS_WDB".InnerColdbox.CBI_DEW_AVG_ACT_TEMP,
               IN_DLT_TEMP:="CBCS_MDB".System.DewTempDelta_Range,
               IN_MINTEMP := "CBCS_WDB".System.Temperature.TargetMinTemp,
               IN_MAXTEMP := "CBCS_WDB".System.Temperature.TargetMaxTemp,
               IN_CMD_COLD := "CTRL_TCYC_GO_COLD",
               IN_CMD_WARM := "CTRL_TCYC_GO_WARM",
               OUT_TARGET_TEMP => "CBCS_WDB".Chiller.Temperature.ActualTargetTemp,
               OUT_ACT_INMINTARGET => "CTRL_TCYC_MINTRG_REACH",
               OUT_ACT_INMAXTARGET => "CTRL_TCYC_MAXTRG_REACH",
               IO_MINTGRT_SET := "CTRL_SYS_MINTRG_SET",
               IO_MAXTGRT_SET := "CTRL_SYS_MAXTRG_SET");

      "AO_UNSCALE_FC"(IN_REAL := "CBCS_WDB".Chiller.Temperature.ActualTargetTemp,
              IN_HI_LIM := "CBCS_MDB".Chiller.Temperature.Chiller_HiLim,
              IN_LO_LIM := "CBCS_MDB".Chiller.Temperature.Chiller_LowLim,
              IN_BIPOLAR := FALSE,
              OUT_Value => "CHL_CTRL_TEMP_CMD",
              IN_OUT_Fault := "CBCS_WDB".Chiller.Temperature.CHILLER_FIELD_FAULT);

      "TIMER_IDLE_STATE"(IN_STA_COUNT := "CTRL_TCYC_MAXTRG_REACH" AND "CTRL_TCYC_GO_WARM",
                 IO_COUNT_RUN := "CBCS_WDB".System.COUNT_IDLE_STAT_RUN,
                 IO_SEC := "CBCS_WDB".System.COUNT_IDLE_STAT_SEC,
                 IO_MIN := "CBCS_WDB".System.COUNT_IDLE_STAT_MIN,
                 IO_HOUR := "CBCS_WDB".System.COUNT_IDLE_STAT_HOUR,
                 TP_COUNTER_IDLE_STATE := #TP_MAN_GO_WARM,
                 F_TRIG_IDLE_STATE := #F_TRIG_MAN_GO_WARM);
```

```
0485
0486        "CTRL_TCYC_RUN" := TRUE;
0487
0488      ELSIF NOT "CTRL_TCYC_GO_WARM" AND NOT "CTRL_TCYC_GO_COLD" THEN
0489        "CTRL_TCYC_RUN" := FALSE;
0490
0491      END_IF;
0492
0493
0494      IF "CTRL_TCYC_MINTRG_REACH" OR "CTRL_TCYC_MAXTRG_REACH" THEN
0495        "CTRL_TCYC_IDLE" := TRUE;
0496      ELSE
0497        "CTRL_TCYC_IDLE" := FALSE;
0498      END_IF;
0499
0500    ELSE
0501      "CTRL_TCYC_GO_WARM" := FALSE;
0502
0503    END_IF;
0504
0505
0506 END_REGION
0507
0508 REGION THERMAL CYCLE IDLE STATE ELAPSED
0509
0510    // THERMAL CYCLE TIME TARGETS REACH
0511    IF "CTRL_TCYC_GO_COLD" AND "CTRL_TCYC_MINTRG_REACH" THEN
0512
0513      IF "CBCS_WDB".System.COUNT_IDLE_STAT_MIN >= "CBCS_WDB".System.COUNT_IDLE_STATMIN_MIN_SET THEN
0514        "CTRL_SYS_TIM_ELAP_MAX" := FALSE;
0515        "CTRL_SYS_TIM_ELAP_MIN" := TRUE;
0516
0517      END_IF;
0518
0519    ELSIF "CTRL_TCYC_GO_WARM" AND "CTRL_TCYC_MAXTRG_REACH" THEN
0520
0521      IF "CBCS_WDB".System.COUNT_IDLE_STAT_MIN >= "CBCS_WDB".System.COUNT_IDLE_STATMAX_MIN_SET THEN
0522        "CTRL_SYS_TIM_ELAP_MIN" := FALSE;
0523        "CTRL_SYS_TIM_ELAP_MAX" := TRUE;
```

```
0524
0525      END_IF;
0526
0527    ELSE
0528
0529      "CTRL_SYS_TIM_ELAP_MIN" := FALSE;
0530      "CTRL_SYS_TIM_ELAP_MAX" := FALSE;
0531
0532    END_IF;
0533
0534
0535
0536 END_REGION
0537
0538
0539 REGION THERMAL CYCLES COUNT AMOUNT
0540
0541    //NUMBER OF CONTINUOS THERMAL CYCLE COUNTER
0542    IF "CTRL_TCYC_RUN" THEN
0543
0544      // COUNT CONTINUOS THERMAL CYCLES
0545      "COUNTER_IDLE_STATE"(IN_EN := TRUE,
0546                  IN_MIN_STAT_ELAPSED := "CTRL_SYS_TIM_ELAP_MIN",
0547                  IN_MAX_STAT_ELAPSED := "CTRL_SYS_TIM_ELAP_MAX",
0548                  IN_RST := FALSE,
0549                  OUT_COUNT => "CBCS_WDB".System.COUNT_TCYC_AMOUNT,
0550                  CTU_TCYC := #CTU_TCYC,
0551                  CTU_INDEX := "CBCS_WDB".System.COUNT_TCYC_INDEX);
0552
0553    ELSE
0554      //RESET COUNTER
0555      "COUNTER_IDLE_STATE"(IN_EN := TRUE,
0556                  IN_MIN_STAT_ELAPSED := "CTRL_SYS_TIM_ELAP_MIN",
0557                  IN_MAX_STAT_ELAPSED := "CTRL_SYS_TIM_ELAP_MAX",
0558                  IN_RST := TRUE,
0559                  OUT_COUNT => "CBCS_WDB".System.COUNT_TCYC_AMOUNT,
0560                  CTU_TCYC := #CTU_TCYC,
0561                  CTU_INDEX := "CBCS_WDB".System.COUNT_TCYC_INDEX);
0562
```

```
0563   END_IF;
0564
0565 END_REGION
0566
0567 REGION THERMAL CYCLES COUNT AMOUNT FOR EACH SESSION
0568
0569   //NUMBER OF CONTINUOS THERMAL CYCLE COUNTER IN ONE SESSION (DOOR CLOSED - DOOP OPENED)
0570   IF NOT "CTRL_TCYC_SETUP" THEN
0571
0572     IF "CTRL_TCYC_RUN" THEN
0573
0574       "COUNTER_IDLE_STATE"(IN_EN := TRUE,
0575                  IN_MIN_STAT_ELAPSED := "CTRL_SYS_TIM_ELAP_MIN",
0576                  IN_MAX_STAT_ELAPSED := "CTRL_SYS_TIM_ELAP_MAX",
0577                  IN_RST := FALSE,
0578                  OUT_COUNT => "CBCS_WDB".System.COUNT_TCYC_AMOUNT_TOT,
0579                  CTU_TCYC := #CTU_TCYC_TOT,
0580                  CTU_INDEX := "CBCS_WDB".System.COUNT_TCYC_INDEX_TOT);
0581     END_IF;
0582
0583   ELSE
0584     //RESET COUNTER
0585     "COUNTER_IDLE_STATE"(IN_EN := TRUE,
0586                IN_MIN_STAT_ELAPSED := "CTRL_SYS_TIM_ELAP_MIN",
0587                IN_MAX_STAT_ELAPSED := "CTRL_SYS_TIM_ELAP_MAX",
0588                IN_RST := TRUE,
0589                OUT_COUNT => "CBCS_WDB".System.COUNT_TCYC_AMOUNT_TOT,
0590                CTU_TCYC := #CTU_TCYC_TOT,
0591                CTU_INDEX := "CBCS_WDB".System.COUNT_TCYC_INDEX_TOT);
0592
0593   END_IF;
0594
0595 END_REGION
0596
0597
0598 (*
0599 ################ ALARM AND STATUS #################
0600 *)
0601
```

```
0602 REGION ALARMS_TO_HMI
0603
0604   "ALRM_GEN_MAIN"(IN_ALRM := "HMI_SYS_ALARM".%X0,
0605           R_TRIG_ALRM := #R_TRIG_ALRM2,
0606           INOUT_HMI_ALRM := "HMI_ALARMS_DB".System.CTRL_SYS_ALRM.%X0); // DOOR OPENED
0607   "ALRM_GEN_MAIN"(IN_ALRM := "HMI_SYS_ALARM".%X1,
0608           R_TRIG_ALRM := #R_TRIG_ALRM,
0609           INOUT_HMI_ALRM := "HMI_ALARMS_DB".System.CTRL_SYS_ALRM.%X1);
0610   "ALRM_GEN_MAIN"(IN_ALRM := "HMI_SYS_ALARM".%X2,
0611           R_TRIG_ALRM := #R_TRIG_ALRM,
0612           INOUT_HMI_ALRM := "HMI_ALARMS_DB".System.CTRL_SYS_ALRM.%X2);
0613   "ALRM_GEN_MAIN"(IN_ALRM := "HMI_SYS_ALARM".%X3,
0614           R_TRIG_ALRM := #R_TRIG_ALRM,
0615           INOUT_HMI_ALRM := "HMI_ALARMS_DB".System.CTRL_SYS_ALRM.%X3);
0616   "ALRM_GEN_MAIN"(IN_ALRM := "HMI_SYS_ALARM".%X4,
0617           R_TRIG_ALRM := #R_TRIG_ALRM,
0618           INOUT_HMI_ALRM := "HMI_ALARMS_DB".System.CTRL_SYS_ALRM.%X4);
0619   "ALRM_GEN_MAIN"(IN_ALRM := "HMI_SYS_ALARM".%X5,
0620           R_TRIG_ALRM := #R_TRIG_ALRM,
0621           INOUT_HMI_ALRM := "HMI_ALARMS_DB".System.CTRL_SYS_ALRM.%X5);
0622   "ALRM_GEN_MAIN"(IN_ALRM := "HMI_SYS_ALARM".%X6,
0623           R_TRIG_ALRM := #R_TRIG_ALRM,
0624           INOUT_HMI_ALRM := "HMI_ALARMS_DB".System.CTRL_SYS_ALRM.%X6);
0625   "ALRM_GEN_MAIN"(IN_ALRM := "HMI_SYS_ALARM".%X7,
0626           R_TRIG_ALRM := #R_TRIG_ALRM,
0627           INOUT_HMI_ALRM := "HMI_ALARMS_DB".System.CTRL_SYS_ALRM.%X7);
0628   "ALRM_GEN_MAIN"(IN_ALRM := "HMI_SYS_ALARM".%X8,
0629           R_TRIG_ALRM := #R_TRIG_ALRM,
0630           INOUT_HMI_ALRM := "HMI_ALARMS_DB".System.CTRL_SYS_ALRM.%X8);
0631   "ALRM_GEN_MAIN"(IN_ALRM := "HMI_SYS_ALARM".%X9,
0632           R_TRIG_ALRM := #R_TRIG_ALRM,
0633           INOUT_HMI_ALRM := "HMI_ALARMS_DB".System.CTRL_SYS_ALRM.%X9);
0634   "ALRM_GEN_MAIN"(IN_ALRM := "HMI_SYS_ALARM".%X10,
0635           R_TRIG_ALRM := #R_TRIG_ALRM,
0636           INOUT_HMI_ALRM := "HMI_ALARMS_DB".System.CTRL_SYS_ALRM.%X10);
0637   "ALRM_GEN_MAIN"(IN_ALRM := "HMI_SYS_ALARM".%X11,
0638           R_TRIG_ALRM := #R_TRIG_ALRM,
0639           INOUT_HMI_ALRM := "HMI_ALARMS_DB".System.CTRL_SYS_ALRM.%X11);
0640   "ALRM_GEN_MAIN"(IN_ALRM := "HMI_SYS_ALARM".%X12,
```

```
0641            R_TRIG_ALRM := #R_TRIG_ALRM,
0642            INOUT_HMI_ALRM := "HMI_ALARMS_DB".System.CTRL_SYS_ALRM.%X12);
0643    "ALRM_GEN_MAIN"(IN_ALRM := "HMI_SYS_ALARM".%X13,
0644            R_TRIG_ALRM := #R_TRIG_ALRM,
0645            INOUT_HMI_ALRM := "HMI_ALARMS_DB".System.CTRL_SYS_ALRM.%X13);
0646    "ALRM_GEN_MAIN"(IN_ALRM := "HMI_SYS_ALARM".%X14,
0647            R_TRIG_ALRM := #R_TRIG_ALRM,
0648            INOUT_HMI_ALRM := "HMI_ALARMS_DB".System.CTRL_SYS_ALRM.%X14);
0649    "ALRM_GEN_MAIN"(IN_ALRM := "HMI_SYS_ALARM".%X15,
0650            R_TRIG_ALRM := #R_TRIG_ALRM,
0651            INOUT_HMI_ALRM := "HMI_ALARMS_DB".System.CTRL_SYS_ALRM.%X15);
0652
0653    // if one system alarm IS active set general FLAG alarm to TRUE
0654    IF "HMI_ALARMS_DB".System.CTRL_SYS_ALRM <> 0 THEN
0655      "CTRL_ALRM_MAIN" := TRUE;
0656      "HMI_ALARMS_DB".System.CTRL_ALRM_MAIN := TRUE;
0657    ELSE
0658      "CTRL_ALRM_MAIN" := FALSE;
0659      "HMI_ALARMS_DB".System.CTRL_ALRM_MAIN := FALSE;
0660    END_IF;
0661
0662
0663 END_REGION
0664
0665 REGION STATUS_TO_HMI
0666
0667    // MAIN SYSTEM STATES TO HMI
0668    "HMI_STATUS_DB".CBCS_SYS_STATES.%X0 := "HMI_SYS_STATES".%X0; //IDLE
0669    "HMI_STATUS_DB".CBCS_SYS_STATES.%X1 := "HMI_SYS_STATES".%X1; //READY
0670    "HMI_STATUS_DB".CBCS_SYS_STATES.%X2 := "HMI_SYS_STATES".%X2; //RUNNING
0671    "HMI_STATUS_DB".CBCS_SYS_STATES.%X3 := "HMI_SYS_STATES".%X3; //WARMING UP
0672    "HMI_STATUS_DB".CBCS_SYS_STATES.%X4 := "HMI_SYS_STATES".%X4; //COOLING DOWN
0673    "HMI_STATUS_DB".CBCS_SYS_STATES.%X5 := "HMI_SYS_STATES".%X5; //WARNING
0674    "HMI_STATUS_DB".CBCS_SYS_STATES.%X6 := "HMI_SYS_STATES".%X6; //FAULT
0675    "HMI_STATUS_DB".CBCS_SYS_STATES.%X7 := "HMI_SYS_STATES".%X7; //SETTING UP
0676    "HMI_STATUS_DB".CBCS_SYS_STATES.%X8 := "HMI_SYS_STATES".%X8; //MANUAL
0677    "HMI_STATUS_DB".CBCS_SYS_STATES.%X9 := "HMI_SYS_STATES".%X9; //AUTO
0678    "HMI_STATUS_DB".CBCS_SYS_STATES.%X10 := "HMI_SYS_STATES".%X10; //MIN TARGET REACH
0679    "HMI_STATUS_DB".CBCS_SYS_STATES.%X11 := "HMI_SYS_STATES".%X11; //MAX TARGET REACH
```

```
0680   "HMI_STATUS_DB".CBCS_SYS_STATES.%X12 := "HMI_SYS_STATES".%X12; //SETTING UP
0681   "HMI_STATUS_DB".CBCS_SYS_STATES.%X13 := "HMI_SYS_STATES".%X13; //SETTING UP
0682   "HMI_STATUS_DB".CBCS_SYS_STATES.%X14 := "HMI_SYS_STATES".%X14; //SETTING UP
0683   "HMI_STATUS_DB".CBCS_SYS_STATES.%X15 := "HMI_SYS_STATES".%X15; //SETTING UP
0684
0685   "HMI_STATUS_DB".CBCS_SYS_RDY := "CTRL_TCYC_RDY"; // SYSTEM READY
0686   "HMI_STATUS_DB".CBCS_PLC_RUN := "Clock_Byte".%X7; // PLC RUNNING
0687 END_REGION
0688
0689 REGION DATAPROCESS_TO_HMI
0690
0691   //UPDATE HMI WITH NEW SET VALUES
0692   IF ("DataSetFromHMI") THEN
0693     "M/R/T_FC"(DATA_IN := "CBCS_WDB".System.Temperature.TargetMinTemp,
0694            OFFSET := 10.0,
0695            DATA_OUT => "HMI_DATAPROC_DB".System.ACT_MIN_TMP_SET);
0696
0697     "M/R/T_FC"(DATA_IN := "CBCS_WDB".System.Temperature.TargetMaxTemp,
0698            OFFSET := 10.0,
0699            DATA_OUT => "HMI_DATAPROC_DB".System.ACT_MAX_TMP_SET);
0700
0701     "M/R/T_FC"(DATA_IN := "CBCS_WDB".System.DewPoint.TargetMinDew,
0702            OFFSET := 10.0,
0703            DATA_OUT => "HMI_DATAPROC_DB".System.ACT_MIN_DEW_SET);
0704
0705     "M/R/T_FC"(DATA_IN := "CBCS_WDB".System.DewPoint.TargetMaxDew,
0706            OFFSET := 10.0,
0707            DATA_OUT => "HMI_DATAPROC_DB".System.ACT_MAX_DEW_SET);
0708
0709     "HMI_DATAPROC_DB".System.COUNT_TCYC_AMOUNT_SET := "CBCS_WDB".System.COUNT_TCYC_AMOUNT_SET;
0710     "HMI_DATAPROC_DB".System.COUNT_IDLE_STATMIN_MIN_SET := "CBCS_WDB".System.COUNT_IDLE_STATMIN_MIN_SET;
0711     "HMI_DATAPROC_DB".System.COUNT_IDLE_STATMAX_MIN_SET := "CBCS_WDB".System.COUNT_IDLE_STATMAX_MIN_SET;
0712
0713     "DataSetFromHMI" := FALSE;
0714   END_IF;
0715
0716   //MAIN SYSTEM VALUES TO HMI
0717   "M/R/T_FC"(DATA_IN := "CBCS_WDB".InnerColdbox.PT1000.CBITR1_PT1000.ActTemp,
0718          OFFSET := 10.0,
```

```
0719            DATA_OUT => "HMI_DATAPROC_DB".InnerColdbox.Temperature.ACT_TMP_TRAY1);
0720
0721    "M/R/T_FC"(DATA_IN := "CBCS_WDB".InnerColdbox.PT1000.CBITR2_PT1000.ActTemp,
0722            OFFSET := 10.0,
0723            DATA_OUT => "HMI_DATAPROC_DB".InnerColdbox.Temperature.ACT_TMP_TRAY2);
0724
0725    "M/R/T_FC"(DATA_IN := "CBCS_WDB".InnerColdbox.PT1000.CBITR3_PT1000.ActTemp,
0726            OFFSET := 10.0,
0727            DATA_OUT => "HMI_DATAPROC_DB".InnerColdbox.Temperature.ACT_TMP_TRAY3);
0728
0729    "M/R/T_FC"(DATA_IN := "CBCS_WDB".InnerColdbox.PT1000.CBITR4_PT1000.ActTemp,
0730            OFFSET := 10.0,
0731            DATA_OUT => "HMI_DATAPROC_DB".InnerColdbox.Temperature.ACT_TMP_TRAY4);
0732
0733    "M/R/T_FC"(DATA_IN := "CBCS_WDB".InnerColdbox.PT1000.CBITR5_PT1000.ActTemp,
0734            OFFSET := 10.0,
0735            DATA_OUT => "HMI_DATAPROC_DB".InnerColdbox.Temperature.ACT_TMP_TRAY5);
0736
0737    "M/R/T_FC"(DATA_IN := "CBCS_WDB".InnerColdbox.PT1000.CBITR6_PT1000.ActTemp,
0738            OFFSET := 10.0,
0739            DATA_OUT => "HMI_DATAPROC_DB".InnerColdbox.Temperature.ACT_TMP_TRAY6);
0740
0741    "M/R/T_FC"(DATA_IN := "CBCS_WDB".InnerColdbox.PT1000.CBITR7_PT1000.ActTemp,
0742            OFFSET := 10.0,
0743            DATA_OUT => "HMI_DATAPROC_DB".InnerColdbox.Temperature.ACT_TMP_TRAY7);
0744
0745    "M/R/T_FC"(DATA_IN := "CBCS_WDB".InnerColdbox.PT1000.CBITR8_PT1000.ActTemp,
0746            OFFSET := 10.0,
0747            DATA_OUT => "HMI_DATAPROC_DB".InnerColdbox.Temperature.ACT_TMP_TRAY8);
0748
0749    "M/R/T_FC"(DATA_IN := "CBCS_WDB".Chiller.Temperature.ActTemp,
0750            OFFSET := 10.0,
0751            DATA_OUT => "HMI_DATAPROC_DB".System.ACT_TEMP);
0752
0753    "M/R/T_FC"(DATA_IN := "CBCS_WDB".Chiller.Temperature.ActualTargetTemp,
0754            OFFSET := 10.0,
0755            DATA_OUT => "HMI_DATAPROC_DB".System.ACT_TARGET_TEMP);
0756
0757    "M/R/T_FC"(DATA_IN := "CBCS_WDB".InnerColdbox.CBI_PT_AVG_ACT_TEMP,
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0758          OFFSET := 10.0,
0759          DATA_OUT => "HMI_DATAPROC_DB".InnerColdbox.Temperature.ACT_TMP_AVG);
0760
0761   "M/R/T_FC"(DATA_IN := "CBCS_WDB".InnerColdbox.DMT143.CBI_DMT1.ActDew,
0762          OFFSET := 10.0,
0763          DATA_OUT => "HMI_DATAPROC_DB".InnerColdbox.Dew.ACT_TMP_DEW1);
0764
0765   "M/R/T_FC"(DATA_IN := "CBCS_WDB".InnerColdbox.DMT143.CBI_DMT2.ActDew,
0766          OFFSET := 10.0,
0767          DATA_OUT => "HMI_DATAPROC_DB".InnerColdbox.Dew.ACT_TMP_DEW2);
0768
0769   "M/R/T_FC"(DATA_IN := "CBCS_WDB".InnerColdbox.CBI_DEW_AVG_ACT_TEMP,
0770          OFFSET := 10.0,
0771          DATA_OUT => "HMI_DATAPROC_DB".InnerColdbox.Dew.ACT_TMP_AVG);
0772
0773   "HMI_DATAPROC_DB".System.COUNT_IDLE_STAT_SEC := "CBCS_WDB".System.COUNT_IDLE_STAT_SEC;
0774   "HMI_DATAPROC_DB".System.COUNT_IDLE_STAT_MIN := "CBCS_WDB".System.COUNT_IDLE_STAT_MIN;
0775   "HMI_DATAPROC_DB".System.COUNT_TCYC_AMOUNT := "CBCS_WDB".System.COUNT_TCYC_AMOUNT;
0776   "HMI_DATAPROC_DB".System.COUNT_TCYC_AMOUNT_TOT := "CBCS_WDB".System.COUNT_TCYC_AMOUNT_TOT;
0777
0778
0779 END_REGION
0780
0781 REGION SETPOINTS_FROM_HMI
0782
0783   //VALUES SET FROM HMI
0784   IF "HMI_CMD_DB".HMI_CMD_DB.ACT_DATA THEN
0785     //temperature working limits
0786     "CBCS_WDB".System.Temperature.TargetMinTemp := (DINT_TO_REAL("HMI_DATASET_DB".DAT_MIN_TMP)) * 0.1;
0787     "CBCS_WDB".System.Temperature.TargetMaxTemp := (DINT_TO_REAL("HMI_DATASET_DB".DAT_MAX_TMP)) * 0.1;
0788     "CBCS_WDB".System.DewPoint.TargetMinDew := (DINT_TO_REAL("HMI_DATASET_DB".DAT_MIN_DEW)) * 0.1;
0789     "CBCS_WDB".System.DewPoint.TargetMaxDew := (DINT_TO_REAL("HMI_DATASET_DB".DAT_MAX_DEW)) * 0.1;
0790     //thermal cycles settings
0791     "CBCS_WDB".System.COUNT_TCYC_AMOUNT_SET := "HMI_DATASET_DB".DAT_COUNT_TCYC_AMOUNT_SET;
0792     "CBCS_WDB".System.COUNT_IDLE_STATMIN_MIN_SET := "HMI_DATASET_DB".DAT_COUNT_IDLEMIN_MIN_SET;
0793     "CBCS_WDB".System.COUNT_IDLE_STATMAX_MIN_SET := "HMI_DATASET_DB".DAT_COUNT_IDLEMAX_MIN_SET;
0794     "DataSetFromHMI" := true;
0795   END_IF;
0796
```

```
END_REGION
```

Symbol	Address	Type	Comment
"CBCS_MDB".Chiller.Temperature.Chiller_HiLim		Real	Chiller temperature - value for 20 mA from SCALE BLOCK
"CBCS_MDB".Chiller.Temperature.Chiller_LowLim		Real	Chiller temperature - value for 4 mA from SCALE BLOCK
"CBCS_MDB".System.DewTempDelta_Range		Real	Defautl delta range between Dew and Temperature
"CBCS_MDB".System.Hysteresis		Real	
"CBCS_MDB".System.MaxTempValue		Real	Tray temperature - value for 20 mA from SCALE BLOCK
"CBCS_MDB".System.SystemMaxTempLimThreshold		Real	System shutdown - max threshold temperature
"CBCS_MDB".System.SystemMinTempLimThreshold		Real	System shutdown - min threshold temperature
"CBCS_WDB".Chiller.Temperature.ActTemp	%DB15.DBD410	Real	Actual temperature from chiller DEG
"CBCS_WDB".Chiller.Temperature.ActualTargetTemp	%DB15.DBD418	Real	Actual Target temperature
"CBCS_WDB".Chiller.Temperature.CHILLER_FIELD_FAULT	%DB15.DBX446.0	Bool	CHILLER sensor error from field
"CBCS_WDB".HMI_CMD.AUTO	%DB15.DBX556.1	Bool	Auto
"CBCS_WDB".HMI_CMD.MAN	%DB15.DBX556.6	Bool	Manual
"CBCS_WDB".HMI_CMD.STA	%DB15.DBX556.0	Bool	Start
"CBCS_WDB".HMI_CMD.STP	%DB15.DBX556.2	Bool	Stop
"CBCS_WDB".InnerColdbox.CBI_DEW_AVG_ACT_TEMP	%DB15.DBD392	Real	Average dew point from DEW sensors
"CBCS_WDB".InnerColdbox.CBI_PT_AVG_ACT_TEMP	%DB15.DBD386	Real	Average temperature from PT sensors
"CBCS_WDB".InnerColdbox.DMT143.CBI_DMT1.ActDew	%DB15.DBD314	Real	Actual dew point of the inner tray
"CBCS_WDB".InnerColdbox.DMT143.CBI_DMT2.ActDew	%DB15.DBD356	Real	Actual dew point of the inner tray
"CBCS_WDB".InnerColdbox.PT1000.CBITR1_PT1000.ActTemp	%DB15.DBD6	Real	Actual temperature of the inner tray DEG
"CBCS_WDB".InnerColdbox.PT1000.CBITR2_PT1000.ActTemp	%DB15.DBD44	Real	Actual temperature of the inner tray DEG

Symbol	Address	Type	Comment
"CBCS_WDB".InnerCold-box.PT1000.CBITR3_PT1000.Act-Temp	%DB15.DBD82	Real	Actual temperature of the inner tray DEG
"CBCS_WDB".InnerCold-box.PT1000.CBITR4_PT1000.Act-Temp	%DB15.DBD120	Real	Actual temperature of the inner tray DEG
"CBCS_WDB".InnerCold-box.PT1000.CBITR5_PT1000.Act-Temp	%DB15.DBD158	Real	Actual temperature of the inner tray DEG
"CBCS_WDB".InnerCold-box.PT1000.CBITR6_PT1000.Act-Temp	%DB15.DBD196	Real	Actual temperature of the inner tray DEG
"CBCS_WDB".InnerCold-box.PT1000.CBITR7_PT1000.Act-Temp	%DB15.DBD234	Real	Actual temperature of the inner tray DEG
"CBCS_WDB".InnerCold-box.PT1000.CBITR8_PT1000.Act-Temp	%DB15.DBD272	Real	Actual temperature of the inner tray DEG
"CBCS_WDB".Sys-tem.COUNT_IDLE_STAT_HOUR	%DB15.DBW534	Int	Hours Counter for Working Time idle state
"CBCS_WDB".Sys-tem.COUNT_IDLE_STAT_MIN	%DB15.DBW532	Int	Minutes Counter for Working Time idle state
"CBCS_WDB".Sys-tem.COUNT_IDLE_STAT_RUN	%DB15.DBX540.0	Bool	Counter for Working Time idle state Running
"CBCS_WDB".Sys-tem.COUNT_IDLE_STAT_SEC	%DB15.DBW530	Int	Seconds Counter for Working Time idle state
"CBCS_WDB".Sys-tem.COUNT_IDLE_STAT-MAX_MIN_SET	%DB15.DBW538	Int	Minutes Counter for Working Time idle warm state set from HMI
"CBCS_WDB".Sys-tem.COUNT_IDLE_STATMIN_MIN_SET	%DB15.DBW536	Int	Minutes Counter for Working Time idle cold state set from HMI
"CBCS_WDB".Sys-tem.COUNT_TCYC_AMOUNT	%DB15.DBW546	Int	amount of thermal cycles
"CBCS_WDB".Sys-tem.COUNT_TCYC_AMOUNT_SET	%DB15.DBW548	Int	amount of thermal cycles set from HMI
"CBCS_WDB".Sys-tem.COUNT_TCYC_AMOUNT_TOT	%DB15.DBW550	Int	Total amount of thermal cycles for session (from close door to open door)
"CBCS_WDB".Sys-tem.COUNT_TCYC_INDEX	%DB15.DBW542	Int	index single cycle

Symbol	Address	Type	Comment
"CBCS_WDB".System.COUNT_TCYC_INDEX_TOT	%DB15.DBW544	Int	total index single cycle
"CBCS_WDB".System.DewPoint.TargetMaxDew	%DB15.DBD514	Real	Target max dew point
"CBCS_WDB".System.DewPoint.TargetMinDew	%DB15.DBD510	Real	Target min dew point
"CBCS_WDB".System.Temperature.TargetMaxTemp	%DB15.DBD472	Real	Max Target temperature
"CBCS_WDB".System.Temperature.TargetMinTemp	%DB15.DBD468	Real	Min Target temperature
"CBS_DR_OS_GN"	%Q0.0	Bool	Coldbox Overhead Semaphore green light
"CBS_DR_OS_OR"	%Q0.1	Bool	Coldbox Overhead Semaphore green orange
"CBS_DR_OS_RD"	%Q0.2	Bool	Coldbox Overhead Semaphore green red
"CBS_DR_STAT_CLS"	%M60.0	Bool	Coldbox door closed
"CBS_DR_SW1"	%I0.0	Bool	Coldbox door structure switch 1
"CBS_DR_SW1_CLS"	%M50.0	Bool	Structure door switch 1 closed
"CBS_DR_SW2"	%I0.1	Bool	Coldbox door structure switch 2
"CBS_DR_SW2_CLS"	%M50.1	Bool	Structure door switch 2 closed
"CHL_CTRL_TEMP_CMD"	%QW112	Int	Chiller out command get Warm/Cold control
"CHL_EN_RUN"	%M50.3	Bool	Chiller is ready and enabled
"CHL_EN_RUN_CMD"	%Q0.3	Bool	Chiller grant command Stand-By input
"CHL_SYS_RDY"	%M50.2	Bool	Chiller is ready state
"CHL_SYS_WRN"	%I0.2	Bool	Chiller system alarm out
"Clock_0.5Hz"	%M30.7	Bool	
"Clock_Byte".%X7		Bool	
"CTRL_ALRM_MAIN"	%M55.0	Bool	General System alarm to disable ready state
"CTRL_SYS_AUTO_SET"	%M102.4	Bool	Auto mode is set
"CTRL_SYS_MAXTRG_SET"	%M102.1	Bool	Max target temperature SET
"CTRL_SYS_MINTRG_SET"	%M102.0	Bool	Min target temperature SET
"CTRL_SYS_TIM_ELAP_MAX"	%M102.3	Bool	Timer warm Idle state elapsed
"CTRL_SYS_TIM_ELAP_MIN"	%M102.2	Bool	Timer cold Idle state elapsed
"CTRL_TCYC_AUTO"	%M101.1	Bool	Control system thermal cycle auto mode
"CTRL_TCYC_FLT"	%M100.6	Bool	Control System fault state
"CTRL_TCYC_GO_COLD"	%M100.4	Bool	Control system is getting colder
"CTRL_TCYC_GO_WARM"	%M100.3	Bool	Control System is getting warmer
"CTRL_TCYC_IDLE"	%M100.0	Bool	Control System in idle state
"CTRL_TCYC_MAN"	%M101.0	Bool	Control system thermal cycle manual mode

Symbol	Address	Type	Comment
"CTRL_TCYC_MAXTRG_REACH"	%M101.3	Bool	Actual temp reach max target
"CTRL_TCYC_MINTRG_REACH"	%M101.2	Bool	Actual temp reach min target
"CTRL_TCYC_RDY"	%M100.1	Bool	Control System Ready
"CTRL_TCYC_RUN"	%M100.2	Bool	Control System is running
"CTRL_TCYC_SETUP"	%M100.7	Bool	Contro System setting up
"CTRL_TCYC_WRN"	%M100.5	Bool	Control System warning state
"DataSetFromHMI"	%M110.0	Bool	Data as been set from the HMI
"HMI_ALARMS_DB".System.CTRL_ALRM_MAIN		Bool	General Alarm
"HMI_ALARMS_DB".System.CTRL_SYS_ALRM		Word	Control System main Alarms
"HMI_ALARMS_DB".System.CTRL_SYS_ALRM.%X0		Bool	Control System main Alarms
"HMI_ALARMS_DB".System.CTRL_SYS_ALRM.%X1		Bool	Control System main Alarms
"HMI_ALARMS_DB".System.CTRL_SYS_ALRM.%X2		Bool	Control System main Alarms
"HMI_ALARMS_DB".System.CTRL_SYS_ALRM.%X3		Bool	Control System main Alarms
"HMI_ALARMS_DB".System.CTRL_SYS_ALRM.%X4		Bool	Control System main Alarms
"HMI_ALARMS_DB".System.CTRL_SYS_ALRM.%X5		Bool	Control System main Alarms
"HMI_ALARMS_DB".System.CTRL_SYS_ALRM.%X6		Bool	Control System main Alarms
"HMI_ALARMS_DB".System.CTRL_SYS_ALRM.%X7		Bool	Control System main Alarms
"HMI_ALARMS_DB".System.CTRL_SYS_ALRM.%X8		Bool	Control System main Alarms
"HMI_ALARMS_DB".System.CTRL_SYS_ALRM.%X9		Bool	Control System main Alarms
"HMI_ALARMS_DB".System.CTRL_SYS_ALRM.%X10		Bool	Control System main Alarms
"HMI_ALARMS_DB".System.CTRL_SYS_ALRM.%X11		Bool	Control System main Alarms
"HMI_ALARMS_DB".System.CTRL_SYS_ALRM.%X12		Bool	Control System main Alarms
"HMI_ALARMS_DB".System.CTRL_SYS_ALRM.%X13		Bool	Control System main Alarms

Symbol	Address	Type	Comment
"HMI_ALARMS_DB".System.CTRL_SYS_ALRM.%X14		Bool	Control System main Alarms
"HMI_ALARMS_DB".System.CTRL_SYS_ALRM.%X15		Bool	Control System main Alarms
"HMI_CMD_DB".HMI_CMD_DB.ACT_DATA		Bool	Actualize Data
"HMI_DATAPROC_DB".InnerColdbox.Dew.ACT_TMP_AVG		DInt	Actual average temperature of all dew sensors
"HMI_DATAPROC_DB".InnerColdbox.Dew.ACT_TMP_DEW1		DInt	actual dew point 1
"HMI_DATAPROC_DB".InnerColdbox.Dew.ACT_TMP_DEW2		DInt	actual dew point 2
"HMI_DATAPROC_DB".InnerColdbox.Temperature.ACT_TMP_AVG		DInt	Actual average temperature of all trays
"HMI_DATAPROC_DB".InnerColdbox.Temperature.ACT_TMP_TRAY1		DInt	actual temperature tray 1
"HMI_DATAPROC_DB".InnerColdbox.Temperature.ACT_TMP_TRAY2		DInt	actual temperature tray 2
"HMI_DATAPROC_DB".InnerColdbox.Temperature.ACT_TMP_TRAY3		DInt	actual temperature tray 3
"HMI_DATAPROC_DB".InnerColdbox.Temperature.ACT_TMP_TRAY4		DInt	actual temperature tray 4
"HMI_DATAPROC_DB".InnerColdbox.Temperature.ACT_TMP_TRAY5		DInt	actual temperature tray 5
"HMI_DATAPROC_DB".InnerColdbox.Temperature.ACT_TMP_TRAY6		DInt	actual temperature tray 6
"HMI_DATAPROC_DB".InnerColdbox.Temperature.ACT_TMP_TRAY7		DInt	actual temperature tray 7
"HMI_DATAPROC_DB".InnerColdbox.Temperature.ACT_TMP_TRAY8		DInt	actual temperature tray 8
"HMI_DATAPROC_DB".System.ACT_MAX_DEW_SET		DInt	Max Dew setpoint
"HMI_DATAPROC_DB".System.ACT_MAX_TMP_SET		DInt	Max temp setpoin
"HMI_DATAPROC_DB".System.ACT_MIN_DEW_SET		DInt	Min Dew setpoint
"HMI_DATAPROC_DB".System.ACT_MIN_TMP_SET		DInt	MIn temp setpoin

Symbol	Address	Type	Comment
"HMI_DATAPROC_DB".System.ACT_TARGET_TEMP		DInt	Actual target temperature
"HMI_DATAPROC_DB".System.ACT_TEMP		DInt	Actual temperature
"HMI_DATAPROC_DB".System.COUNT_IDLE_STAT_MIN		Int	Minutes Counter for Working Time idle state
"HMI_DATAPROC_DB".System.COUNT_IDLE_STAT_SEC		Int	Seconds Counter for Working Time idle state
"HMI_DATAPROC_DB".System.COUNT_IDLE_STAT-MAX_MIN_SET		Int	Minutes Counter for Working Time warm idle state SET
"HMI_DATAPROC_DB".System.COUNT_IDLE_STATMIN_MIN_SET		Int	Minutes Counter for Working Time cold idle state SET
"HMI_DATAPROC_DB".System.COUNT_TCYC_AMOUNT		Int	current amount of t cycles
"HMI_DATAPROC_DB".System.COUNT_TCYC_AMOUNT_SET		Int	current amount of t cycles set
"HMI_DATAPROC_DB".System.COUNT_TCYC_AMOUNT_TOT		Int	Session amount of t cycles
"HMI_DATA-SET_DB".DAT_COUNT_IDLE-MAX_MIN_SET		Int	Minutes set from HMI as idle state warm
"HMI_DATA-SET_DB".DAT_COUNT_IDLE-MIN_MIN_SET		Int	Minutes set from HMI as idle state cold
"HMI_DATA-SET_DB".DAT_COUNT_TCYC_AMOUNT_SET		Int	Amount of thermal cycles set from HMI
"HMI_DATASET_DB".DAT_MAX_DEW		DInt	Max Dew set from HMI
"HMI_DATASET_DB".DAT_MAX_TMP		DInt	Max temp set from HMI
"HMI_DATASET_DB".DAT_MIN_DEW		DInt	Min Dew set from HMI
"HMI_DATASET_DB".DAT_MIN_TMP		DInt	MIn temp set from HMI
"HMI_STATUS_DB".CBCS_PLC_RUN		Bool	PLC is running
"HMI_STATUS_DB".CBCS_SYS_RDY		Bool	Control system Ready
"HMI_STA-TUS_DB".CBCS_SYS_STATES.%X0		Bool	ControlSystem Machine States bits (0=idle, 1=enable, 2=running, 3=warming, 4=cooling, 5=warning, 6=fault
"HMI_STA-TUS_DB".CBCS_SYS_STATES.%X1		Bool	ControlSystem Machine States bits (0=idle, 1=enable, 2=running, 3=warming, 4=cooling, 5=warning, 6=fault

Symbol	Address	Type	Comment
"HMI_STA-TUS_DB".CBCS_SYS_STATES.%X2		Bool	ControlSystem Machine States bits (0=idle, 1=enable, 2=running, 3=warm-ing, 4=cooling, 5=warning, 6=fault
"HMI_STA-TUS_DB".CBCS_SYS_STATES.%X3		Bool	ControlSystem Machine States bits (0=idle, 1=enable, 2=running, 3=warm-ing, 4=cooling, 5=warning, 6=fault
"HMI_STA-TUS_DB".CBCS_SYS_STATES.%X4		Bool	ControlSystem Machine States bits (0=idle, 1=enable, 2=running, 3=warm-ing, 4=cooling, 5=warning, 6=fault
"HMI_STA-TUS_DB".CBCS_SYS_STATES.%X5		Bool	ControlSystem Machine States bits (0=idle, 1=enable, 2=running, 3=warm-ing, 4=cooling, 5=warning, 6=fault
"HMI_STA-TUS_DB".CBCS_SYS_STATES.%X6		Bool	ControlSystem Machine States bits (0=idle, 1=enable, 2=running, 3=warm-ing, 4=cooling, 5=warning, 6=fault
"HMI_STA-TUS_DB".CBCS_SYS_STATES.%X7		Bool	ControlSystem Machine States bits (0=idle, 1=enable, 2=running, 3=warm-ing, 4=cooling, 5=warning, 6=fault
"HMI_STA-TUS_DB".CBCS_SYS_STATES.%X8		Bool	ControlSystem Machine States bits (0=idle, 1=enable, 2=running, 3=warm-ing, 4=cooling, 5=warning, 6=fault
"HMI_STA-TUS_DB".CBCS_SYS_STATES.%X9		Bool	ControlSystem Machine States bits (0=idle, 1=enable, 2=running, 3=warm-ing, 4=cooling, 5=warning, 6=fault
"HMI_STA-TUS_DB".CBCS_SYS_STATES.%X10		Bool	ControlSystem Machine States bits (0=idle, 1=enable, 2=running, 3=warm-ing, 4=cooling, 5=warning, 6=fault
"HMI_STA-TUS_DB".CBCS_SYS_STATES.%X11		Bool	ControlSystem Machine States bits (0=idle, 1=enable, 2=running, 3=warm-ing, 4=cooling, 5=warning, 6=fault
"HMI_STA-TUS_DB".CBCS_SYS_STATES.%X12		Bool	ControlSystem Machine States bits (0=idle, 1=enable, 2=running, 3=warm-ing, 4=cooling, 5=warning, 6=fault
"HMI_STA-TUS_DB".CBCS_SYS_STATES.%X13		Bool	ControlSystem Machine States bits (0=idle, 1=enable, 2=running, 3=warm-ing, 4=cooling, 5=warning, 6=fault
"HMI_STA-TUS_DB".CBCS_SYS_STATES.%X14		Bool	ControlSystem Machine States bits (0=idle, 1=enable, 2=running, 3=warm-ing, 4=cooling, 5=warning, 6=fault
"HMI_STA-TUS_DB".CBCS_SYS_STATES.%X15		Bool	ControlSystem Machine States bits (0=idle, 1=enable, 2=running, 3=warm-ing, 4=cooling, 5=warning, 6=fault
"HMI_SYS_ALARM".%X0		Bool	Control system machine Alarms to HMI
"HMI_SYS_ALARM".%X1		Bool	Control system machine Alarms to HMI
"HMI_SYS_ALARM".%X2		Bool	Control system machine Alarms to HMI
"HMI_SYS_ALARM".%X3		Bool	Control system machine Alarms to HMI
"HMI_SYS_ALARM".%X4		Bool	Control system machine Alarms to HMI
"HMI_SYS_ALARM".%X5		Bool	Control system machine Alarms to HMI
"HMI_SYS_ALARM".%X6		Bool	Control system machine Alarms to HMI
"HMI_SYS_ALARM".%X7		Bool	Control system machine Alarms to HMI
"HMI_SYS_ALARM".%X8		Bool	Control system machine Alarms to HMI
"HMI_SYS_ALARM".%X9		Bool	Control system machine Alarms to HMI

Symbol	Address	Type	Comment
"HMI_SYS_ALARM".%X10		Bool	Control system machine Alarms to HMI
"HMI_SYS_ALARM".%X11		Bool	Control system machine Alarms to HMI
"HMI_SYS_ALARM".%X12		Bool	Control system machine Alarms to HMI
"HMI_SYS_ALARM".%X13		Bool	Control system machine Alarms to HMI
"HMI_SYS_ALARM".%X14		Bool	Control system machine Alarms to HMI
"HMI_SYS_ALARM".%X15		Bool	Control system machine Alarms to HMI
"HMI_SYS_STATES"	%MW105	Word	Control system machine states to HMI
"HMI_SYS_STATES".%X0		Bool	Control system machine states to HMI
"HMI_SYS_STATES".%X1		Bool	Control system machine states to HMI
"HMI_SYS_STATES".%X2		Bool	Control system machine states to HMI
"HMI_SYS_STATES".%X3		Bool	Control system machine states to HMI
"HMI_SYS_STATES".%X4		Bool	Control system machine states to HMI
"HMI_SYS_STATES".%X5		Bool	Control system machine states to HMI
"HMI_SYS_STATES".%X6		Bool	Control system machine states to HMI
"HMI_SYS_STATES".%X7		Bool	Control system machine states to HMI
"HMI_SYS_STATES".%X8		Bool	Control system machine states to HMI
"HMI_SYS_STATES".%X9		Bool	Control system machine states to HMI
"HMI_SYS_STATES".%X10		Bool	Control system machine states to HMI
"HMI_SYS_STATES".%X11		Bool	Control system machine states to HMI
"HMI_SYS_STATES".%X12		Bool	Control system machine states to HMI
"HMI_SYS_STATES".%X13		Bool	Control system machine states to HMI
"HMI_SYS_STATES".%X14		Bool	Control system machine states to HMI
"HMI_SYS_STATES".%X15		Bool	Control system machine states to HMI
#CTU_TCYC		IEC_Counter	Cycle count
#CTU_TCYC_TOT		IEC_Counter	Session cycle count
#F_TRIG_MAN_GO_COLD		Multi_FB	
#F_TRIG_MAN_GO_WARM		Multi_FB	
#R_TRIG_ALRM		Multi_FB	
#R_TRIG_ALRM2		Multi_FB	
#R_TRIG_MAN_STA		Multi_FB	Manual Start cmd rising edge detect
#R_TRIG_MAN_STP		Multi_FB	Manual Stop cmd rising edge detect
#TMP_MAN_STA_PREV		Bool	Previous manual start state
#TMP_MAN_STP_PREV		Bool	Previous manual stop state
#TMP_R_TRIG_MAN_STA		Bool	Manual Start cmd rising edge
#TMP_R_TRIG_MAN_STP		Bool	Manual Stop cmd rising edge
#TOFF_CHL_SYS_WRN		IEC_Timer	Timer filter chiller alarm out
#TOFF_SW1		IEC_Timer	Timer filter door switch 1

Symbol	Address	Type	Comment
#TOFF_SW2		IEC_Timer	Timer filter door switch 2
#TP_MAN_GO_COLD		IEC_Timer	
#TP_MAN_GO_WARM		IEC_Timer	

PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly] / Program blocks

CBI_FB_DAQ [FB20]

CBI_FB_DAQ Properties							
General							
Name	CBI_FB_DAQ	**Number**	20	**Type**	FB	**Language**	SCL
Numbering	Manual						
Information							
Title	CBI_FB_DAQ	**Author**	MicheleBaschiera	**Comment**	Main FB data acquisition	**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Default value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
Input									
Output									
InOut									
Static									
▼ Temp									
▼ TMP_PT_SEL	Array[0..7] of Bool								Array to selected Temp Sensors
TMP_PT_SEL[0]	Bool								Array to selected Temp Sensors
TMP_PT_SEL[1]	Bool								Array to selected Temp Sensors
TMP_PT_SEL[2]	Bool								Array to selected Temp Sensors
TMP_PT_SEL[3]	Bool								Array to selected Temp Sensors
TMP_PT_SEL[4]	Bool								Array to selected Temp Sensors

Name	Data type	Default value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
TMP_PT_SEL[5]	Bool								Array to selected Temp Sensors
TMP_PT_SEL[6]	Bool								Array to selected Temp Sensors
TMP_PT_SEL[7]	Bool								Array to selected Temp Sensors
▼ TMP_DEW_SEL	Array[0..7] of Bool								Array to selected Dew Sensors
TMP_DEW_SEL[0]	Bool								Array to selected Dew Sensors
TMP_DEW_SEL[1]	Bool								Array to selected Dew Sensors
TMP_DEW_SEL[2]	Bool								Array to selected Dew Sensors
TMP_DEW_SEL[3]	Bool								Array to selected Dew Sensors
TMP_DEW_SEL[4]	Bool								Array to selected Dew Sensors
TMP_DEW_SEL[5]	Bool								Array to selected Dew Sensors
TMP_DEW_SEL[6]	Bool								Array to selected Dew Sensors
TMP_DEW_SEL[7]	Bool								Array to selected Dew Sensors
Constant									

```
0001 (*
0002 //################### FUNCTION TO READ ALL TEMPERATURE AND DEW SENSORS INSTALLED INSIDE THE COLDBOX
     #######################//
0003 *)
0004
0005 REGION TEMP_SENSORS_ACTIVE
```

```
0006
0007   "CBI_PT_SEL_TR1" := TRUE;
0008   "CBI_PT_SEL_TR2" := TRUE;
0009   "CBI_PT_SEL_TR3" := "HMI_CMD_DB".HMI_CMD_DB.PTSEL3;
0010   "CBI_PT_SEL_TR4" := "HMI_CMD_DB".HMI_CMD_DB.PTSEL4;
0011   "CBI_PT_SEL_TR5" := "HMI_CMD_DB".HMI_CMD_DB.PTSEL5;
0012   "CBI_PT_SEL_TR6" := "HMI_CMD_DB".HMI_CMD_DB.PTSEL6;
0013   "CBI_PT_SEL_TR7" := "HMI_CMD_DB".HMI_CMD_DB.PTSEL7;
0014   "CBI_PT_SEL_TR8" := "HMI_CMD_DB".HMI_CMD_DB.PTSEL8;
0015
0016   #TMP_PT_SEL[0] := TRUE;
0017   #TMP_PT_SEL[1] := TRUE;
0018   #TMP_PT_SEL[2] := "HMI_CMD_DB".HMI_CMD_DB.PTSEL3;
0019   #TMP_PT_SEL[3] := "HMI_CMD_DB".HMI_CMD_DB.PTSEL4;
0020   #TMP_PT_SEL[4] := "HMI_CMD_DB".HMI_CMD_DB.PTSEL5;
0021   #TMP_PT_SEL[5] := "HMI_CMD_DB".HMI_CMD_DB.PTSEL6;
0022   #TMP_PT_SEL[6] := "HMI_CMD_DB".HMI_CMD_DB.PTSEL7;
0023   #TMP_PT_SEL[7] := "HMI_CMD_DB".HMI_CMD_DB.PTSEL8;
0024
0025    "IN_CNT_BOOL"(IN_MB:=#TMP_PT_SEL,
0026             INOUT_CNT:="CBCS_WDB".InnerColdbox.CBI_PT_AMT_SEL);
0027
0028 END_REGION
0029
0030
0031 REGION READ_TEMP_SENSORS
0032
0033   //TO DO ADD ALARM RESET
0034
0035   //TRAY 1
0036   "REF_PT_INT_REAL"(IN_PTActive := TRUE,
0037             IN_Safe_OUT := 0.0,
0038             IN_ACK := FALSE,
0039             IN_PIW_Address := "CBI_PT1000_TR1_RAW",
0040             OUT_Value => "CBCS_WDB".InnerColdbox.PT1000.CBITR1_PT1000.ActTemp,
0041             IN_OUT_Field_Fault := "CBCS_WDB".InnerColdbox.PT1000.CBITR1_PT1000.PT_FIELD_FAULT,
0042             IN_OUT_AI_CH_Fault := "CBCS_WDB".InnerColdbox.PT1000.CBITR1_PT1000.PT_CH_FAULT);
0043   //TRAY 2
0044   "REF_PT_INT_REAL"(IN_PTActive := TRUE,
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0045            IN_Safe_OUT := 0.0,
0046            IN_ACK := FALSE,
0047            IN_PIW_Address := "CBI_PT1000_TR2_RAW",
0048            OUT_Value => "CBCS_WDB".InnerColdbox.PT1000.CBITR2_PT1000.ActTemp,
0049            IN_OUT_Field_Fault := "CBCS_WDB".InnerColdbox.PT1000.CBITR2_PT1000.PT_FIELD_FAULT,
0050            IN_OUT_AI_CH_Fault := "CBCS_WDB".InnerColdbox.PT1000.CBITR2_PT1000.PT_CH_FAULT);
0051   //TRAY 3
0052   "REF_PT_INT_REAL"(IN_PTActive := "CBI_PT_SEL_TR3",
0053            IN_Safe_OUT := 0.0,
0054            IN_ACK := FALSE,
0055            IN_PIW_Address := "CBI_PT1000_TR3_RAW",
0056            OUT_Value => "CBCS_WDB".InnerColdbox.PT1000.CBITR3_PT1000.ActTemp,
0057            IN_OUT_Field_Fault := "CBCS_WDB".InnerColdbox.PT1000.CBITR3_PT1000.PT_FIELD_FAULT,
0058            IN_OUT_AI_CH_Fault := "CBCS_WDB".InnerColdbox.PT1000.CBITR3_PT1000.PT_CH_FAULT);
0059   //TRAY 4
0060   "REF_PT_INT_REAL"(IN_PTActive := "CBI_PT_SEL_TR4",
0061            IN_Safe_OUT := 0.0,
0062            IN_ACK := FALSE,
0063            IN_PIW_Address := "CBI_PT1000_TR4_RAW",
0064            OUT_Value => "CBCS_WDB".InnerColdbox.PT1000.CBITR4_PT1000.ActTemp,
0065            IN_OUT_Field_Fault := "CBCS_WDB".InnerColdbox.PT1000.CBITR4_PT1000.PT_FIELD_FAULT,
0066            IN_OUT_AI_CH_Fault := "CBCS_WDB".InnerColdbox.PT1000.CBITR4_PT1000.PT_CH_FAULT);
0067   //TRAY 5
0068   "REF_PT_INT_REAL"(IN_PTActive := "CBI_PT_SEL_TR5",
0069            IN_Safe_OUT := 0.0,
0070            IN_ACK := FALSE,
0071            IN_PIW_Address := "CBI_PT1000_TR5_RAW",
0072            OUT_Value => "CBCS_WDB".InnerColdbox.PT1000.CBITR5_PT1000.ActTemp,
0073            IN_OUT_Field_Fault := "CBCS_WDB".InnerColdbox.PT1000.CBITR5_PT1000.PT_FIELD_FAULT,
0074            IN_OUT_AI_CH_Fault := "CBCS_WDB".InnerColdbox.PT1000.CBITR5_PT1000.PT_CH_FAULT);
0075   //TRAY 6
0076   "REF_PT_INT_REAL"(IN_PTActive := "CBI_PT_SEL_TR6",
0077            IN_Safe_OUT := 0.0,
0078            IN_ACK := FALSE,
0079            IN_PIW_Address := "CBI_PT1000_TR6_RAW",
0080            OUT_Value => "CBCS_WDB".InnerColdbox.PT1000.CBITR6_PT1000.ActTemp,
0081            IN_OUT_Field_Fault := "CBCS_WDB".InnerColdbox.PT1000.CBITR6_PT1000.PT_FIELD_FAULT,
0082            IN_OUT_AI_CH_Fault := "CBCS_WDB".InnerColdbox.PT1000.CBITR6_PT1000.PT_CH_FAULT);
0083   //TRAY 7
```

```
0084   "REF_PT_INT_REAL"(IN_PTActive := "CBI_PT_SEL_TR7",
0085             IN_Safe_OUT := 0.0,
0086             IN_ACK := FALSE,
0087             IN_PIW_Address := "CBI_PT1000_TR7_RAW",
0088             OUT_Value => "CBCS_WDB".InnerColdbox.PT1000.CBITR7_PT1000.ActTemp,
0089             IN_OUT_Field_Fault := "CBCS_WDB".InnerColdbox.PT1000.CBITR7_PT1000.PT_FIELD_FAULT,
0090             IN_OUT_AI_CH_Fault := "CBCS_WDB".InnerColdbox.PT1000.CBITR7_PT1000.PT_CH_FAULT);
0091   //TRAY 8
0092   "REF_PT_INT_REAL"(IN_PTActive := "CBI_PT_SEL_TR8",
0093             IN_Safe_OUT := 0.0,
0094             IN_ACK := FALSE,
0095             IN_PIW_Address := "CBI_PT1000_TR8_RAW",
0096             OUT_Value => "CBCS_WDB".InnerColdbox.PT1000.CBITR8_PT1000.ActTemp,
0097             IN_OUT_Field_Fault := "CBCS_WDB".InnerColdbox.PT1000.CBITR8_PT1000.PT_FIELD_FAULT,
0098             IN_OUT_AI_CH_Fault := "CBCS_WDB".InnerColdbox.PT1000.CBITR8_PT1000.PT_CH_FAULT);
0099
0100 END_REGION
0101
0102 REGION AVERAGE TEMP READOUT
0103
0104   "CBCS_WDB".InnerColdbox.CBI_PT_AVG_ACT_TEMP := (
0105   "CBCS_WDB".InnerColdbox.PT1000.CBITR1_PT1000.ActTemp +
0106   "CBCS_WDB".InnerColdbox.PT1000.CBITR2_PT1000.ActTemp +
0107   "CBCS_WDB".InnerColdbox.PT1000.CBITR3_PT1000.ActTemp +
0108   "CBCS_WDB".InnerColdbox.PT1000.CBITR4_PT1000.ActTemp +
0109   "CBCS_WDB".InnerColdbox.PT1000.CBITR5_PT1000.ActTemp +
0110   "CBCS_WDB".InnerColdbox.PT1000.CBITR6_PT1000.ActTemp +
0111   "CBCS_WDB".InnerColdbox.PT1000.CBITR7_PT1000.ActTemp +
0112   "CBCS_WDB".InnerColdbox.PT1000.CBITR8_PT1000.ActTemp) / "CBCS_WDB".InnerColdbox.CBI_PT_AMT_SEL;
0113
0114 END_REGION
0115
0116 REGION DEW SENSORS ACTIVE
0117
0118   "CBI_DEW_SEL_S1" := TRUE;
0119   "CBI_DEW_SEL_S2" := "HMI_CMD_DB".HMI_CMD_DB.DEWSEL2;
0120
0121   IF "CBI_DEW_SEL_S2" THEN
0122     "CBCS_WDB".InnerColdbox.CMI_DEW_AMT_SEL := 2;
```

```
0123
0124   ELSE
0125     "CBCS_WDB".InnerColdbox.CMI_DEW_AMT_SEL := 1;
0126   END_IF;
0127
0128 END_REGION
0129
0130
0131 REGION READ_DEW_SENSORS
0132
0133   "AI_SCALE_FC"(IN_PTActive:=TRUE,
0134           IN_PIW_Address := "CBI_DMT143_I1_RAW",
0135           IN_HI_LIM := "CBCS_MDB".InnerColdbox.DMT143.Dew_HiLim,
0136           IN_LO_LIM := "CBCS_MDB".InnerColdbox.DMT143.Dew_LowLim,
0137           IN_Safe_OUT := 0.0,
0138           IN_BIPOLAR := FALSE,
0139           IN_ACK := FALSE,
0140           OUT_Value => "CBCS_WDB".InnerColdbox.DMT143.CBI_DMT1.ActDew,
0141           IN_OUT_Field_Fault := "CBCS_WDB".InnerColdbox.DMT143.CBI_DMT1.DMT_FIELD_FAULT,
0142           IN_OUT_AI_CH_Fault := "CBCS_WDB".InnerColdbox.DMT143.CBI_DMT1.DMT_CH_FAULT);
0143
0144
0145   "AI_SCALE_FC"(IN_PTActive:="CBI_DEW_SEL_S2",
0146           IN_PIW_Address := "CBI_DMT143_I2_RAW",
0147           IN_HI_LIM := "CBCS_MDB".InnerColdbox.DMT143.Dew_HiLim,
0148           IN_LO_LIM := "CBCS_MDB".InnerColdbox.DMT143.Dew_LowLim,
0149           IN_Safe_OUT := 0.0,
0150           IN_BIPOLAR := FALSE,
0151           IN_ACK := FALSE,
0152           OUT_Value => "CBCS_WDB".InnerColdbox.DMT143.CBI_DMT2.ActDew,
0153           IN_OUT_Field_Fault := "CBCS_WDB".InnerColdbox.DMT143.CBI_DMT2.DMT_FIELD_FAULT,
0154           IN_OUT_AI_CH_Fault := "CBCS_WDB".InnerColdbox.DMT143.CBI_DMT2.DMT_CH_FAULT);
0155
0156 END_REGION
0157
0158 REGION AVERAGE DEW POINT TEMP READOUT
0159
0160   "CBCS_WDB".InnerColdbox.CBI_DEW_AVG_ACT_TEMP := (
0161   "CBCS_WDB".InnerColdbox.DMT143.CBI_DMT1.ActDew +
```

```
0162   "CBCS_WDB".InnerColdbox.DMT143.CBI_DMT2.ActDew) / "CBCS_WDB".InnerColdbox.CMI_DEW_AMT_SEL;
0163
0164 END_REGION
0165
0166
0167 REGION READ_CHILLER_TEMP
0168
0169   //TO DO : ADD ALARM RESET
0170
0171   "AI_SCALE_FC"(IN_PTActive:=TRUE,
0172           IN_PIW_Address := "CHL_SYS_TEMP_RAW",
0173           IN_HI_LIM := "CBCS_MDB".Chiller.Temperature.Chiller_HiLim,
0174           IN_LO_LIM := "CBCS_MDB".Chiller.Temperature.Chiller_LowLim,
0175           IN_Safe_OUT := 0.0,
0176           IN_BIPOLAR := FALSE,
0177           IN_ACK := FALSE,
0178           OUT_Value => "CBCS_WDB".Chiller.Temperature.ActTemp,
0179           IN_OUT_Field_Fault := "CBCS_WDB".Chiller.Temperature.CHILLER_FIELD_FAULT,
0180           IN_OUT_AI_CH_Fault := "CBCS_WDB".Chiller.Temperature.CHILLER_CH_FAULT);
0181
0182 END_REGION
0183
0184 REGION TRAYS TEMPERATURE LIMITS SET FOR SHOOTDOWN
0185
0186   "TEMP_LIMT_FC"(IN_TR_ActTemp := "CBCS_WDB".InnerColdbox.PT1000.CBITR1_PT1000.ActTemp,
0187            OUT_TempLimit_Hi => "CBI_PT_HITEMPLIMT_TR1",
0188            OUT_TempLimit_Low => "CBI_PT_LOWTEMPLIMT_TR1");
0189
0190   "TEMP_LIMT_FC"(IN_TR_ActTemp := "CBCS_WDB".InnerColdbox.PT1000.CBITR2_PT1000.ActTemp,
0191            OUT_TempLimit_Hi => "CBI_PT_HITEMPLIMT_TR2",
0192            OUT_TempLimit_Low => "CBI_PT_LOWTEMPLIMT_TR2");
0193
0194   "TEMP_LIMT_FC"(IN_TR_ActTemp := "CBCS_WDB".InnerColdbox.PT1000.CBITR3_PT1000.ActTemp,
0195            OUT_TempLimit_Hi => "CBI_PT_HITEMPLIMT_TR3",
0196            OUT_TempLimit_Low => "CBI_PT_LOWTEMPLIMT_TR3");
0197
0198   "TEMP_LIMT_FC"(IN_TR_ActTemp := "CBCS_WDB".InnerColdbox.PT1000.CBITR4_PT1000.ActTemp,
0199            OUT_TempLimit_Hi => "CBI_PT_HITEMPLIMT_TR4",
0200            OUT_TempLimit_Low => "CBI_PT_LOWTEMPLIMT_TR4");
```

```
0201
0202   "TEMP_LIMT_FC"(IN_TR_ActTemp := "CBCS_WDB".InnerColdbox.PT1000.CBITR5_PT1000.ActTemp,
0203            OUT_TempLimit_Hi => "CBI_PT_HITEMPLIMT_TR5",
0204            OUT_TempLimit_Low => "CBI_PT_LOWTEMPLIMT_TR5");
0205
0206   "TEMP_LIMT_FC"(IN_TR_ActTemp := "CBCS_WDB".InnerColdbox.PT1000.CBITR6_PT1000.ActTemp,
0207            OUT_TempLimit_Hi => "CBI_PT_HITEMPLIMT_TR6",
0208            OUT_TempLimit_Low => "CBI_PT_LOWTEMPLIMT_TR6");
0209
0210   "TEMP_LIMT_FC"(IN_TR_ActTemp := "CBCS_WDB".InnerColdbox.PT1000.CBITR7_PT1000.ActTemp,
0211            OUT_TempLimit_Hi => "CBI_PT_HITEMPLIMT_TR7",
0212            OUT_TempLimit_Low => "CBI_PT_LOWTEMPLIMT_TR7");
0213
0214   "TEMP_LIMT_FC"(IN_TR_ActTemp := "CBCS_WDB".InnerColdbox.PT1000.CBITR8_PT1000.ActTemp,
0215            OUT_TempLimit_Hi => "CBI_PT_HITEMPLIMT_TR8",
0216            OUT_TempLimit_Low => "CBI_PT_LOWTEMPLIMT_TR8");
0217 END_REGION
0218
0219 REGION TIMEOUT CHILLER TEMPERATURE
0220
0221   // KEEPS MONITORING THE TEMP OF CHILLER CHANGING. OF IS NOT CHANGINNG OVER SOME TIME TRIGGER ALLARM
0222
0223 END_REGION
0224
0225 (*
0226 ################ ALARM AND STATUS #################
0227 *)
0228
0229 REGION ALARMS_TO_HMI
0230   // Statement section REGION
0231
0232 END_REGION
0233
0234
0235
```

Symbol	Address	Type	Comment
"CBCS_MDB".Chiller.Tempera-ture.Chiller_HiLim		Real	Chiller temperature - value for 20 mA from SCALE BLOCK

Symbol	Address	Type	Comment
"CBCS_MDB".Chiller.Tempera-ture.Chiller_LowLim		Real	Chiller temperature - value for 4 mA from SCALE BLOCK
"CBCS_MDB".InnerCold-box.DMT143.Dew_HiLim		Real	Tray dew point - value for 20 mA from SCALE BLOCK
"CBCS_MDB".InnerCold-box.DMT143.Dew_LowLim		Real	Tray dew point - value for 4 mA from SCALE BLOCK
"CBCS_WDB".Chiller.Tempera-ture.ActTemp	%DB15.DBD410	Real	Actual temperature from chiller DEG
"CBCS_WDB".Chiller.Tempera-ture.CHILLER_CH_FAULT	%DB15.DBX446.1	Bool	CHILLER AIN channel fault
"CBCS_WDB".Chiller.Tempera-ture.CHILLER_FIELD_FAULT	%DB15.DBX446.0	Bool	CHILLER sensor error from field
"CBCS_WDB".InnerCold-box.CBI_DEW_AVG_ACT_TEMP	%DB15.DBD392	Real	Average dew point from DEW sensors
"CBCS_WDB".InnerCold-box.CBI_PT_AMT_SEL	%DB15.DBW390	Int	PT amount selected
"CBCS_WDB".InnerCold-box.CBI_PT_AVG_ACT_TEMP	%DB15.DBD386	Real	Average temperature from PT sensors
"CBCS_WDB".InnerCold-box.CMI_DEW_AMT_SEL	%DB15.DBW396	Int	DEW amount selected
"CBCS_WDB".InnerCold-box.DMT143.CBI_DMT1.ActDew	%DB15.DBD314	Real	Actual dew point of the inner tray
"CBCS_WDB".InnerCold-box.DMT143.CBI_DMT1.DMT_CH_FA ULT	%DB15.DBX342.1	Bool	DMT AIN channel fault
"CBCS_WDB".InnerCold-box.DMT143.CBI_DMT1.DMT_FIELD_ FAULT	%DB15.DBX342.0	Bool	DMT sensor error from field
"CBCS_WDB".InnerCold-box.DMT143.CBI_DMT2.ActDew	%DB15.DBD356	Real	Actual dew point of the inner tray
"CBCS_WDB".InnerCold-box.DMT143.CBI_DMT2.DMT_CH_FA ULT	%DB15.DBX384.1	Bool	DMT AIN channel fault
"CBCS_WDB".InnerCold-box.DMT143.CBI_DMT2.DMT_FIELD_ FAULT	%DB15.DBX384.0	Bool	DMT sensor error from field
"CBCS_WDB".InnerCold-box.PT1000.CBITR1_PT1000.Act-Temp	%DB15.DBD6	Real	Actual temperature of the inner tray DEG

Symbol	Address	Type	Comment
"CBCS_WDB".InnerCold-box.PT1000.CBITR1_PT1000.PT_CH_FAULT	%DB15.DBX34.1	Bool	PT AIN channel fault
"CBCS_WDB".InnerCold-box.PT1000.CBITR1_PT1000.PT_FIELD_FAULT	%DB15.DBX34.0	Bool	PT sensor error from field
"CBCS_WDB".InnerCold-box.PT1000.CBITR2_PT1000.Act-Temp	%DB15.DBD44	Real	Actual temperature of the inner tray DEG
"CBCS_WDB".InnerCold-box.PT1000.CBITR2_PT1000.PT_CH_FAULT	%DB15.DBX72.1	Bool	PT AIN channel fault
"CBCS_WDB".InnerCold-box.PT1000.CBITR2_PT1000.PT_FIELD_FAULT	%DB15.DBX72.0	Bool	PT sensor error from field
"CBCS_WDB".InnerCold-box.PT1000.CBITR3_PT1000.Act-Temp	%DB15.DBD82	Real	Actual temperature of the inner tray DEG
"CBCS_WDB".InnerCold-box.PT1000.CBITR3_PT1000.PT_CH_FAULT	%DB15.DBX110.1	Bool	PT AIN channel fault
"CBCS_WDB".InnerCold-box.PT1000.CBITR3_PT1000.PT_FIELD_FAULT	%DB15.DBX110.0	Bool	PT sensor error from field
"CBCS_WDB".InnerCold-box.PT1000.CBITR4_PT1000.Act-Temp	%DB15.DBD120	Real	Actual temperature of the inner tray DEG
"CBCS_WDB".InnerCold-box.PT1000.CBITR4_PT1000.PT_CH_FAULT	%DB15.DBX148.1	Bool	PT AIN channel fault
"CBCS_WDB".InnerCold-box.PT1000.CBITR4_PT1000.PT_FIELD_FAULT	%DB15.DBX148.0	Bool	PT sensor error from field
"CBCS_WDB".InnerCold-box.PT1000.CBITR5_PT1000.Act-Temp	%DB15.DBD158	Real	Actual temperature of the inner tray DEG
"CBCS_WDB".InnerCold-box.PT1000.CBITR5_PT1000.PT_CH_FAULT	%DB15.DBX186.1	Bool	PT AIN channel fault

Symbol	Address	Type	Comment
"CBCS_WDB".InnerCold-box.PT1000.CBITR5_PT1000.PT_FIELD_FAULT	%DB15.DBX186.0	Bool	PT sensor error from field
"CBCS_WDB".InnerCold-box.PT1000.CBITR6_PT1000.Act-Temp	%DB15.DBD196	Real	Actual temperature of the inner tray DEG
"CBCS_WDB".InnerCold-box.PT1000.CBITR6_PT1000.PT_CH_FAULT	%DB15.DBX224.1	Bool	PT AIN channel fault
"CBCS_WDB".InnerCold-box.PT1000.CBITR6_PT1000.PT_FIELD_FAULT	%DB15.DBX224.0	Bool	PT sensor error from field
"CBCS_WDB".InnerCold-box.PT1000.CBITR7_PT1000.Act-Temp	%DB15.DBD234	Real	Actual temperature of the inner tray DEG
"CBCS_WDB".InnerCold-box.PT1000.CBITR7_PT1000.PT_CH_FAULT	%DB15.DBX262.1	Bool	PT AIN channel fault
"CBCS_WDB".InnerCold-box.PT1000.CBITR7_PT1000.PT_FIELD_FAULT	%DB15.DBX262.0	Bool	PT sensor error from field
"CBCS_WDB".InnerCold-box.PT1000.CBITR8_PT1000.Act-Temp	%DB15.DBD272	Real	Actual temperature of the inner tray DEG
"CBCS_WDB".InnerCold-box.PT1000.CBITR8_PT1000.PT_CH_FAULT	%DB15.DBX300.1	Bool	PT AIN channel fault
"CBCS_WDB".InnerCold-box.PT1000.CBITR8_PT1000.PT_FIELD_FAULT	%DB15.DBX300.0	Bool	PT sensor error from field
"CBI_DEW_SEL_S1"	%M41.0	Bool	Sensor selected DEW1
"CBI_DEW_SEL_S2"	%M41.1	Bool	Sensor selected DEW2
"CBI_DMT143_I1_RAW"	%IW112	Int	Inner coldbox dew sensor raw 1
"CBI_DMT143_I2_RAW"	%IW114	Int	Inner coldbox dew sensor raw 2
"CBI_PT_HITEMPLIMT_TR1"	%M51.0	Bool	Temperature tray 1 high limit
"CBI_PT_HITEMPLIMT_TR2"	%M51.1	Bool	Temperature tray 2 high limit
"CBI_PT_HITEMPLIMT_TR3"	%M51.2	Bool	Temperature tray 3 high limit
"CBI_PT_HITEMPLIMT_TR4"	%M51.3	Bool	Temperature tray 4 high limit
"CBI_PT_HITEMPLIMT_TR5"	%M51.4	Bool	Temperature tray 5 high limit

Symbol	Address	Type	Comment
"CBI_PT_HITEMPLIMT_TR6"	%M51.5	Bool	Temperature tray 6 high limit
"CBI_PT_HITEMPLIMT_TR7"	%M51.6	Bool	Temperature tray 7 high limit
"CBI_PT_HITEMPLIMT_TR8"	%M51.7	Bool	Temperature tray 8 high limit
"CBI_PT_LOWTEMPLIMT_TR1"	%M52.0	Bool	Temperature tray 1 Low limit
"CBI_PT_LOWTEMPLIMT_TR2"	%M52.1	Bool	Temperature tray 2 Low limit
"CBI_PT_LOWTEMPLIMT_TR3"	%M52.2	Bool	Temperature tray 3 Low limit
"CBI_PT_LOWTEMPLIMT_TR4"	%M52.3	Bool	Temperature tray 4 Low limit
"CBI_PT_LOWTEMPLIMT_TR5"	%M52.4	Bool	Temperature tray 5 Low limit
"CBI_PT_LOWTEMPLIMT_TR6"	%M52.5	Bool	Temperature tray 6 Low limit
"CBI_PT_LOWTEMPLIMT_TR7"	%M52.6	Bool	Temperature tray 7 Low limit
"CBI_PT_LOWTEMPLIMT_TR8"	%M52.7	Bool	Temperature tray 8 Low limit
"CBI_PT_SEL_TR1"	%M40.0	Bool	Sensor selected PT1
"CBI_PT_SEL_TR2"	%M40.1	Bool	Sensor selected PT2
"CBI_PT_SEL_TR3"	%M40.2	Bool	Sensor selected PT3
"CBI_PT_SEL_TR4"	%M40.3	Bool	Sensor selected PT4
"CBI_PT_SEL_TR5"	%M40.4	Bool	Sensor selected PT5
"CBI_PT_SEL_TR6"	%M40.5	Bool	Sensor selected PT6
"CBI_PT_SEL_TR7"	%M40.6	Bool	Sensor selected PT7
"CBI_PT_SEL_TR8"	%M40.7	Bool	Sensor selected PT8
"CBI_PT1000_TR1_RAW"	%IW96	Int	Inner coldbox temperature sensor raw tray 1
"CBI_PT1000_TR2_RAW"	%IW98	Int	Inner coldbox temperature sensor raw tray 2
"CBI_PT1000_TR3_RAW"	%IW100	Int	Inner coldbox temperature sensor raw tray 3
"CBI_PT1000_TR4_RAW"	%IW102	Int	Inner coldbox temperature sensor raw tray 4
"CBI_PT1000_TR5_RAW"	%IW104	Int	Inner coldbox temperature sensor raw tray 5
"CBI_PT1000_TR6_RAW"	%IW106	Int	Inner coldbox temperature sensor raw tray 6
"CBI_PT1000_TR7_RAW"	%IW108	Int	Inner coldbox temperature sensor raw tray 7
"CBI_PT1000_TR8_RAW"	%IW110	Int	Inner coldbox temperature sensor raw tray 8
"CHL_SYS_TEMP_RAW"	%IW116	Int	Chiller temperature recorder output raw
"HMI_CMD_DB".HMI_CMD_DB.DEW-SEL2		Bool	Dew selection 2
"HMI_CMD_DB".HMI_CMD_DB.PTSEL 3		Bool	PT Selection 3
"HMI_CMD_DB".HMI_CMD_DB.PTSEL 4		Bool	PT Selection 4
"HMI_CMD_DB".HMI_CMD_DB.PTSEL 5		Bool	PT Selection 5

Symbol	Address	Type	Comment
"HMI_CMD_DB".HMI_CMD_DB.PTSEL 6		Bool	PT Selection 6
"HMI_CMD_DB".HMI_CMD_DB.PTSEL 7		Bool	PT Selection 7
"HMI_CMD_DB".HMI_CMD_DB.PTSEL 8		Bool	PT Selection 8
#TMP_PT_SEL		Array	Array to selected Temp Sensors
#TMP_PT_SEL[0]		Bool	Array to selected Temp Sensors
#TMP_PT_SEL[1]		Bool	Array to selected Temp Sensors
#TMP_PT_SEL[2]		Bool	Array to selected Temp Sensors
#TMP_PT_SEL[3]		Bool	Array to selected Temp Sensors
#TMP_PT_SEL[4]		Bool	Array to selected Temp Sensors
#TMP_PT_SEL[5]		Bool	Array to selected Temp Sensors
#TMP_PT_SEL[6]		Bool	Array to selected Temp Sensors
#TMP_PT_SEL[7]		Bool	Array to selected Temp Sensors

PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly] / Program blocks

CBCS_FB_CTRL_DB [DB21]

CBCS_FB_CTRL_DB Properties							
General							
Name	CBCS_FB_CTRL_DB	**Number**	21	**Type**	DB	**Language**	DB
Numbering	Manual						
Information							
Title		**Author**	MicheleBaschiera	**Comment**		**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writ-able from HMI/ OPC UA/ Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion	Comment
Input									
Output									
InOut									
▼ Static									
▼ TOFF_SW1	TOF_TIME		False	True	True	True	True		Timer filter door switch 1
PT	Time	T#0ms	False	True	True	True	False		
ET	Time	T#0ms	False	True	False	True	False		
IN	Bool	false	False	True	True	True	False		
Q	Bool	false	False	True	False	True	False		
▼ TOFF_SW2	TOF_TIME		False	True	True	True	True		Timer filter door switch 2
PT	Time	T#0ms	False	True	True	True	False		
ET	Time	T#0ms	False	True	False	True	False		
IN	Bool	false	False	True	True	True	False		
Q	Bool	false	False	True	False	True	False		
▼ TOFF_CHL_SYS_WRN	TOF_TIME		False	True	True	True	True		Timer filter chiller alarm out

Name	Data type	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writ-able from HMI/ OPC UA/ Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion	Comment
PT	Time	T#0ms	False	True	True	True	False		
ET	Time	T#0ms	False	True	False	True	False		
IN	Bool	false	False	True	True	True	False		
Q	Bool	false	False	True	False	True	False		
R_TRIG_MAN_STA	R_TRIG		False	True	True	True	True		Manual Start cmd rising edge detect
Input									
CLK	Bool	false	False	True	True	True	False		
Output									
Q	Bool	false	False	True	True	True	False		
InOut									
Static									
Stat_Bit	Bool	false	False	True	True	True	False		
R_TRIG_MAN_STP	R_TRIG		False	True	True	True	True		Manual Stop cmd rising edge detect
Input									
CLK	Bool	false	False	True	True	True	False		
Output									
Q	Bool	false	False	True	True	True	False		
InOut									
Static									
Stat_Bit	Bool	false	False	True	True	True	False		
R_TRIG_ALRM	R_TRIG		False	True	True	True	True		
Input									
CLK	Bool	false	False	True	True	True	False		
Output									
Q	Bool	false	False	True	True	True	False		

Name	Data type	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writ-able from HMI/ OPC UA/ Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion	Comment
InOut									
Static									
Stat_Bit	Bool	false	False	True	True	True	False		
R_TRIG_ALRM2	R_TRIG		False	True	True	True	True		
Input									
CLK	Bool	false	False	True	True	True	False		
Output									
Q	Bool	false	False	True	True	True	False		
InOut									
Static									
Stat_Bit	Bool	false	False	True	True	True	False		
TP_MAN_GO_COLD	TP_TIME		False	True	True	True	True		
PT	Time	T#0ms	False	True	True	True	False		
ET	Time	T#0ms	False	True	False	True	False		
IN	Bool	false	False	True	True	True	False		
Q	Bool	false	False	True	False	True	False		
TP_MAN_GO_WARM	TP_TIME		False	True	True	True	True		
PT	Time	T#0ms	False	True	True	True	False		
ET	Time	T#0ms	False	True	False	True	False		
IN	Bool	false	False	True	True	True	False		
Q	Bool	false	False	True	False	True	False		
F_TRIG_MAN_GO_COLD	F_TRIG		False	True	True	True	True		
Input									
CLK	Bool	false	False	True	True	True	False		
Output									
Q	Bool	false	False	True	True	True	False		

Name	Data type	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writ-able from HMI/ OPC UA/ Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion	Comment
InOut									
Static									
Stat_Bit	Bool	false	False	True	True	True	False		
F_TRIG_MAN_GO_WARM	F_TRIG		False	True	True	True	True		
Input									
CLK	Bool	false	False	True	True	True	False		
Output									
Q	Bool	false	False	True	True	True	False		
InOut									
Static									
Stat_Bit	Bool	false	False	True	True	True	False		
CTU_TCYC	CTU_INT		True	True	True	True	True		Cycle count
CU	Bool	false	True	True	True	True	False		
CD	Bool	false	True	True	True	True	False		
R	Bool	false	True	True	True	True	False		
LD	Bool	false	True	True	True	True	False		
QU	Bool	false	True	True	True	True	False		
QD	Bool	false	True	True	True	True	False		
PV	Int	0	True	True	True	True	False		
CV	Int	0	True	True	True	True	False		
CTU_TCYC_TOT	CTU_INT		True	True	True	True	True		Session cycle count
CU	Bool	false	True	True	True	True	False		
CD	Bool	false	True	True	True	True	False		
R	Bool	false	True	True	True	True	False		
LD	Bool	false	True	True	True	True	False		
QU	Bool	false	True	True	True	True	False		

Name	Data type	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writ-able from HMI/ OPC UA/ Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion				Comment
QD	Bool	false	True	True	True	True	False					
PV	Int	0	True	True	True	True	False					
CV	Int	0	True	True	True	True	False					

PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly] / Program blocks

CBCS_MDB [DB16]

CBCS_MDB Properties							
General							
Name	CBCS_MDB	**Number**	16	**Type**	DB	**Language**	DB
Numbering	Manual						
Information							
Title	CBCS_MDB	**Author**	MicheleBaschiera	**Comment**	ColdboxControlSystem MachineDataBlock - Contains all default machine parameters	**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
Static									
InnerColdbox	Struct		False	True	True	True	False		#######CONFIGURATION DATA#########
PT1000	Struct		False	True	True	True	False		
MaxTempValue	Real	20.0	False	True	True	True	False		Tray temperature - value for 20 mA from SCALE BLOCK
MInTempValue	Real	-30.0	False	True	True	True	False		Tray temperature - value for 4 mA from SCALE BLOCK
Hysteresis	Real	2.5	False	True	True	True	False		
SystemMinTempStartConsent	Real	0.0	False	True	True	True	False		HMI temperature - min setpoint
SystemMaxTempStartConsent	Real	0.0	False	True	True	True	False		HMI temperature - max setpoint
SystemMinTempThreshold	Real	-40.0	False	True	True	True	False		System shutdown - min threshold temperature

Name	Data type	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
SystemMaxTempThreshold	Real	30.0	False	True	True	True	False		System shutdown - max threshold temperature
DMT143	Struct		False	True	True	True	False		
Dew_LowLim	Real	-80.0	False	True	True	True	False		Tray dew point - value for 4 mA from SCALE BLOCK
Dew_HiLim	Real	20.0	False	True	True	True	False		Tray dew point - value for 20 mA from SCALE BLOCK
Hysteresis	Real	0.0	False	True	True	True	False		
SystemMinDewStartConsent	Real	0.0	False	True	True	True	False		HMI dew point - min setpoin
SystemMaxDewStartConsent	Real	0.0	False	True	True	True	False		HMI dew point - max setpoin
SistemMinDewThreshold	Real	0.0	False	True	True	True	False		System shutdown - min threshold dew
SistemMaxDewThreshold	Real	0.0	False	True	True	True	False		System shutdown - max threshold dew
Chiller	Struct		False	True	True	True	False		
Temperature	Struct		False	True	True	True	False		
Chiller_HiLim	Real	20.0	False	True	True	True	False		Chiller temperature - value for 20 mA from SCALE BLOCK
Chiller_LowLim	Real	-80.0	False	True	True	True	False		Chiller temperature - value for 4 mA from SCALE BLOCK
System	Struct		False	True	True	True	False		
MaxTempValue	Real	20.0	False	True	True	True	False		Tray temperature - value for 20 mA from SCALE BLOCK
MinTempValue	Real	-30.0	False	True	True	True	False		Tray temperature - value for 4 mA from SCALE BLOCK
Hysteresis	Real	2.5	False	True	True	True	False		
SystemMinTempLimThreshold	Real	-50.0	False	True	True	True	False		System shutdown - min threshold temperature

Name	Data type	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
SystemMaxTempLimThreshold	Real	50.0	False	True	True	True	False		System shutdown - max threshold temperature
MinDewValue	Real	40.7	False	True	True	True	False		System defaults - min dew point
MaxDewValue	Real	60.2	False	True	True	True	False		System defaults - max dew point
SystemMinDewStartConsent	Real	0.0	False	True	True	True	False		HMI dew point - min setpoin
SystemMaxDewStartConsent	Real	0.0	False	True	True	True	False		HMI dew point - max setpoin
SistemMinDewThreshold	Real	0.0	False	True	True	True	False		System shutdown - min threshold dew
SistemMaxDewThreshold	Real	0.0	False	True	True	True	False		System shutdown - max threshold dew
Count_TCYC_Amount_Default	Int	5	False	True	True	True	False		Default amount of Thermal cycles
Interval_TCYC_MIN	Int	2	False	True	True	True	False		Default amount of minutes set to stay in idle state
DewTempDelta_Range	Real	3.0	False	True	True	True	False		Defautl delta range between Dew and Temperature

PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly] / Program blocks

CBCS_WDB [DB15]

CBCS_WDB Properties							
General							
Name	CBCS_WDB	**Number**	15	**Type**	DB	**Language**	DB
Numbering	Manual						
Information							
Title	CBCS_WDB	**Author**	MicheleBaschiera	**Comment**	Working DataBlock	**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Offset	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
Static										
InnerColdbox	Struct	0.0		False	True	True	True	False		
PT1000	Struct	0.0		False	True	True	True	False		
CBITR1_PT1000	Struct	0.0		False	True	True	True	False		
ActRawData	Int	0.0	0	False	True	True	True	False		Actual data raw temp from card
SPARE1	DWord	2.0	16#0	False	True	True	True	False		
ActTemp	Real	6.0	0.0	False	True	True	True	False		Actual temperature of the inner tray DEG
SPARE3	DInt	10.0	0	False	True	True	True	False		
SPARE4	Real	14.0	0.0	False	True	True	True	False		
SPARE5	Real	18.0	0.0	False	True	True	True	False		
SIM_TEMP	DInt	22.0	0	False	True	True	True	False		Simulated temperature of the inner tray
SPARE6	Real	26.0	0.0	False	True	True	True	False		
SPARE7	Real	30.0	0.0	False	True	True	True	False		
PT_FIELD_FAULT	Bool	34.0	false	False	True	True	True	False		PT sensor error from field

Name	Data type	Offset	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
PT_CH_FAULT	Bool	34.1	false	False	True	True	True	False		PT AIN channel fault
▼ CBITR2_PT1000	Struct	36.0		False	True	True	True	False		
ActRawData	DInt	36.0	0	False	True	True	True	False		Actual data raw temp from card
SPARE1	DWord	40.0	16#0	False	True	True	True	False		
ActTemp	Real	44.0	0.0	False	True	True	True	False		Actual temperature of the inner tray DEG
SPARE3	DInt	48.0	0	False	True	True	True	False		
SPARE4	Real	52.0	0.0	False	True	True	True	False		
SPARE5	Real	56.0	0.0	False	True	True	True	False		
SIM_TEMP	DInt	60.0	0	False	True	True	True	False		Simulated temperature of the inner tray
SPARE6	Real	64.0	0.0	False	True	True	True	False		
SPARE7	Real	68.0	0.0	False	True	True	True	False		
PT_FIELD_FAULT	Bool	72.0	false	False	True	True	True	False		PT sensor error from field
PT_CH_FAULT	Bool	72.1	false	False	True	True	True	False		PT AIN channel fault
▼ CBITR3_PT1000	Struct	74.0		False	True	True	True	False		
ActRawData	DInt	74.0	0	False	True	True	True	False		Actual data raw temp from card
SPARE1	DWord	78.0	16#0	False	True	True	True	False		
ActTemp	Real	82.0	0.0	False	True	True	True	False		Actual temperature of the inner tray DEG
SPARE3	DInt	86.0	0	False	True	True	True	False		
SPARE4	Real	90.0	0.0	False	True	True	True	False		
SPARE5	Real	94.0	0.0	False	True	True	True	False		
SIM_TEMP	DInt	98.0	0	False	True	True	True	False		Simulated temperature of the inner tray
SPARE6	Real	102.0	0.0	False	True	True	True	False		
SPARE7	Real	106.0	0.0	False	True	True	True	False		
PT_FIELD_FAULT	Bool	110.0	false	False	True	True	True	False		PT sensor error from field
PT_CH_FAULT	Bool	110.1	false	False	True	True	True	False		PT AIN channel fault

Name	Data type	Offset	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
CBITR4_PT1000	Struct	112.0		False	True	True	True	False		
ActRawData	DInt	112.0	0	False	True	True	True	False		Actual data raw temp from card
SPARE1	DWord	116.0	16#0	False	True	True	True	False		
ActTemp	Real	120.0	0.0	False	True	True	True	False		Actual temperature of the inner tray DEG
SPARE3	DInt	124.0	0	False	True	True	True	False		
SPARE4	Real	128.0	0.0	False	True	True	True	False		
SPARE5	Real	132.0	0.0	False	True	True	True	False		
SIM_TEMP	DInt	136.0	0	False	True	True	True	False		Simulated temperature of the inner tray
SPARE6	Real	140.0	0.0	False	True	True	True	False		
SPARE7	Real	144.0	0.0	False	True	True	True	False		
PT_FIELD_FAULT	Bool	148.0	false	False	True	True	True	False		PT sensor error from field
PT_CH_FAULT	Bool	148.1	false	False	True	True	True	False		PT AIN channel fault
CBITR5_PT1000	Struct	150.0		False	True	True	True	False		
ActRawData	DInt	150.0	0	False	True	True	True	False		Actual data raw temp from card
SPARE1	DWord	154.0	16#0	False	True	True	True	False		
ActTemp	Real	158.0	0.0	False	True	True	True	False		Actual temperature of the inner tray DEG
SPARE3	DInt	162.0	0	False	True	True	True	False		
SPARE4	Real	166.0	0.0	False	True	True	True	False		
SPARE5	Real	170.0	0.0	False	True	True	True	False		
SIM_TEMP	DInt	174.0	0	False	True	True	True	False		Simulated temperature of the inner tray
SPARE6	Real	178.0	0.0	False	True	True	True	False		
SPARE7	Real	182.0	0.0	False	True	True	True	False		
PT_FIELD_FAULT	Bool	186.0	false	False	True	True	True	False		PT sensor error from field
PT_CH_FAULT	Bool	186.1	false	False	True	True	True	False		PT AIN channel fault
CBITR6_PT1000	Struct	188.0		False	True	True	True	False		

Name	Data type	Offset	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision				Comment
ActRawData	DInt	188.0	0	False	True	True	True	False					Actual data raw temp from card
SPARE1	DWord	192.0	16#0	False	True	True	True	False					
ActTemp	Real	196.0	0.0	False	True	True	True	False					Actual temperature of the inner tray DEG
SPARE3	DInt	200.0	0	False	True	True	True	False					
SPARE4	Real	204.0	0.0	False	True	True	True	False					
SPARE5	Real	208.0	0.0	False	True	True	True	False					
SIM_TEMP	DInt	212.0	0	False	True	True	True	False					Simulated temperature of the inner tray
SPARE6	Real	216.0	0.0	False	True	True	True	False					
SPARE7	Real	220.0	0.0	False	True	True	True	False					
PT_FIELD_FAULT	Bool	224.0	false	False	True	True	True	False					PT sensor error from field
PT_CH_FAULT	Bool	224.1	false	False	True	True	True	False					PT AIN channel fault
▼ CBITR7_PT1000	Struct	226.0		False	True	True	True	False					
ActRawData	DInt	226.0	0	False	True	True	True	False					Actual data raw temp from card
SPARE1	DWord	230.0	16#0	False	True	True	True	False					
ActTemp	Real	234.0	0.0	False	True	True	True	False					Actual temperature of the inner tray DEG
SPARE3	DInt	238.0	0	False	True	True	True	False					
SPARE4	Real	242.0	0.0	False	True	True	True	False					
SPARE5	Real	246.0	0.0	False	True	True	True	False					
SIM_TEMP	DInt	250.0	0	False	True	True	True	False					Simulated temperature of the inner tray
SPARE6	Real	254.0	0.0	False	True	True	True	False					
SPARE7	Real	258.0	0.0	False	True	True	True	False					
PT_FIELD_FAULT	Bool	262.0	false	False	True	True	True	False					PT sensor error from field
PT_CH_FAULT	Bool	262.1	false	False	True	True	True	False					PT AIN channel fault
▼ CBITR8_PT1000	Struct	264.0		False	True	True	True	False					
ActRawData	DInt	264.0	0	False	True	True	True	False					Actual data raw temp from card

Name	Data type	Offset	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
SPARE1	DWord	268.0	16#0	False	True	True	True	False		
ActTemp	Real	272.0	0.0	False	True	True	True	False		Actual temperature of the inner tray DEG
SPARE3	DInt	276.0	0	False	True	True	True	False		
SPARE4	Real	280.0	0.0	False	True	True	True	False		
SPARE5	Real	284.0	0.0	False	True	True	True	False		
SIM_TEMP	DInt	288.0	0	False	True	True	True	False		Simulated temperature of the inner tray
SPARE6	Real	292.0	0.0	False	True	True	True	False		
SPARE7	Real	296.0	0.0	False	True	True	True	False		
PT_FIELD_FAULT	Bool	300.0	false	False	True	True	True	False		PT sensor error from field
PT_CH_FAULT	Bool	300.1	false	False	True	True	True	False		PT AIN channel fault
DMT143	Struct	302.0		False	True	True	True	False		
CBI_DMT1	Struct	302.0		False	True	True	True	False		
ActRawData	DInt	302.0	0	False	True	True	True	False		Actual data raw dew from card
SPARE1	DWord	306.0	16#0	False	True	True	True	False		
SPARE2	Real	310.0	0.0	False	True	True	True	False		
ActDew	Real	314.0	0.0	False	True	True	True	False		Actual dew point of the inner tray
SPARE3	DInt	318.0	0	False	True	True	True	False		
SPARE4	Real	322.0	0.0	False	True	True	True	False		
SPARE5	Real	326.0	0.0	False	True	True	True	False		
SIM_DEW	DInt	330.0	0	False	True	True	True	False		Simulated dew of the inner coldbox
SPARE6	Real	334.0	0.0	False	True	True	True	False		
SPARE7	Real	338.0	0.0	False	True	True	True	False		
DMT_FIELD_FAULT	Bool	342.0	false	False	True	True	True	False		DMT sensor error from field
DMT_CH_FAULT	Bool	342.1	false	False	True	True	True	False		DMT AIN channel fault
CBI_DMT2	Struct	344.0		False	True	True	True	False		
ActRawData	DInt	344.0	0	False	True	True	True	False		Actual data raw dew from card

Name	Data type	Offset	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
SPARE1	DWord	348.0	16#0	False	True	True	True	False		
SPARE2	Real	352.0	0.0	False	True	True	True	False		
ActDew	Real	356.0	0.0	False	True	True	True	False		Actual dew point of the inner tray
SPARE3	DInt	360.0	0	False	True	True	True	False		
SPARE4	Real	364.0	0.0	False	True	True	True	False		
SPARE5	Real	368.0	0.0	False	True	True	True	False		
SIM_DEW	DInt	372.0	0	False	True	True	True	False		Simulated dew of the inner coldbox
SPARE6	Real	376.0	0.0	False	True	True	True	False		
SPARE7	Real	380.0	0.0	False	True	True	True	False		
DMT_FIELD_FAULT	Bool	384.0	false	False	True	True	True	False		DMT sensor error from field
DMT_CH_FAULT	Bool	384.1	false	False	True	True	True	False		DMT AIN channel fault
CBI_PT_AVG_ACT_TEMP	Real	386.0	0.0	False	True	True	True	False		Average temperature from PT sensors
CBI_PT_AMT_SEL	Int	390.0	0	False	True	True	True	False		PT amount selected
CBI_DEW_AVG_ACT_TEMP	Real	392.0	0.0	False	True	True	True	False		Average dew point from DEW sensors
CMI_DEW_AMT_SEL	Int	396.0	0	False	True	True	True	False		DEW amount selected
Chiller	Struct	398.0		False	True	True	True	False		
Temperature	Struct	398.0		False	True	True	True	False		
ActRawData	DInt	398.0	0	False	True	True	True	False		Actual data raw temp from card
SPARE1	DWord	402.0	16#0	False	True	True	True	False		
SPARE2	Real	406.0	0.0	False	True	True	True	False		
ActTemp	Real	410.0	0.0	False	True	True	True	False		Actual temperature from chiller DEG
SPARE3	DInt	414.0	0	False	True	True	True	False		
ActualTargetTemp	Real	418.0	0.0	False	True	True	True	False		Actual Target temperature
PrevTargetTemp	Real	422.0	0.0	False	True	True	True	False		Previous Target temperature
SPARE4	Real	426.0	0.0	False	True	True	True	False		
SPARE5	Real	430.0	0.0	False	True	True	True	False		

Name	Data type	Offset	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
SIM_TEMP	DInt	434.0	0	False	True	True	True	False		Simulated temperature of the chiller
SPARE6	Real	438.0	0.0	False	True	True	True	False		
SPARE7	Real	442.0	0.0	False	True	True	True	False		
CHILLER_FIELD_FAULT	Bool	446.0	false	False	True	True	True	False		CHILLER sensor error from field
CHILLER_CH_FAULT	Bool	446.1	false	False	True	True	True	False		CHILLER AIN channel fault
System	Struct	448.0		False	True	True	True	False		
Temperature	Struct	448.0		False	True	True	True	False		
ActRawData	DInt	448.0	0	False	True	True	True	False		Actual data raw temp from card
SPARE1	DWord	452.0	16#0	False	True	True	True	False		
SPARE2	Real	456.0	0.0	False	True	True	True	False		
ActTemp	Real	460.0	0.0	False	True	True	True	False		Actual temperature from chiller DEG
SPARE3	DInt	464.0	0	False	True	True	True	False		
TargetMinTemp	Real	468.0	0.0	False	True	True	True	False		Min Target temperature
TargetMaxTemp	Real	472.0	0.0	False	True	True	True	False		Max Target temperature
SIM_TEMP	DInt	476.0	0	False	True	True	True	False		Simulated temperature of the chiller
SPARE6	Real	480.0	0.0	False	True	True	True	False		
SPARE7	Real	484.0	0.0	False	True	True	True	False		
OUT_FAULT	Bool	488.0	false	False	True	True	True	False		AO value exceeded
DewPoint	Struct	490.0		False	True	True	True	False		
ActRawData	DInt	490.0	0	False	True	True	True	False		Actual data raw temp from card
SPARE1	DWord	494.0	16#0	False	True	True	True	False		
SPARE2	Real	498.0	0.0	False	True	True	True	False		
ActDew	Real	502.0	0.0	False	True	True	True	False		Actual temperature from chiller DEG
SPARE3	DInt	506.0	0	False	True	True	True	False		
TargetMinDew	Real	510.0	0.0	False	True	True	True	False		Target min dew point

Name	Data type	Offset	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
TargetMaxDew	Real	514.0	0.0	False	True	True	True	False		Target max dew point
SIM_TEMP	DInt	518.0	0	False	True	True	True	False		Simulated temperature of the chiller
SPARE6	Real	522.0	0.0	False	True	True	True	False		
SPARE7	Real	526.0	0.0	False	True	True	True	False		
COUNT_IDLE_STAT_SEC	Int	530.0	0	False	True	True	True	False		Seconds Counter for Working Time idle state
COUNT_IDLE_STAT_MIN	Int	532.0	0	False	True	True	True	False		Minutes Counter for Working Time idle state
COUNT_IDLE_STAT_HOUR	Int	534.0	0	False	True	True	True	False		Hours Counter for Working Time idle state
COUNT_IDLE_STAT-MIN_MIN_SET	Int	536.0	0	False	True	True	True	False		Minutes Counter for Working Time idle cold state set from HMI
COUNT_IDLE_STAT-MAX_MIN_SET	Int	538.0	0	False	True	True	True	False		Minutes Counter for Working Time idle warm state set from HMI
COUNT_IDLE_STAT_RUN	Bool	540.0	false	False	True	True	True	False		Counter for Working Time idle state Running
COUNT_TCYC_INDEX	Int	542.0	0	False	True	True	True	False		index single cycle
COUNT_TCYC_INDEX_TOT	Int	544.0	0	False	True	True	True	False		total index single cycle
COUNT_TCYC_AMOUNT	Int	546.0	0	False	True	True	True	False		amount of thermal cycles
COUNT_TCYC_AMOUNT_SET	Int	548.0	0	False	True	True	True	False		amount of thermal cycles set from HMI
COUNT_TCYC_AMOUNT_TOT	Int	550.0	0	False	True	True	True	False		Total amount of thermal cycles for session (from close door to open door)
ACT_DEWTEMPDLT_RNG	Real	552.0	0.0	False	True	True	True	False		Actual delta range between Dew and Temperature
▼ HMI_CMD	"UDT01_HMI_CMD"	556.0		False	True	True	True	True		
STA	Bool	556.0	false	False	True	True	True	False		Start
AUTO	Bool	556.1	true	False	True	True	True	False		Auto

Name	Data type	Offset	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
STP	Bool	556.2	false	False	True	True	True	False		Stop
ON	Bool	556.3	false	False	True	True	True	False		On
OFF	Bool	556.4	false	False	True	True	True	False		Off
ALM_RES	Bool	556.5	false	False	True	True	True	False		Alarm reset
MAN	Bool	556.6	false	False	True	True	True	False		Manual
LAMP_TST	Bool	556.7	false	False	True	True	True	False		Lamp test
DR_LK	Bool	557.0	false	False	True	True	True	False		Door lock
DR_ULK	Bool	557.1	false	False	True	True	True	False		Door Unlock
REST	Bool	557.2	false	False	True	True	True	False		Restart
INCR_UP	Bool	557.3	false	False	True	True	True	False		Increase/Up
DECR_DW	Bool	557.4	false	False	True	True	True	False		Decrease/Down
ACT_DATA	Bool	557.5	false	False	True	True	True	False		Actualize Data
SYNCH	Bool	557.6	false	False	True	True	True	False		Synchronization
RES	Bool	557.7	false	False	True	True	True	False		Reset
PTSEL1	Bool	558.0	true	False	True	True	True	False		PT Selection 1
PTSEL2	Bool	558.1	true	False	True	True	True	False		PT Selection 2
PTSEL3	Bool	558.2	true	False	True	True	True	False		PT Selection 3
PTSEL4	Bool	558.3	true	False	True	True	True	False		PT Selection 4
PTSEL5	Bool	558.4	true	False	True	True	True	False		PT Selection 5
PTSEL6	Bool	558.5	true	False	True	True	True	False		PT Selection 6
PTSEL7	Bool	558.6	true	False	True	True	True	False		PT Selection 7
PTSEL8	Bool	558.7	true	False	True	True	True	False		PT Selection 8
DEWSEL1	Bool	559.0	true	False	True	True	True	False		Dew selection 1
DEWSEL2	Bool	559.1	true	False	True	True	True	False		Dew selection 2

PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly] / Program blocks

CBI_FB_DAQ_DB [DB20]

CBI_FB_DAQ_DB Properties							
General							
Name	CBI_FB_DAQ_DB	**Number**	20	**Type**	DB	**Language**	DB
Numbering	Manual						
Information							
Title		**Author**	MicheleBaschiera	**Comment**		**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writ-able from HMI/ OPC UA/ Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion	Comment
Input									
Output									
InOut									
Static									

PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly] / Program blocks

HMI_ALARMS_DB [DB14]

HMI_ALARMS_DB Properties							
General							
Name	HMI_ALARMS_DB	**Number**	14	**Type**	DB	**Language**	DB
Numbering	Manual						
Information							
Title	HMI_ALARMS_DB	**Author**	MicheleBaschiera	**Comment**	Datablock of alarms to be send to HMI	**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
Static									
DAQ	Struct		False	True	True	True	False		
CBI_MIN_TMP_TR1	Bool	FALSE	False	True	True	True	False		Min temperator TR1
CBI_MAX_TMP_TR1	Bool	FALSE	False	True	True	True	False		Max temperature TR1
CBI_SNS_FLT_TR1	Bool	FALSE	False	True	True	True	False		Fault from sensor TR1
CBI_MIN_TMP_TR2	Bool	FALSE	False	True	True	True	False		Min temperator TR2
CBI_MAX_TMP_TR2	Bool	FALSE	False	True	True	True	False		Max temperature TR2
CBI_SNS_FLT_TR2	Bool	FALSE	False	True	True	True	False		Fault from sensor TR2
CBI_MIN_TMP_TR3	Bool	FALSE	False	True	True	True	False		Min temperator TR3
CBI_MAX_TMP_TR3	Bool	FALSE	False	True	True	True	False		Max temperature TR3
CBI_SNS_FLT_TR3	Bool	FALSE	False	True	True	True	False		Fault from sensor TR3
CBI_MIN_TMP_TR4	Bool	FALSE	False	True	True	True	False		Min temperator TR4
CBI_MAX_TMP_TR4	Bool	FALSE	False	True	True	True	False		Max temperature TR4
CBI_SNS_FLT_TR4	Bool	FALSE	False	True	True	True	False		Fault from sensor TR4
CBI_MIN_TMP_TR5	Bool	FALSE	False	True	True	True	False		Min temperator TR5

Name	Data type	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writ-able from HMI/ OPC UA/ Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion	Comment
CBI_MAX_TMP_TR5	Bool	FALSE	False	True	True	True	False		Max temperature TR5
CBI_SNS_FLT_TR5	Bool	FALSE	False	True	True	True	False		Fault from sensor TR5
CBI_MIN_TMP_TR6	Bool	FALSE	False	True	True	True	False		Min temperator TR6
CBI_MAX_TMP_TR6	Bool	FALSE	False	True	True	True	False		Max temperature TR6
CBI_SNS_FLT_TR6	Bool	FALSE	False	True	True	True	False		Fault from sensor TR6
CBI_MIN_TMP_TR7	Bool	FALSE	False	True	True	True	False		Min temperator TR7
CBI_MAX_TMP_TR7	Bool	FALSE	False	True	True	True	False		Max temperature TR7
CBI_SNS_FLT_TR7	Bool	FALSE	False	True	True	True	False		Fault from sensor TR7
CBI_MIN_TMP_TR8	Bool	FALSE	False	True	True	True	False		Min temperator TR8
CBI_MAX_TMP_TR8	Bool	FALSE	False	True	True	True	False		Max temperature TR8
CBI_SNS_FLT_TR8	Bool	FALSE	False	True	True	True	False		Fault from sensor TR8
CBI_MIN_DEW_DMT1	Bool	FALSE	False	True	True	True	False		Min dew point DMT1
CBI_MAX_DEW_DMT1	Bool	FALSE	False	True	True	True	False		Max dew point DMT1
CBI_SNS_FLT_DMT1	Bool	FALSE	False	True	True	True	False		Fault from sensor DMT1
CBI_MIN_DEW_DMT2	Bool	FALSE	False	True	True	True	False		Min dew point DMT2
CBI_MAX_DEW_DMT2	Bool	FALSE	False	True	True	True	False		Max dew point DMT2
CBI_SNS_FLT_DMT2	Bool	FALSE	False	True	True	True	False		Fault from sensor DMT2
▼ System	Struct		False	True	True	True	False		
CHL_MIN_TMP	Bool	FALSE	False	True	True	True	False		Min temperature chiller
CHL_MAX_TMP	Bool	FALSE	False	True	True	True	False		Max temperature chiller
CHL_SYS_WRN	Bool	false	False	True	True	True	False		Chiller alarm out
CBS_DR_OPN_SW1	Bool	false	False	True	True	True	False		Door open from switch 1
CBS_DR_OPN_SW2	Bool	false	False	True	True	True	False		Door open from switch 2
CBS_DR_STAT_CLS	Bool	false	False	True	True	True	False		Coldbox door is open alarm
CTRL_ALRM_MAIN	Bool	false	False	True	True	True	False		General Alarm
CTRL_SYS_ALRM	Word	16#0	False	True	True	True	False		Control System main Alarms

PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly] / Program blocks

HMI_CMD_DB [DB10]

HMI_CMD_DB Properties							
General							
Name	HMI_CMD_DB	**Number**	10	**Type**	DB	**Language**	DB
Numbering	Manual						
Information							
Title	HMI_COMMAND_DB	**Author**	MicheleBaschiera	**Comment**	All commands send by the HMI to PLC	**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
Static									
SpareSlot	Array[0..5] of Int		False	True	True	True	False		
SpareSlot[0]	Int	0	False	True	True	True	False		
SpareSlot[1]	Int	0	False	True	True	True	False		
SpareSlot[2]	Int	0	False	True	True	True	False		
SpareSlot[3]	Int	0	False	True	True	True	False		
SpareSlot[4]	Int	0	False	True	True	True	False		
SpareSlot[5]	Int	0	False	True	True	True	False		
HMI_CMD_DB	"UDT01_HMI_CMD"		False	True	True	True	True		
STA	Bool	false	False	True	True	True	False		Start
AUTO	Bool	true	False	True	True	True	False		Auto
STP	Bool	false	False	True	True	True	False		Stop
ON	Bool	false	False	True	True	True	False		On
OFF	Bool	false	False	True	True	True	False		Off

Name	Data type	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writ-able from HMI/ OPC UA/ Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion	Comment
ALM_RES	Bool	false	False	True	True	True	False		Alarm reset
MAN	Bool	false	False	True	True	True	False		Manual
LAMP_TST	Bool	false	False	True	True	True	False		Lamp test
DR_LK	Bool	false	False	True	True	True	False		Door lock
DR_ULK	Bool	false	False	True	True	True	False		Door Unlock
REST	Bool	false	False	True	True	True	False		Restart
INCR_UP	Bool	false	False	True	True	True	False		Increase/Up
DECR_DW	Bool	false	False	True	True	True	False		Decrease/Down
ACT_DATA	Bool	false	False	True	True	True	False		Actualize Data
SYNCH	Bool	false	False	True	True	True	False		Synchronization
RES	Bool	false	False	True	True	True	False		Reset
PTSEL1	Bool	true	False	True	True	True	False		PT Selection 1
PTSEL2	Bool	true	False	True	True	True	False		PT Selection 2
PTSEL3	Bool	true	False	True	True	True	False		PT Selection 3
PTSEL4	Bool	true	False	True	True	True	False		PT Selection 4
PTSEL5	Bool	true	False	True	True	True	False		PT Selection 5
PTSEL6	Bool	true	False	True	True	True	False		PT Selection 6
PTSEL7	Bool	true	False	True	True	True	False		PT Selection 7
PTSEL8	Bool	true	False	True	True	True	False		PT Selection 8
DEWSEL1	Bool	true	False	True	True	True	False		Dew selection 1
DEWSEL2	Bool	true	False	True	True	True	False		Dew selection 2

PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly] / Program blocks

HMI_DATAPROC_DB [DB8]

HMI_DATAPROC_DB Properties							
General							
Name	HMI_DATAPROC_DB	**Number**	8	**Type**	DB	**Language**	DB
Numbering	Manual						
Information							
Title	HMI_DATAPROC_DB	**Author**	MicheleBaschiera	**Comment**	Data processed to HMI	**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writ-able from HMI/ OPC UA/ Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion	Comment
Static									
System	Struct		False	True	True	True	False		
ACT_MIN_TMP_SET	DInt	-100	False	True	True	True	False		MIn temp setpoin
ACT_MAX_TMP_SET	DInt	200	False	True	True	True	False		Max temp setpoin
ACT_MIN_DEW_SET	DInt	400	False	True	True	True	False		Min Dew setpoint
ACT_MAX_DEW_SET	DInt	600	False	True	True	True	False		Max Dew setpoint
ACT_TEMP	DInt	0	False	True	True	True	False		Actual temperature
ACT_TARGET_TEMP	DInt	0	False	True	True	True	False		Actual target temperature
Spare_DBW000	Int	0	False	True	True	True	False		
Spare_DBW67	Int	0	False	True	True	True	False		
COUNT_TCYC_AMOUNT	Int	0	False	True	True	True	False		current amount of t cycles
COUNT_TCYC_AMOUNT_SET	Int	0	False	True	True	True	False		current amount of t cycles set
COUNT_TCYC_AMOUNT_TOT	Int	0	False	True	True	True	False		Session amount of t cycles

Name	Data type	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
COUNT_IDLE_STAT_SEC	Int	0	False	True	True	True	False		Seconds Counter for Working Time idle state
COUNT_IDLE_STAT_MIN	Int	0	False	True	True	True	False		Minutes Counter for Working Time idle state
COUNT_IDLE_STAT-MAX_MIN_SET	Int	0	False	True	True	True	False		Minutes Counter for Working Time warm idle state SET
COUNT_IDLE_STAT-MIN_MIN_SET	Int	0	False	True	True	True	False		Minutes Counter for Working Time cold idle state SET
COUNT_IDLE_STAT_RUN	Bool	false	False	True	True	True	False		Counter for Working Time idle state Running
Chiller	Struct		False	True	True	True	False		
ACT_MIN_TMP	DInt	-100	False	True	True	True	False		MIn temp set from PLC
Spare_DBW000	Int	0	False	True	True	True	False		
ACT_MAX_TMP	DInt	200	False	True	True	True	False		Max temp set from PLC
Spare_DBW002	Int	0	False	True	True	True	False		
InnerColdbox	Struct		False	True	True	True	False		
Temperature	Struct		False	True	True	True	False		
ACT_TMP_TRAY1	DInt	0	False	True	True	True	False		actual temperature tray 1
ACT_TMP_TRAY2	DInt	0	False	True	True	True	False		actual temperature tray 2
ACT_TMP_TRAY3	DInt	0	False	True	True	True	False		actual temperature tray 3
ACT_TMP_TRAY4	DInt	0	False	True	True	True	False		actual temperature tray 4
ACT_TMP_TRAY5	DInt	0	False	True	True	True	False		actual temperature tray 5
ACT_TMP_TRAY6	DInt	0	False	True	True	True	False		actual temperature tray 6
ACT_TMP_TRAY7	DInt	0	False	True	True	True	False		actual temperature tray 7
ACT_TMP_TRAY8	DInt	0	False	True	True	True	False		actual temperature tray 8
ACT_TMP_AVG	DInt	0	False	True	True	True	False		Actual average temperature of all trays
Dew	Struct		False	True	True	True	False		
ACT_TMP_DEW1	DInt	0	False	True	True	True	False		actual dew point 1

Name	Data type	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writ-able from HMI/ OPC UA/ Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion	Comment
ACT_TMP_DEW2	DInt	0	False	True	True	True	False		actual dew point 2
ACT_TMP_AVG	DInt	0	False	True	True	True	False		Actual average temperature of all dew sensors

PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly] / Program blocks

HMI_DATASET_DB [DB9]

HMI_DATASET_DB Properties							
General							
Name	HMI_DATASET_DB	**Number**	9	**Type**	DB	**Language**	DB
Numbering	Manual						
Information							
Title	HMI_DATASET_DB	**Author**	MicheleBaschiera	**Comment**	Dataset from HMI	**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
Static									
DAT_COUNT_IDLE-MIN_MIN_SET	Int	0	False	True	True	True	False		Minutes set from HMI as idle state cold
DAT_COUNT_IDLE-MAX_MIN_SET	Int	0	False	True	True	True	False		Minutes set from HMI as idle state warm
DAT_COUNT_TCYC_AMOUNT_SET	Int	0	False	True	True	True	False		Amount of thermal cycles set from HMI
Spare_DBW010	Int	0	False	True	True	True	False		
Spare_DBW012	Int	0	False	True	True	True	False		
Spare_DBW014	Int	0	False	True	True	True	False		
Spare_DBW016	Int	0	False	True	True	True	False		
Spare_DBW018	Int	0	False	True	True	True	False		
Spare_DBX020_0	Bool	FALSE	False	True	True	True	False		
Spare_DBX020_1	Bool	FALSE	False	True	True	True	False		
Spare_DBX020_2	Bool	FALSE	False	True	True	True	False		
Spare_DBX020_3	Bool	FALSE	False	True	True	True	False		

Name	Data type	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writ-able from HMI/ OPC UA/ Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion	Comment
Spare_DBX020_4	Bool	FALSE	False	True	True	True	False		
Spare_DBX020_5	Bool	FALSE	False	True	True	True	False		
Spare_DBX020_6	Bool	FALSE	False	True	True	True	False		
Spare_DBX020_7	Bool	FALSE	False	True	True	True	False		
Spare_DBX021_0	Bool	FALSE	False	True	True	True	False		
Spare_DBX021_1	Bool	FALSE	False	True	True	True	False		
Spare_DBX021_2	Bool	FALSE	False	True	True	True	False		
Spare_DBX021_3	Bool	FALSE	False	True	True	True	False		
Spare_DBX021_4	Bool	FALSE	False	True	True	True	False		
Spare_DBX021_5	Bool	FALSE	False	True	True	True	False		
Spare_DBX021_6	Bool	FALSE	False	True	True	True	False		
Spare_DBX021_7	Bool	FALSE	False	True	True	True	False		
Spare_DBW022	Int	0	False	True	True	True	False		
Spare_DBW024	Int	0	False	True	True	True	False		
Spare_DBW026	Int	0	False	True	True	True	False		
Spare_DBW028	Int	0	False	True	True	True	False		
Spare_DBW030	Int	0	False	True	True	True	False		
Spare_DBW032	Int	0	False	True	True	True	False		
Spare_DBW034	Int	0	False	True	True	True	False		
Spare_DBW036	Int	0	False	True	True	True	False		
Spare_DBW038	Int	0	False	True	True	True	False		
Spare_DBW040	Int	0	False	True	True	True	False		
Spare_DBW042	Int	0	False	True	True	True	False		
Spare_DBW044	Int	0	False	True	True	True	False		
Spare_DBW046	Int	0	False	True	True	True	False		
Spare_DBW048	Int	0	False	True	True	True	False		
DAT_MIN_TMP	DInt	L#0	False	True	True	True	False		MIn temp set from HMI
Spare_DBW054	Int	0	False	True	True	True	False		

Name	Data type	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writ-able from HMI/ OPC UA/ Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion	Comment
DAT_MAX_TMP	DInt	L#0	False	True	True	True	False		Max temp set from HMI
Spare_DBW060	Int	0	False	True	True	True	False		
DAT_MIN_DEW	DInt	L#0	False	True	True	True	False		Min Dew set from HMI
Spare_DBW066	Int	0	False	True	True	True	False		
DAT_MAX_DEW	DInt	L#0	False	True	True	True	False		Max Dew set from HMI
Spare_DBW072	Int	0	False	True	True	True	False		
Spare_DBW074	Int	0	False	True	True	True	False		
DAT_HYS_TEMP	DInt	L#50	False	True	True	True	False		Temp Hysteresis set from HMI
Spare_DBW080	DInt	L#0	False	True	True	True	False		
Spare_DBW084	Int	0	False	True	True	True	False		
Spare_DBW086	Int	0	False	True	True	True	False		
DAT_HYS_DEW	DInt	L#0	False	True	True	True	False		Dew hysteresis set from HMI
Spare_DBW092	DInt	L#0	False	True	True	True	False		
Spare_DBW096	Int	0	False	True	True	True	False		
Spare_DBW098	Int	0	False	True	True	True	False		
Spare_DBW100	DInt	L#0	False	True	True	True	False		

PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly] / Program blocks

HMI_STATUS_DB [DB13]

HMI_STATUS_DB Properties							
General							
Name	HMI_STATUS_DB	**Number**	13	**Type**	DB	**Language**	DB
Numbering	Manual						
Information							
Title	HMI_STATUS_DB	**Author**	MicheleBaschiera	**Comment**	Machine states To HMI	**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
Static									
Spare_DBW000	Int	0	False	True	True	True	False		
Spare_DBW002	Int	0	False	True	True	True	False		
Spare_DBW004	Int	0	False	True	True	True	False		
Spare_DBW006	Int	0	False	True	True	True	False		
Spare_DBW008	Int	0	False	True	True	True	False		
CBCS_PLC_RUN	Bool	FALSE	False	True	True	True	False		PLC is running
Spare_DBX010_1	Bool	FALSE	False	True	True	True	False		
Spare_DBX010_2	Bool	FALSE	False	True	True	True	False		
Spare_DBX010_3	Bool	FALSE	False	True	True	True	False		
Spare_DBX010_4	Bool	FALSE	False	True	True	True	False		
Spare_DBX010_5	Bool	FALSE	False	True	True	True	False		
Spare_DBX010_6	Bool	FALSE	False	True	True	True	False		
Spare_DBX010_7	Bool	FALSE	False	True	True	True	False		
GEN_QS_EMER	Bool	FALSE	False	True	True	True	False		QuickStop general safety
Spare_DBX010_8	Bool	FALSE	False	True	True	True	False		

Name	Data type	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writ-able from HMI/ OPC UA/ Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion	Comment
Spare_DBX010_9	Bool	FALSE	False	True	True	True	False		
Spare_DBX010_10	Bool	FALSE	False	True	True	True	False		
Spare_DBX010_11	Bool	false	False	True	True	True	False		
Spare_DBX010_12	Bool	false	False	True	True	True	False		
Spare_DBX010_13	Bool	false	False	True	True	True	False		
Spare_DBX010_14	Bool	false	False	True	True	True	False		
Spare_DBX010_15	Bool	false	False	True	True	True	False		
Spare_DBX010_16	Bool	false	False	True	True	True	False		
Spare_DBX010_17	Bool	false	False	True	True	True	False		
Spare_DBX010_18	Bool	false	False	True	True	True	False		
Spare_DBX010_19	Bool	false	False	True	True	True	False		
CHL_RUN_CONF	Bool	false	False	True	True	True	False		Chiller is running
Spare_DBX010_21	Bool	false	False	True	True	True	False		
CHL_SYS_RDY	Bool	false	False	True	True	True	False		Chiller system is ready
Spare_DBX010_23	Bool	false	False	True	True	True	False		
Spare_DBX010_24	Bool	false	False	True	True	True	False		
Spare_DBX010_25	Bool	false	False	True	True	True	False		
Spare_DBX010_26	Bool	false	False	True	True	True	False		
Spare_DBX010_27	Bool	false	False	True	True	True	False		
Spare_DBX010_28	Bool	false	False	True	True	True	False		
Spare_DBX010_29	Bool	false	False	True	True	True	False		
CBCS_SYS_RDY	Bool	false	False	True	True	True	False		Control system Ready
Spare_DBX010_31	Bool	false	False	True	True	True	False		
CBCS_SYS_EN	Bool	false	False	True	True	True	False		Control system enabled
Spare_DBX010_33	Bool	false	False	True	True	True	False		
CBCS_SYS_RUN	Bool	false	False	True	True	True	False		Control system running
Spare_DBX010_35	Bool	false	False	True	True	True	False		
Spare_DBX010_36	Bool	false	False	True	True	True	False		

Name	Data type	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writ-able from HMI/ OPC UA/ Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion				Comment
Spare_DBX010_37	Bool	false	False	True	True	True	False					
Spare_DBX010_38	Bool	false	False	True	True	True	False					
CBCS_SYS_STATES	Word	16#0	False	True	True	True	False					ControlSystem Machine States bits (0=idle, 1=enable, 2=running, 3=warming, 4=cooling, 5=warning, 6=fault

PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly] / Program blocks / System blocks / Program resources

R_TRIG [FB1001]

R_TRIG Properties							
General							
Name	R_TRIG	**Number**	1001	**Type**	FB	**Language**	SCL
Numbering	Automatic						
Information							
Title		**Author**	SIMATIC	**Comment**		**Family**	BIT
Version	1.0	**User-defined ID**	R_TRIG				

Name	Data type	Default value	Retain	Accessible from HMI/OPC UA/Web API	Writ-able from HMI/ OPC UA/ Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion	Comment
Input									
CLK	Bool	false	Non-retain	True	True	True	False		
Output									
Q	Bool	false	Non-retain	True	True	True	False		
InOut									
Static									
Stat_Bit	Bool	false	Non-retain	True	True	True	False		

PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly] / Program blocks / System blocks / Program resources

F_TRIG [FB1002]

F_TRIG Properties							
General							
Name	F_TRIG	**Number**	1002	**Type**	FB	**Language**	SCL
Numbering	Automatic						
Information							
Title		**Author**	SIMATIC	**Comment**		**Family**	BIT
Version	1.0	**User-defined ID**	F_TRIG				

Name	Data type	Default value	Retain	Accessible from HMI/OPC UA/Web API	Writ-able from HMI/ OPC UA/ Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion	Comment
Input									
CLK	Bool	false	Non-retain	True	True	True	False		
Output									
Q	Bool	false	Non-retain	True	True	True	False		
InOut									
Static									
Stat_Bit	Bool	false	Non-retain	True	True	True	False		

PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly]

Technology objects

This folder is empty.

PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly] / PLC tags / Default tag table [111]

PLC tags

PLC tags

	Name	Data type	Address	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Supervision				Comment
	CBS_DR_SW1	Bool	%I0.0	False	True	True	True					Coldbox door structure switch 1
	CBS_DR_SW2	Bool	%I0.1	False	True	True	True					Coldbox door structure switch 2
	CBS_DR_OS_GN	Bool	%Q0.0	False	True	True	True					Coldbox Overhead Semaphore green light
	CBS_DR_OS_OR	Bool	%Q0.1	False	True	True	True					Coldbox Overhead Semaphore green orange
	CBS_DR_OS_RD	Bool	%Q0.2	False	True	True	True					Coldbox Overhead Semaphore green red
	CHL_SYS_WRN	Bool	%I0.2	False	True	True	True					Chiller system alarm out
	CHL_EN_RUN_CMD	Bool	%Q0.3	False	True	True	True					Chiller grant command Stand-By input
	CBI_PT1000_TR1_RAW	Int	%IW96	False	True	True	True					Inner coldbox temperature sensor raw tray 1
	CBI_PT1000_TR2_RAW	Int	%IW98	False	True	True	True					Inner coldbox temperature sensor raw tray 2
	CBI_PT1000_TR3_RAW	Int	%IW100	False	True	True	True					Inner coldbox temperature sensor raw tray 3
	CBI_PT1000_TR4_RAW	Int	%IW102	False	True	True	True					Inner coldbox temperature sensor raw tray 4
	CBI_PT1000_TR5_RAW	Int	%IW104	False	True	True	True					Inner coldbox temperature sensor raw tray 5
	CBI_PT1000_TR6_RAW	Int	%IW106	False	True	True	True					Inner coldbox temperature sensor raw tray 6
	CBI_PT1000_TR7_RAW	Int	%IW108	False	True	True	True					Inner coldbox temperature sensor raw tray 7
	CBI_PT1000_TR8_RAW	Int	%IW110	False	True	True	True					Inner coldbox temperature sensor raw tray 8

	Name	Data type	Address	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Supervision				Comment
	CBI_DMT143_I1_RAW	Int	%IW112	False	True	True	True					Inner coldbox dew sensor raw 1
	CBI_DMT143_I2_RAW	Int	%IW114	False	True	True	True					Inner coldbox dew sensor raw 2
	CHL_SYS_TEMP_RAW	Int	%IW116	False	True	True	True					Chiller temperature recorder output raw
	CHL_CTRL_TEMP_CMD	Int	%QW112	False	True	True	True					Chiller out command get Warm/Cold control
	Clock_Byte	Byte	%MB30	False	True	True	True					
	Clock_10Hz	Bool	%M30.0	False	True	True	True					
	Clock_5Hz	Bool	%M30.1	False	True	True	True					
	Clock_2.5Hz	Bool	%M30.2	False	True	True	True					
	Clock_2Hz	Bool	%M30.3	False	True	True	True					
	Clock_1.25Hz	Bool	%M30.4	False	True	True	True					
	Clock_1Hz	Bool	%M30.5	False	True	True	True					
	Clock_0.625Hz	Bool	%M30.6	False	True	True	True					
	Clock_0.5Hz	Bool	%M30.7	False	True	True	True					
	CBS_DR_SW1_CLS	Bool	%M50.0	False	True	True	True					Structure door switch 1 closed
	CBS_DR_SW2_CLS	Bool	%M50.1	False	True	True	True					Structure door switch 2 closed
	CHL_SYS_RDY	Bool	%M50.2	False	True	True	True					Chiller is ready state
	CBS_DR_STAT_CLS	Bool	%M60.0	False	True	True	True					Coldbox door closed
	CTRL_TCYC_RDY	Bool	%M100.1	False	True	True	True					Control System Ready
	CTRL_TCYC_RUN	Bool	%M100.2	False	True	True	True					Control System is running
	DataSetFromHMI	Bool	%M110.0	False	True	True	True					Data as been set from the HMI
	CBI_PT_HITEMPLIMT_TR1	Bool	%M51.0	False	True	True	True					Temperature tray 1 high limit
	CBI_PT_LOWTEMPLIMT_TR1	Bool	%M52.0	False	True	True	True					Temperature tray 1 Low limit
	CBI_PT_HITEMPLIMT_TR2	Bool	%M51.1	False	True	True	True					Temperature tray 2 high limit
	CBI_PT_LOWTEMPLIMT_TR2	Bool	%M52.1	False	True	True	True					Temperature tray 2 Low limit

	Name	Data type	Address	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Supervision				Comment
	CBI_PT_HITEMPLIMT_TR3	Bool	%M51.2	False	True	True	True					Temperature tray 3 high limit
	CBI_PT_LOWTEMPLIMT_TR3	Bool	%M52.2	False	True	True	True					Temperature tray 3 Low limit
	CBI_PT_HITEMPLIMT_TR4	Bool	%M51.3	False	True	True	True					Temperature tray 4 high limit
	CBI_PT_LOWTEMPLIMT_TR4	Bool	%M52.3	False	True	True	True					Temperature tray 4 Low limit
	CBI_PT_HITEMPLIMT_TR5	Bool	%M51.4	False	True	True	True					Temperature tray 5 high limit
	CBI_PT_LOWTEMPLIMT_TR5	Bool	%M52.4	False	True	True	True					Temperature tray 5 Low limit
	CBI_PT_HITEMPLIMT_TR6	Bool	%M51.5	False	True	True	True					Temperature tray 6 high limit
	CBI_PT_LOWTEMPLIMT_TR6	Bool	%M52.5	False	True	True	True					Temperature tray 6 Low limit
	CBI_PT_HITEMPLIMT_TR7	Bool	%M51.6	False	True	True	True					Temperature tray 7 high limit
	CBI_PT_LOWTEMPLIMT_TR7	Bool	%M52.6	False	True	True	True					Temperature tray 7 Low limit
	CBI_PT_HITEMPLIMT_TR8	Bool	%M51.7	False	True	True	True					Temperature tray 8 high limit
	CBI_PT_LOWTEMPLIMT_TR8	Bool	%M52.7	False	True	True	True					Temperature tray 8 Low limit
	CTRL_TCYC_IDLE	Bool	%M100.0	False	True	True	True					Control System in idle state
	CTRL_TCYC_GO_WARM	Bool	%M100.3	False	True	True	True					Control System is getting warmer
	CTRL_TCYC_GO_COLD	Bool	%M100.4	False	True	True	True					Control system is getting colder
	CTRL_TCYC_WRN	Bool	%M100.5	False	True	True	True					Control System warning state
	CTRL_TCYC_FLT	Bool	%M100.6	False	True	True	True					Control System fault state
	CHL_EN_RUN	Bool	%M50.3	False	True	True	True					Chiller is ready and enabled
	CTRL_TCYC_AUTO	Bool	%M101.1	False	True	True	True					Control system thermal cycle auto mode
	CTRL_TCYC_MAN	Bool	%M101.0	False	True	True	True					Control system thermal cycle manual mode
	CTRL_TCYC_SETUP	Bool	%M100.7	False	True	True	True					Contro System setting up
	CTRL_TCYC_MINTRG_REACH	Bool	%M101.2	False	True	True	True					Actual temp reach min target
	CTRL_TCYC_MAXTRG_REACH	Bool	%M101.3	False	True	True	True					Actual temp reach max target
	CTRL_SYS_MAXTRG_SET	Bool	%M102.1	False	True	True	True					Max target temperature SET

	Name	Data type	Address	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Supervision				Comment
	CTRL_SYS_MINTRG_SET	Bool	%M102.0	False	True	True	True					Min target temperature SET
	CTRL_TCYC_STATES	Word	%MW100	False	True	True	True					Control system machine states
	HMI_SYS_STATES	Word	%MW105	False	True	True	True					Control system machine states to HMI
	CTRL_SYS_TIM_ELAP_MIN	Bool	%M102.2	False	True	True	True					Timer cold Idle state elapsed
	CTRL_SYS_TIM_ELAP_MAX	Bool	%M102.3	False	True	True	True					Timer warm Idle state elapsed
	CTRL_SYS_AUTO_SET	Bool	%M102.4	False	True	True	True					Auto mode is set
	CTRL_ALRM_MAIN	Bool	%M55.0	False	True	True	True					General System alarm to disable ready state
	HMI_SYS_ALARM	Word	%MW53	False	True	True	True					Control system machine Alarms to HMI
	CBI_PT_SEL_TR1	Bool	%M40.0	False	True	True	True					Sensor selected PT1
	CBI_PT_SEL_TR2	Bool	%M40.1	False	True	True	True					Sensor selected PT2
	CBI_PT_SEL_TR3	Bool	%M40.2	False	True	True	True					Sensor selected PT3
	CBI_PT_SEL_TR4	Bool	%M40.3	False	True	True	True					Sensor selected PT4
	CBI_PT_SEL_TR5	Bool	%M40.4	False	True	True	True					Sensor selected PT5
	CBI_PT_SEL_TR6	Bool	%M40.5	False	True	True	True					Sensor selected PT6
	CBI_PT_SEL_TR7	Bool	%M40.6	False	True	True	True					Sensor selected PT7
	CBI_PT_SEL_TR8	Bool	%M40.7	False	True	True	True					Sensor selected PT8
	CBI_DEW_SEL_S1	Bool	%M41.0	False	True	True	True					Sensor selected DEW1
	CBI_DEW_SEL_S2	Bool	%M41.1	False	True	True	True					Sensor selected DEW2
	CBI_PT_SEL_AMT	Byte	%MB40	False	True	True	True					PT selected amount

PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly] / PLC tags / Default tag table [111]

User constants

User constants				
	Name	Data type	Value	Comment

PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly] / PLC data types

UDT01_HMI_CMD

UDT01_HMI_CMD Properties							
General							
Name	UDT01_HMI_CMD	**Number**	1	**Type**	UDT	**Language**	
Numbering							
Information							
Title		**Author**		**Comment**		**Family**	
Version		**User-defined ID**					

Name	Data type	Default value	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
STA	Bool	false	True	True	True	False		Start
AUTO	Bool	true	True	True	True	False		Auto
STP	Bool	false	True	True	True	False		Stop
ON	Bool	false	True	True	True	False		On
OFF	Bool	false	True	True	True	False		Off
ALM_RES	Bool	false	True	True	True	False		Alarm reset
MAN	Bool	false	True	True	True	False		Manual
LAMP_TST	Bool	false	True	True	True	False		Lamp test
DR_LK	Bool	false	True	True	True	False		Door lock
DR_ULK	Bool	false	True	True	True	False		Door Unlock
REST	Bool	false	True	True	True	False		Restart
INCR_UP	Bool	false	True	True	True	False		Increase/Up
DECR_DW	Bool	false	True	True	True	False		Decrease/Down
ACT_DATA	Bool	false	True	True	True	False		Actualize Data
SYNCH	Bool	false	True	True	True	False		Synchronization
RES	Bool	false	True	True	True	False		Reset

Name	Data type	Default value	Accessible from HMI/OPC UA/Web API	Writ-able from HMI/ OPC UA/ Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion				Comment
PTSEL1	Bool	true	True	True	True	False					PT Selection 1
PTSEL2	Bool	true	True	True	True	False					PT Selection 2
PTSEL3	Bool	true	True	True	True	False					PT Selection 3
PTSEL4	Bool	true	True	True	True	False					PT Selection 4
PTSEL5	Bool	true	True	True	True	False					PT Selection 5
PTSEL6	Bool	true	True	True	True	False					PT Selection 6
PTSEL7	Bool	true	True	True	True	False					PT Selection 7
PTSEL8	Bool	true	True	True	True	False					PT Selection 8
DEWSEL1	Bool	true	True	True	True	False					Dew selection 1
DEWSEL2	Bool	true	True	True	True	False					Dew selection 2

PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly] / PLC data types

System data types

This folder is empty.

PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly] / Watch and force tables

Force table

Name	Address	Display format	Force value	Comment

PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly] / Watch and force tables

Watch table_1

Name	Address	Display format	Modify value	Comment
"CBI_PT1000_TR1_RAW"	%IW96	DEC+/-		

PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly]

Traces

Name

PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly] / Traces

Measurements

This folder is empty.

PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly] / Traces

Combined measurements

Name

PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly] / OPC UA communication

Server interfaces

This folder is empty.

PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly]

PLC alarm text lists

This folder is empty.

PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly] / Local modules

PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly]

PLC_DC/DC/RLY					
General\Project information					
Name	PLC_DC/DC/RLY	**Author**	MicheleBaschiera	**Comment**	This project is protected by CERN copyright, ensuring that all original content, including text, images, and designs, are legally safeguarded. Unauthorized reproduction, distribution, or use of any part of this project without explicit written permission from the copyright holder is strictly prohibited and may result in legal action.
Slot	1	**Rack**	0		
General\Catalog information					
Short designation	CPU 1214C DC/DC/Rly	**Description**	Work memory 150 KB; 24VDC power supply with DI14 x 24VDC SINK/SOURCE, DQ10 x relay and AI2 on board; 6 high-speed counters and 4 pulse outputs on-board; signal board expands on-board I/O; up to 3 communication modules for serial communication; up to 8 signal modules for I/O expansion; PROFINET IO controller, I-device, transport protocol TCP/IP, secure Open User Communication, S7 communication, Web server, OPC UA: Server DA	**Article number**	6ES7 214-1HG40-0XB0
Firmware version	V4.6		False		
General\Identification & Maintenance					
Plant designation		**Location identifier**		**Installation date**	2024-08-08 09:33:41.107
Additional information					
General\Checksums					
Text lists	FA 70 E8 75 1D 5A 8E 29	**Software**	BD 89 9D 84 69 26 21 A8		
PROFINET interface [X1]\General					
Name	PROFINET interface_1	**Author**	mbaschie	**Comment**	

PROFINET interface [X1]\General\Project information					
Name	DI 14/DQ 10_1	**Comment**		**Name**	AI 2_1
Comment					
PROFINET interface [X1]\Ethernet addresses\Interface networked with					
Subnet:	PN/IE_1				
PROFINET interface [X1]\Ethernet addresses\Internet protocol version 4 (IPv4)					
IP configuration	Set IP address in the project	**IP address:**	192.168.0.1	**Subnet mask:**	255.255.255.0
Use router	False				
PROFINET interface [X1]\Ethernet addresses\PROFINET					
PROFINET device name is set directly at the device	False	**Generate PROFINET device name automatically**	True	**PROFINET device name:**	plc_dc/dc/rly
Converted name:	plcxbdcxrdcxrrly34b4	**Device number:**	0		
PROFINET interface [X1]\Time synchronization					
Enable time synchronization via NTP server	Enable time synchronization via NTP server		IP addresses	**Server 1**	0.0.0.0
Server 2	0.0.0.0	**Server 3**	0.0.0.0	**Server 4**	0.0.0.0
Update interval	10sec			**CPU synchronizes the modules of the device.**	No synchronization
PROFINET interface [X1]\Digital inputs\Channel0					
Channel address	I0.0	**Input filters**	6.4 millisec	**Enable pulse catch**	0
PROFINET interface [X1]\Digital inputs\Channel0					
Enable rising edge detection	0	**Prefix Event Rising Edge**	49152	**Event name:**	0
Hardware interrupt:	0	**Rising edge0**	Rising edge0		
PROFINET interface [X1]\Digital inputs\Channel0					
Enable falling edge detection	0	**Prefix Event Falling Edge**	49280	**Event name:**	0
Hardware interrupt:	0	**Falling edge0**	Falling edge0		
PROFINET interface [X1]\Digital inputs\Channel1					
Channel address	I0.1	**Input filters**	6.4 millisec	**Enable pulse catch**	0
PROFINET interface [X1]\Digital inputs\Channel1					
Enable rising edge detection	0	**Prefix Event Rising Edge**	49153	**Event name:**	0
Hardware interrupt:	0	**Rising edge1**	Rising edge1		

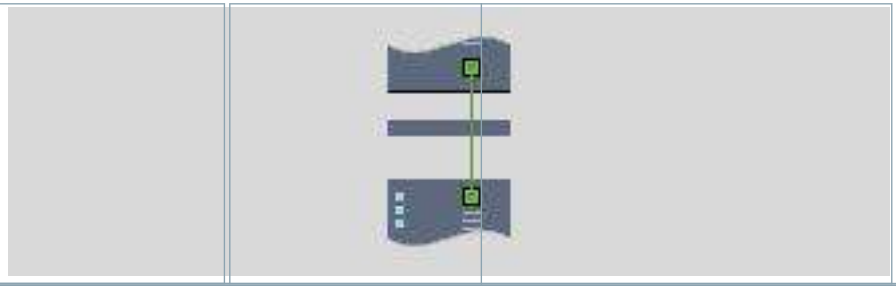
PROFINET interface [X1]\Digital inputs\Channel1					
Enable falling edge detection	0	**Prefix Event Falling Edge**	49281	**Event name:**	0
Hardware interrupt:	0	**Falling edge1**	Falling edge1		
PROFINET interface [X1]\Digital inputs\Channel2					
Channel address	I0.2	**Input filters**	6.4 millisec	**Enable pulse catch**	0
PROFINET interface [X1]\Digital inputs\Channel2					
Enable rising edge detection	0	**Prefix Event Rising Edge**	49154	**Event name:**	0
Hardware interrupt:	0	**Rising edge2**	Rising edge2		
PROFINET interface [X1]\Digital inputs\Channel2					
Enable falling edge detection	0	**Prefix Event Falling Edge**	49282	**Event name:**	0
Hardware interrupt:	0	**Falling edge2**	Falling edge2		
PROFINET interface [X1]\Digital inputs\Channel3					
Channel address	I0.3	**Input filters**	6.4 millisec	**Enable pulse catch**	0
PROFINET interface [X1]\Digital inputs\Channel3					
Enable rising edge detection	0	**Prefix Event Rising Edge**	49155	**Event name:**	0
Hardware interrupt:	0	**Rising edge3**	Rising edge3		
PROFINET interface [X1]\Digital inputs\Channel3					
Enable falling edge detection	0	**Prefix Event Falling Edge**	49283	**Event name:**	0
Hardware interrupt:	0	**Falling edge3**	Falling edge3		
PROFINET interface [X1]\Digital inputs\Channel4					
Channel address	I0.4	**Input filters**	6.4 millisec	**Enable pulse catch**	0
PROFINET interface [X1]\Digital inputs\Channel4					
Enable rising edge detection	0	**Prefix Event Rising Edge**	49156	**Event name:**	0
Hardware interrupt:	0	**Rising edge4**	Rising edge4		
PROFINET interface [X1]\Digital inputs\Channel4					
Enable falling edge detection	0	**Prefix Event Falling Edge**	49284	**Event name:**	0
Hardware interrupt:	0	**Falling edge4**	Falling edge4		
PROFINET interface [X1]\Digital inputs\Channel5					
Channel address	I0.5	**Input filters**	6.4 millisec	**Enable pulse catch**	0
PROFINET interface [X1]\Digital inputs\Channel5					
Enable rising edge detection	0	**Prefix Event Rising Edge**	49157	**Event name:**	0

Hardware interrupt:	0	**Rising edge5**	Rising edge5		
PROFINET interface [X1]\Digital inputs\Channel5					
Enable falling edge detection	0	**Prefix Event Falling Edge**	49285	**Event name:**	0
Hardware interrupt:	0	**Falling edge5**	Falling edge5		
PROFINET interface [X1]\Digital inputs\Channel6					
Channel address	I0.6	**Input filters**	6.4 millisec	**Enable pulse catch**	0
PROFINET interface [X1]\Digital inputs\Channel6					
Enable rising edge detection	0	**Prefix Event Rising Edge**	49158	**Event name:**	0
Hardware interrupt:	0	**Rising edge6**	Rising edge6		
PROFINET interface [X1]\Digital inputs\Channel6					
Enable falling edge detection	0	**Prefix Event Falling Edge**	49286	**Event name:**	0
Hardware interrupt:	0	**Falling edge6**	Falling edge6		
PROFINET interface [X1]\Digital inputs\Channel7					
Channel address	I0.7	**Input filters**	6.4 millisec	**Enable pulse catch**	0
PROFINET interface [X1]\Digital inputs\Channel7					
Enable rising edge detection	0	**Prefix Event Rising Edge**	49159	**Event name:**	0
Hardware interrupt:	0	**Rising edge7**	Rising edge7		
PROFINET interface [X1]\Digital inputs\Channel7					
Enable falling edge detection	0	**Prefix Event Falling Edge**	49287	**Event name:**	0
Hardware interrupt:	0	**Falling edge7**	Falling edge7		
PROFINET interface [X1]\Digital inputs\Channel8					
Channel address	I1.0	**Input filters**	6.4 millisec	**Enable pulse catch**	0
PROFINET interface [X1]\Digital inputs\Channel8					
Enable rising edge detection	0	**Prefix Event Rising Edge**	49160	**Event name:**	0
Hardware interrupt:	0	**Rising edge8**	Rising edge8		
PROFINET interface [X1]\Digital inputs\Channel8					
Enable falling edge detection	0	**Prefix Event Falling Edge**	49288	**Event name:**	0
Hardware interrupt:	0	**Falling edge8**	Falling edge8		
PROFINET interface [X1]\Digital inputs\Channel9					
Channel address	I1.1	**Input filters**	6.4 millisec	**Enable pulse catch**	0

PROFINET interface [X1]\Digital inputs\Channel9					
Enable rising edge detection	0	**Prefix Event Rising Edge**	49161	**Event name:**	0
Hardware interrupt:	0	**Rising edge9**	Rising edge9		
PROFINET interface [X1]\Digital inputs\Channel9					
Enable falling edge detection	0	**Prefix Event Falling Edge**	49289	**Event name:**	0
Hardware interrupt:	0	**Falling edge9**	Falling edge9		
PROFINET interface [X1]\Digital inputs\Channel10					
Channel address	I1.2	**Input filters**	6.4 millisec	**Enable pulse catch**	0
PROFINET interface [X1]\Digital inputs\Channel10					
Enable rising edge detection	0	**Prefix Event Rising Edge**	49162	**Event name:**	0
Hardware interrupt:	0	**Rising edge10**	Rising edge10		
PROFINET interface [X1]\Digital inputs\Channel10					
Enable falling edge detection	0	**Prefix Event Falling Edge**	49290	**Event name:**	0
Hardware interrupt:	0	**Falling edge10**	Falling edge10		
PROFINET interface [X1]\Digital inputs\Channel11					
Channel address	I1.3	**Input filters**	6.4 millisec	**Enable pulse catch**	0
PROFINET interface [X1]\Digital inputs\Channel11					
Enable rising edge detection	0	**Prefix Event Rising Edge**	49163	**Event name:**	0
Hardware interrupt:	0	**Rising edge11**	Rising edge11		
PROFINET interface [X1]\Digital inputs\Channel11					
Enable falling edge detection	0	**Prefix Event Falling Edge**	49291	**Event name:**	0
Hardware interrupt:	0	**Falling edge11**	Falling edge11		
PROFINET interface [X1]\Digital inputs\Channel12					
Channel address	I1.4	**Input filters**	6.4 millisec	**Enable pulse catch**	0
PROFINET interface [X1]\Digital inputs\Channel13					
Channel address	I1.5	**Input filters**	6.4 millisec	**Enable pulse catch**	0
PROFINET interface [X1]\Analog inputs\Noise reduction					
Integration time	50 Hz (20 ms)				
PROFINET interface [X1]\Analog inputs\Channel0					
Channel address	IW64	**Measurement type**	Voltage	**Voltage range**	0..10 V
Smoothing	Weak (4 cycles)			**Enable overflow diag-nostics**	1

PROFINET interface [X1]\Analog inputs\Channel1					
Channel address	IW66	**Measurement type**	Voltage	**Voltage range**	0..10 V
Smoothing	Weak (4 cycles)			**Enable overflow diagnostics**	1
PROFINET interface [X1]\Digital outputs					
Reaction to CPU STOP	Use substitute value				
PROFINET interface [X1]\Digital outputs\Channel0					
Channel address	Q0.0	**Substitute a value of 1 on a change from RUN to STOP.**	0		
PROFINET interface [X1]\Digital outputs\Channel1					
Channel address	Q0.1	**Substitute a value of 1 on a change from RUN to STOP.**	0		
PROFINET interface [X1]\Digital outputs\Channel2					
Channel address	Q0.2	**Substitute a value of 1 on a change from RUN to STOP.**	0		
PROFINET interface [X1]\Digital outputs\Channel3					
Channel address	Q0.3	**Substitute a value of 1 on a change from RUN to STOP.**	0		
PROFINET interface [X1]\Digital outputs\Channel4					
Channel address	Q0.4	**Substitute a value of 1 on a change from RUN to STOP.**	0		
PROFINET interface [X1]\Digital outputs\Channel5					
Channel address	Q0.5	**Substitute a value of 1 on a change from RUN to STOP.**	0		
PROFINET interface [X1]\Digital outputs\Channel6					
Channel address	Q0.6	**Substitute a value of 1 on a change from RUN to STOP.**	0		
PROFINET interface [X1]\Digital outputs\Channel7					
Channel address	Q0.7	**Substitute a value of 1 on a change from RUN to STOP.**	0		

PROFINET interface [X1]\Digital outputs\Channel8					
Channel address	Q1.0	**Substitute a value of 1 on a change from RUN to STOP.**	0		
PROFINET interface [X1]\Digital outputs\Channel9					
Channel address	Q1.1	**Substitute a value of 1 on a change from RUN to STOP.**	0		
PROFINET interface [X1]\Operating mode					
IO controller	True	**IO system**		**Device number**	0
IO device	False				
PROFINET interface [X1]\I/O addresses\Input addresses					
Start address	0.0	**End address**	1.7	**Organization block**	0
Process image	0				
PROFINET interface [X1]\I/O addresses\Input addresses					
Start address	64	**End address**	67	**Organization block**	0
Process image	0				
PROFINET interface [X1]\I/O addresses\Output addresses					
Start address	0.0	**End address**	1.7	**Organization block**	0
Process image	0				
PROFINET interface [X1]\Advanced options\Interface options					
Support device replacement without exchangeable medium	True	**Permit overwriting of device names of all assigned IO devices**	False	**Use IEC V2.2 LLDP mode**	False
Keep-Alive connection monitoring:	30s				
PROFINET interface [X1]\Advanced options\Real time settings\IO communication					
Send clock:	1.000ms				
PROFINET interface [X1]\Advanced options\Real time settings\Real time options					
Calculated bandwidth for cyclic IO data:	0.000ms	**Calculated bandwidth for cyclic IO data:**	0.000%		
PROFINET interface [X1]\Advanced options\Port [X1 P1]\General					
Name	Port_1	**Author**	mbaschie	**Comment**	
PROFINET interface [X1]\Advanced options\Port [X1 P1]\Port interconnection\Local port:					
Local port:	PLC_DC/DC/RLY\PROFINET interface_1 [X1]\Port_1 [X1 P1]	**Medium:**	Copper	**Cable name:**	---



PROFINET interface [X1]\Advanced options\Port [X1 P1]\Port interconnection\Partner port:					
	Monitoring of partner port is executed	**Partner port:**	HMI01.IE_CP_1\PROFINET Interface_1 [X1]\Port_1 [X1 P1 R]	**Medium:**	Copper
Cable length:	< 100 m				
PROFINET interface [X1]\Advanced options\Port [X1 P1]\Port options\Activate					
Activate this port for use	True				
PROFINET interface [X1]\Advanced options\Port [X1 P1]\Port options\Connection					
Transmission rate / duplex:	Automatic	**Monitor**	False	**Enable autonegotia-tion**	True
PROFINET interface [X1]\Advanced options\Port [X1 P1]\Port options\Boundaries					
End of detection of accessible devices	False	**End of topology dis-covery**	False	**End of the sync do-main**	False
PROFINET interface [X1]\Web server access					
Enable Web server for the IP address of this interface	False	**The Web server must also be activated in the properties of the PLC.**			
High speed counters (HSC)\HSC1\General\Enable					
Enable this high speed counter	0	**Enable this high speed counter**	0	**Enable this high speed counter**	0
Enable this high speed counter	0	**Enable this high speed counter**	0	**Enable this high speed counter**	0
High speed counters (HSC)\HSC1\General\Project information					
Name	HSC_1	**Comment**		**Name**	HSC_2
Comment		**Name**	HSC_3	**Comment**	
Name	HSC_4	**Comment**		**Name**	HSC_5
Comment		**Name**	HSC_6	**Comment**	
High speed counters (HSC)\HSC1\I/O addresses\Input addresses					
Start address	1000.0	**End address**	1003.7	**Start address**	1004.0

End address	1007.7	**Organization block**	0	**Start address**	1008.0
End address	1011.7	**Organization block**	0	**Process image**	0
Start address	1012.0	**End address**	1015.7	**Organization block**	0
Process image	0	**Start address**	1016.0	**End address**	1019.7
Organization block	0	**Process image**	0	**Start address**	1020.0
End address	1023.7	**Organization block**	0	**Process image**	0
Organization block	0	**Process image**	0	**Process image**	0
Pulse generators (PTO/PWM)\PTO1/PWM1\General\Enable					
Enable this pulse generator	0	**Enable this pulse generator**	0		
Pulse generators (PTO/PWM)\PTO1/PWM1\General\Project information					
Name	Pulse_1	**Comment**		**Name**	Pulse_2
Comment					
Pulse generators (PTO/PWM)\PTO1/PWM1\I/O addresses\Output addresses					
Start address	1000.0	**End address**	1001.7	**Start address**	1002.0
End address	1003.7	**Organization block**	0	**Organization block**	0
Process image	0	**Process image**	0		
Startup					
Startup after POWER ON	Warm restart - mode before POWER OFF	**Comparison preset to actual configuration**	Startup CPU even if mismatch	**Configuration time**	60000ms
OBs should be interruptible	0				
Cycle					
Cycle monitoring time [ms]	150ms			**Enable minimum cycle time for cyclic OBs**	0
Minimum cycle time	1ms				
Communication load					
Cycle load due to communication [%]	20%				
System and clock memory\System memory bits					
Enable the use of system memory byte	0	**Address of system memory byte (MBx)**	1	**First cycle**	
Diagnostic status changed		**Always 1 (high)**		**Always 0 (low)**	
System and clock memory\Clock memory bits					
Enable the use of clock memory byte	1	**Address of clock memory byte (MBx)**	30	**10 Hz clock**	%M30.0 (Clock_10Hz)
5 Hz clock	%M30.1 (Clock_5Hz)	**2.5 Hz clock**	%M30.2 (Clock_2.5Hz)	**2 Hz clock**	%M30.3 (Clock_2Hz)
1.25 Hz clock	%M30.4 (Clock_1.25Hz)	**1 Hz clock**	%M30.5 (Clock_1Hz)	**0.625 Hz clock**	%M30.6 (Clock_0.625Hz)

0.5 Hz clock	%M30.7 (Clock_0.5Hz)				
Web server\General					
Activate Web server on all modules of this device	False	**Permit access only with HTTPS**	False		
Web server\Automatic update					
Enable automatic update	True	**Update interval**	0s		
Web server\User management					
User name			**User rights**		
Everybody					
Web server\User-defined web pages					
Application name	**HTML source path**	**Default HTML page**	**Files with dynamic content**	**Web DB number**	**Fragment DB number**
		index.htm	.htm;.html	333	334
Web server\Overview of interfaces					
Device		**Interface**		**Enabled web server access**	
PLC_DC/DC/RLY		PROFINET interface_1		False	
User interface languages					
Assign project language			**User interface languages**		
None			German		
None			English		
None			French		
None			Spanish		
None			Italian		
None			Chinese (simplified)		
Time of day\Local time					
Time zone	(UTC +01:00) Berlin, Bern, Brussels, Rome, Stockholm, Vienna				
Time of day\Daylight saving time					
Activate daylight saving time	0	**Difference between standard and daylight saving time**	60min		
Time of day\Daylight saving time\Start of daylight saving time					
Starting week of the month:	Last		Sunday	**in**	March
at	1:00 a.m.				
Time of day\Daylight saving time\Start of standard time					
	Last		Sunday	**in**	October

at	2:00 a.m.				
Protection & Security					
Level of protection	No protection				
Protection & Security\Connection mechanisms					
Permit access with PUT/GET communication from remote partner	True				
Protection & Security\Security event					
Summarize diagnostics in case of high message volume	True	**Length of an interval**	20	**Unit**	seconds
Protection & Security\External load memory					
Disable copying from internal load memory to external load memory	False				
Advanced configuration\DnsParameterConfigurationMenu					
The Tree-Node DnsConfigurationMenu was not filled by some ACF					
Advanced configuration\Configuration control\Configuration control for central configuration					
Allow to reconfigure the device via the user program	0				
Advanced configuration\SNMP\SNMP configuration (Simple Network Management Protocol).					
Activate SNMP	True				
Configuration control\Configuration control for central configuration					
Allow to reconfigure the device via the user program	0				

Connection resources\				
	Station resources - Reserved - Maximum	**Station resources - Reserved - Configured**	**Station resources - Dynamic - Configured**	**Module resources - PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly] - Configured**
Maximum number of resources:		34	34	68
	Maximum	Configured	Configured	Configured
PG communication:	4	-	-	-
HMI communication:	12	2	0	2
S7 communication:	8	0	0	0
Open user communication:	8	0	0	0
Web communication:	2	-	-	-
OPC UA client/server communication:	0	-	-	-
Other communication:	-	-	0	0
Total resources used:		2	0	2
Available resources:		32	34	66

Overview of addresses\Overview of addresses\Overview of addresses					
Inputs	True	**Outputs**	True	**Address gaps**	False
Slot	True				

Type	Addr. from	Addr. to	Module	PIP	Device name	Device number	Size	Master / IO system	Rack	Slot
I	0	1	DI 14/DQ 10_1	Automatic update	PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly]	-	2 Bytes	-	0	1 1
O	0	1	DI 14/DQ 10_1	Automatic update	PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly]	-	2 Bytes	-	0	1 1
I	64	67	AI 2_1	Automatic update	PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly]	-	4 Bytes	-	0	1 2
I	1000	1003	HSC_1	Automatic update	PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly]	-	4 Bytes	-	0	1 16
I	1004	1007	HSC_2	Automatic update	PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly]	-	4 Bytes	-	0	1 17
I	1008	1011	HSC_3	Automatic update	PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly]	-	4 Bytes	-	0	1 18
I	1012	1015	HSC_4	Automatic update	PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly]	-	4 Bytes	-	0	1 19
I	1016	1019	HSC_5	Automatic update	PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly]	-	4 Bytes	-	0	1 20
I	1020	1023	HSC_6	Automatic update	PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly]	-	4 Bytes	-	0	1 21
O	1000	1001	Pulse_1	Automatic update	PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly]	-	2 Bytes	-	0	1 32
O	1002	1003	Pulse_2	Automatic update	PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly]	-	2 Bytes	-	0	1 33
O	1004	1005	Pulse_3	Automatic update	PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly]	-	2 Bytes	-	0	1 34
O	1006	1007	Pulse_4	Automatic update	PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly]	-	2 Bytes	-	0	1 35

Type	Addr. from	Addr. to	Module	PIP	Device name	Device number	Size	Master / IO system	Rack	Slot
I	96	111	A100	Automatic update	PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly]	-	16 Bytes	-	0	2
I	112	119	A101	Automatic update	PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly]	-	8 Bytes	-	0	3
O	112	115	A101	Automatic update	PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly]	-	4 Bytes	-	0	3

PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly] / Local modules

A100

A100					
General\Project information					
Name	A100	**Author**	mbaschie	**Comment**	
Slot	2				
General\Catalog information					
Short designation	SM 1231 AI8 x RTD	**Description**	Analog input module AI8 x RTD	**Article number**	6ES7 231-5PF32-0XB0
Firmware version	V2.0		False		
AI 8xRTD\Project information					
Name	A100	**Comment**			
AI 8xRTD\Module diagnostics					
Enable power supply diagnostics	1	**Additional diagnostics may be selected for each input/output.**			
AI 8xRTD\Analog inputs\Noise reduction					
Integration time	50 Hz (20 ms)				
AI 8xRTD\Analog inputs\Channel0					
Channel address	IW96	**Measurement type**	Thermal resistor (4-wire)	**Thermal resistor**	Pt 1000 standard range
Temperature coefficient	Pt 0.00385055 ohms/ohms/°C (DIN EN 60751)	**Temperature scale**	Celsius	**Smoothing**	Weak (4 cycles)
		Enable broken wire diagnostics	0	**Enable overflow diagnostics**	1
Enable underflow diagnostics	1				
AI 8xRTD\Analog inputs\Channel1					
Channel address	IW98	**Measurement type**	Thermal resistor (4-wire)	**Thermal resistor**	Pt 1000 standard range
Temperature coefficient	Pt 0.00385055 ohms/ohms/°C (DIN EN 60751)	**Temperature scale**	Celsius	**Smoothing**	Weak (4 cycles)
		Enable broken wire diagnostics	0	**Enable overflow diagnostics**	1
Enable underflow diagnostics	1				
AI 8xRTD\Analog inputs\Channel2					
Channel address	IW100	**Measurement type**	Thermal resistor (4-wire)	**Thermal resistor**	Pt 1000 standard range

Temperature coefficient	Pt 0.00385055 ohms/ohms/°C (DIN EN 60751)	**Temperature scale**	Celsius	**Smoothing**	Weak (4 cycles)
		Enable broken wire diagnostics	0	**Enable overflow diagnostics**	1
Enable underflow diagnostics	1				
AI 8xRTD\Analog inputs\Channel3					
Channel address	IW102	**Measurement type**	Thermal resistor (4-wire)	**Thermal resistor**	Pt 1000 standard range
Temperature coefficient	Pt 0.00385055 ohms/ohms/°C (DIN EN 60751)	**Temperature scale**	Celsius	**Smoothing**	Weak (4 cycles)
		Enable broken wire diagnostics	0	**Enable overflow diagnostics**	1
Enable underflow diagnostics	1				
AI 8xRTD\Analog inputs\Channel4					
Channel address	IW104	**Measurement type**	Thermal resistor (4-wire)	**Thermal resistor**	Pt 1000 standard range
Temperature coefficient	Pt 0.00385055 ohms/ohms/°C (DIN EN 60751)	**Temperature scale**	Celsius	**Smoothing**	Weak (4 cycles)
		Enable broken wire diagnostics	0	**Enable overflow diagnostics**	1
Enable underflow diagnostics	1				
AI 8xRTD\Analog inputs\Channel5					
Channel address	IW106	**Measurement type**	Thermal resistor (4-wire)	**Thermal resistor**	Pt 1000 standard range
Temperature coefficient	Pt 0.00385055 ohms/ohms/°C (DIN EN 60751)	**Temperature scale**	Celsius	**Smoothing**	Weak (4 cycles)
		Enable broken wire diagnostics	0	**Enable overflow diagnostics**	1
Enable underflow diagnostics	1				
AI 8xRTD\Analog inputs\Channel6					
Channel address	IW108	**Measurement type**	Thermal resistor (4-wire)	**Thermal resistor**	Pt 1000 standard range
Temperature coefficient	Pt 0.00385055 ohms/ohms/°C (DIN EN 60751)	**Temperature scale**	Celsius	**Smoothing**	Weak (4 cycles)
		Enable broken wire diagnostics	0	**Enable overflow diagnostics**	1
Enable underflow diagnostics	1				
AI 8xRTD\Analog inputs\Channel7					
Channel address	IW110	**Measurement type**	Thermal resistor (4-wire)	**Thermal resistor**	Pt 1000 standard range

Temperature coefficient	Pt 0.00385055 ohms/ohms/°C (DIN EN 60751)	**Temperature scale**	Celsius	**Smoothing**	Weak (4 cycles)
		Enable broken wire diagnostics	0	**Enable overflow diagnostics**	1
Enable underflow diagnostics	1				
AI 8xRTD\I/O addresses\Input addresses					
Start address	96	**End address**	111	**Organization block**	0
Process image	0				

PLC_DC/DC/RLY [CPU 1214C DC/DC/Rly] / Local modules

A101

A101					
General\Project information					
Name	A101	**Author**	mbaschie	**Comment**	
Slot	3				
General\Catalog information					
Short designation	SM 1234 AI4/AQ2	**Description**	Analog input/output module AI4 + AQ2; plug-in terminal blocks; inputs: 13 bits, 2.5V, 5V, 10V and 0 to 20 mA; selectable frequency suppression; selectable smoothing; configurable diagnostics; outputs: +/-10V and 0 to 20 mA; configurable diagnostics; configurable substitute value for output	**Article number**	6ES7 234-4HE30-0XB0
Firmware version	V1.0		False		
AI 4/AQ 2\Project information					
Name	AI 4x13BIT/AQ 2x14BIT_1	**Comment**			
AI 4/AQ 2\Module diagnostics					
Enable power supply diagnostics	1	**Additional diagnostics may be selected for each input/output.**			
AI 4/AQ 2\Analog inputs\Noise reduction					
Integration time	50 Hz (20 ms)				
AI 4/AQ 2\Analog inputs\Channel0					
Channel address	IW112	**Measurement type**	Current	**Current range**	0..20 mA
Smoothing	Weak (4 cycles)			**Enable overflow diagnostics**	1
Enable underflow diagnostics	1				
AI 4/AQ 2\Analog inputs\Channel1					
Channel address	IW114	**Measurement type**	Current	**Current range**	0..20 mA
Smoothing	Weak (4 cycles)			**Enable overflow diagnostics**	1
Enable underflow diagnostics	1				

AI 4/AQ 2\Analog inputs\Channel2					
Channel address	IW116	**Measurement type**	Current	**Current range**	0..20 mA
Smoothing	Weak (4 cycles)			**Enable overflow diagnostics**	1
Enable underflow diagnostics	1				
AI 4/AQ 2\Analog inputs\Channel3					
Channel address	IW118	**Measurement type**	Current	**Current range**	0..20 mA
Smoothing	Weak (4 cycles)			**Enable overflow diagnostics**	1
Enable underflow diagnostics	1				
AI 4/AQ 2\Analog outputs					
Reaction to CPU STOP	Use substitute value				
AI 4/AQ 2\Analog outputs\Channel0					
Channel address	QW112	**Analog output type**	Current	**Current range**	0..20 mA
Substitute value for channel on a change from RUN to STOP	15.000mA			**Enable broken wire diagnostics**	0
Enable overflow diagnostics	1	**Enable underflow diagnostics**	1		
AI 4/AQ 2\Analog outputs\Channel1					
Channel address	QW114	**Analog output type**	Current	**Current range**	0..20 mA
Substitute value for channel on a change from RUN to STOP	15.000mA			**Enable broken wire diagnostics**	0
Enable overflow diagnostics	1	**Enable underflow diagnostics**	1		
AI 4/AQ 2\I/O addresses\Input addresses					
Start address	112	**End address**	119	**Organization block**	0
Process image	0				
AI 4/AQ 2\I/O addresses\Output addresses					
Start address	112	**End address**	115	**Organization block**	0
Process image	0				